

**GETULIO VARGAS FOUNDATION
SCHOOL OF APPLIED MATHEMATICS
GRADUATE PROGRAM IN APPLIED MATHEMATICS**

**Stochastic Supply Curves and Liquidity Costs:
Estimation for Brazilian Equities**

Guilherme Hideo Assaoka Hossaka

Rio de Janeiro - RJ

June 2018

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Guilherme Hideo Assaoka Hossaka

Dissertation presented as a partial requirement for the title of Master in Mathematics

Supervisor: Prof. Dr. Margaret Armstrong

Rio de Janeiro - RJ

June 2018

Hossaka, Guilherme Hideo Assaoka

Stochastic supply curves and liquidity costs: estimation for Brazilian equities /
Guilherme Hideo Assaoka Hossaka. – 2018.

117 f.

Dissertação (mestrado) – Fundação Getulio Vargas, Escola de Matemática
Aplicada.

Orientadora: Margaret Armstrong.

Inclui bibliografia.

1. Liquidez (Economia). 2. Risco (Economia). 3. Análise estocástica. 4.
Finanças. 5. Modelagem de dados. I. Armstrong, Margaret. II. Fundação Getulio
Vargas. Escola de Matemática Aplicada. III. Título.

CDD – 519.22

GUILHERME HIDEO ASSAOKA HOSSAKA

**“STOCHASTIC SUPPLY CURVES AND LIQUIDITY COSTS: ESTIMATION FOR BRAZILIAN
EQUITIES”.**

Dissertação apresentado(a) ao Curso de Mestrado em Modelagem Aplicada da
Informação do(a) Escola de Matemática Aplicada para obtenção do grau de Mestre(a)
em Modelagem Aplicada da Informação.

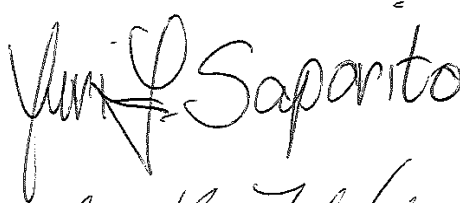
Data da defesa: 26/06/2018

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Acknowledgements

Pursuing this masters as a partial time student was far from being the life path that I dreamed: it demanded concessions in both professional and academic aspects, specially after taking the recent position of manager of the Budget Division of Dataprev. Although, talking about it when I finally present this very work seems pity, I do so to make a point: it would not be possible without counting on so much friends, coworkers and professors.

By mentioning the former Chief Controller of Dataprev, MSc. Márcio Souza Paula and, my great predecessor at the position of Budget Manager of Dataprev, Mr. Renato Sérgio Vieira, I express my gratitude to so many remarkable coworkers and chiefs from the Controlling Department of Dataprev. I am also thankful for the support of the former Chief Accountant of Dataprev, Mr. Jorge Sebastião Gomes da Costa, and the former Manager of Performance and Capacity Planning of IT, Mr. Ideraldo Dias de Figueiredo.

Professor Luiz Alberto Esteves, for always helping and believing me and for being my role model as an economist, academic and pragmatic progressive intellectual.

Lucilena and Silas: such an unique Carioca family we are! I could not do it without the home you made to me here.

Rodrigo Moreira D'Áuria and Dr. Ricardo Zortéa Vieira: in this land so far from my hometown you guys are two brothers beyond the bonds of blood that I am fortunate to have.

To Amanda Telesse, for her care and our time together; neither you nor I to blame when all is said and done: it really was infinite while it lasted.

To “Aunt” Marli: you raised me like a child of yours and I am thankful for your care as a son.

To my parents, Lila Tiemi Assaoka Hossaka and Carlos Hideo Hossaka: if there is something to be proud on me it all started because of you.

And, finally, I am most thankful to my advisor, Professor Margaret Armstrong: for her wisdom and guidance, her comprehension due to my struggling circumstances, and the trust on me I will always be in debt.

I

*Do not cry, my son;
Do not cry, because life
Is a relentless fight:
To live is to fight.
Life is a battle,
That culls the weak,
And the stronger, the brave
Can only glorify.
(...)*

VII

*The grip in these houses,
Wishing to silence
The children raised
Under fear and restrain;
Tell them your name,
Cause the enemies' people
May do not hear
without tears, without fear!
(...)*

X

*Straighten your weapons,
Strike through life:
Heavy or beloved,
To live is to fight.
If the hard battle,
Culls the weak,
The stronger, the brave
Can only glorify.*

— PIECES FROM THE POEM “CANÇÃO DO TAMOIO” (TAMOIO’S SONG),
BY ANTÔNIO GONÇALVES DIAS.

THESE EXCERPTS WERE CITED BY NELSON TEIXEIRA LOTT, EX-MEMBER OF
NATIONAL LIBERATION ACTION AND GRANDSON OF MARSHAL HENRIQUE
DUFFLES TEIXEIRA LOTT, IN THE DOCUMENTARY FILM
“MILITARES DA DEMOCRACIA: OS MILITARES QUE DISSERAM NÃO”
(“DEMOCRATIC MILITARY: THE MILITARY WHO SAID NO”)

“To all the colors of Aline...”

Abstract

Market Liquidity is characterized by the easiness and freedom to trade assets at desired volumes and for prices perceived as representative of their values. When there is a scarcity of bid and ask offers at those terms, traders face the so called Market Liquidity Risk and they must offer concessions on their original offers, leading to additional costs. Approaches to model this phenomena exist in broad variety but a common component of most Market Liquidity models is an instantaneous cost component, also known as transaction/execution costs or realized/instantaneous impact. This element, here the Liquidity Cost, gives the actual trading prices faced by a trader, frequently a deviation from the unobservable “true price”, normally represented as a GBM with the mid-price as a proxy for modeling purposes. Although it is clear that Liquidity Costs are a relevant aspect of Market Liquidity Risk and it is present in many models, it is relegated to a more simplistic treatment, being though as well-behaved, deterministic, smooth and static. The main point of this work is to follow a different approach by evaluating Liquidity Costs at a microstructural level by estimating the Stochastic Supply Curve from Çetin-Jarrow-Protter Model for Brazilian equities. To do so, high-frequency-data from B3’s ftp is used and to build Limit Order Books for several stocks at intra-day periods. The empirical findings support the existence of non-trivial Stochastic Supply Curves as a representation for Liquidity Costs in several equities on Brazilian Markets. Additionally, there is evidence that Liquidity Costs may behave in contrast with some of the literature, being stochastic with time-varying functional representations on the LOB and with liquidity parameters that could be represented as mean-reverting stochastic process.

KEYWORDS: Market Liquidity Risk, Stochastic Supply Curve, Limit Order Book

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Chapter 1

Introduction

Liquidity by itself, (i.e. everything else remaining constant), is a desirable quality of an asset: it refers to the ability to quickly and cost-effectively trade it on the market and, from an asset-liability management perspective, to convert it in cash. Although convincing, this statement raises several theoretical and practical problems coming from the key concepts and assumptions employed to do so: two main concepts of risk, namely Market Liquidity Risk and Funding Liquidity Risk, are embedded in it.

Market Liquidity Risk appears in the context of trading. An agent seeking to buy or sell an asset at a target price and volume is only able to do so under current supply and demand conditions (i.e. quantities available at different price levels) that may not fulfill a proper execution. A scarcity of available volume at reasonable bid/ask prices imposes a need to increase/decrease the original target price. This is the notion of Liquidity Costs: trading at an actual price that differs from the price perceived as the best expression of an asset's value generates additional costs.

Funding Liquidity Risk is the event that, despite economic surpluses¹, there is a failure to meet expiring liabilities, mainly by lack of immediate cash for disbursement or indirectly by a lack of general funding sources. As a last resource, it may be necessary to quickly unwind portfolio positions no matter how unfavorable market conditions are.

As [Levy \(2015\)](#) states, if there are no funding restrictions it is possible to wait indefinitely until optimal trading conditions arise, which makes Market Liquidity Risk practically immaterial. Conversely, if there is no Market Liquidity Risk manifested on transaction costs one's asset holdings may be simply freely shifted between securities, thus avoiding a funding bind.

¹Profits on the income statement are accounted by accrual basis and do not imply positive cash generation.

Although Market Liquidity Risk has a close relation with Funding Liquidity Risk, the main point of this dissertation is that Market Liquidity is a phenomena happening at the microstructural level.

To this purpose, this work studies Market Liquidity on Brazilian equities by employing the Çetin-Jarrow-Protter Model (Cetin et al., 2004, 2006, Çetin et al., 2010, Jarrow and Protter, 2015, Jarrow et al., 2012) and the Stochastic Supply Curve on the Limit Order Book, regarded as the “ultimate microscopic level of description of financial markets” (Bouchaud et al., 2002, p.2). “the ultimate microscopic level of description of financial markets” (Bouchaud et al., 2002, p.2).

Chapter 2 presents Market Liquidity Risk literature and Çetin-Jarrow-Protter Model and the Stochastic Supply Curve as the main approach on this work. This model is also compared with a framework very opposite in nature, but with its own merits: the Almgren-Chriss framework for Option Execution Strategies (Almgren and Chriss, 1999, 2001). In the Almgren-Chriss’ framework the focus relies on finding and execution strategy for a trader large enough to move trading prices and the solution to the problem relies on assumption on Liquidity Costs and on the nature of the price formation very different from Çetin-Jarrow-Protter Model.

Since Stochastic Supply Curve estimation relies on order data, **Chapter 3** discusses how to build the Limit Order Book (LOB from now on) for Brazilian Equities using data publicly available at B3 (2017b). By combining trades and orders of High-Frequency Data it was possible to implement an “Event-Based” data architecture that handles each execution type and order status in such a way that it will generate the exact number and type of entries that modify the LOB through time, level and depth.

The LOBs built on Chapter 3 are employed for estimations for the Stochastic Supply Curve on **Chapter 4** following the Blais and Protter (2010) method for Brazilian equities, focusing on the MULT3 stock and the results on intra-day behavior and the stochastic nature of Liquidity Costs are discussed.

Chapter 5 aims to raise some theoretical possibilities for general and wider studies on Market Liquidity commonalities for Brazilian markets in the context of option hedging. The Chapter tries to link imminent exercises of ATM options of physical settlement as an analogy for funding binds in the context of Market Liquidity Risk taking the “Saw-Tooth” Pattern as an example in the context of option hedging in a “Gamma Week”².

Liquidity Cost, in CJP Model terminology, appears in a variety of names like instantaneous impact, transaction/realized/execution costs and so on, but its existence and relevancy is undeniable. Such component is ubiquitous in Market

²The current week of options expiration.

Liquidity formulations, as it represents the critical threshold between accruals³ and cash.

Counter-intuitively, though, some literature on Optimal Execution Strategies, including those based on market-orders, relegate Liquidity Costs as of secondary importance by avoiding incorporating its dynamic and stochastic nature.

Then, **Chapter 6** takes Almgren-Chriss ([Almgren and Chriss, 1999, 2001](#)) stochastic portfolio process as a ruling benchmark to start discussing theoretical and practical implications of empirical findings from Chapters 4 and 5.

Finally, **Chapter 7** ends with final remarks, relevant findings and topics for further research.

³The term “accruals” refers to provisions, deferrals, future expenses and incomes that are registered in the balance sheet under the matching principle on the “accrual-based” accounting: values of accrued incomes are registered before they are cashed in and accrued expenses before they are cashed out.

Chapter 2

Models on Market Liquidity Risk

The mocking remark “Perhaps I could never succeed in precisely defining what is liquidity risk in the stock market. But I know it when I see it” (BLAIS & PROTTER, 2010) shows how to modeling Market Liquidity Risk remains an open problem and there is no definitive answer for problems coming from Optimal Trading Strategies to Option Pricing Theory.

Modeling Market Liquidity Risk is not straight forward as there is not an easy way to incorporate so many ways in which general macroeconomic and trading conditions integrate with things happenings at the very level of the order book data and there is also the problem of how such things affect and helps to design a reasonable modeling of a trader behavior.

About two different approaches on Market Liquidity Risk, Gökay and Soner (2012, p.251-255) identifies at least two different classes of models in the literature of Market Liquidity Risk in option pricing theory.

The first class of models are concerned with the effects of a trader hedging a portfolio large enough to affect market conditions and the optimal hedging strategy itself. This class are referred as “models with feedback effects”. Examples would be Platen and Schweizer (1998), where a large scale hedging has affects underlying asset’s volatility, and Frey and Stremme (1997), where the GBM prices of the underlying process may change to an Ito process dependent on time and order size. Although mainly focused on Optimal Execution, the Almgren-Chriss framework has extensions applied to option pricing theory, as in Li and Almgren (2011) and (Almgren and Li, 2016).

The second class of models focuses on the super-replication problem and the equalization of the supply and demand locally, in time, so that trade volumes does not have a lasting impact on the asset price. A flexible approach to model this phenomena is the so-called “Stochastic Supply Curve” (Cetin et al., 2004). Starting from the replicating portfolio argument from option pricing theory, if

there are liquidity costs affecting actual trading prices and, therefore, the portfolio value, they must be accounted into option prices that differs from the classical Black-Merton-Scholes formula.

This dissertation studies Market Liquidity by two models of those classes: the Almgren-Chriss framework (Almgren and Chriss, 1999, 2001) as a reference in Optimal Execution Strategies from Guéant (2016) perspective and the literature on Cetin et al. (2004, 2006), Çetin et al. (2010) Stochastic Supply Curve, particularly the LOB/microstructural approach of Blais and Protter (2010).

2.1 Çetin-Jarrow-Protter Stochastic Supply Curve

Çetin-Jarrow-Protter Model (CJP Model from now on) main concern is on super-replication of a contingency claim under an economy with liquidity costs represented as a Stochastic Supply Curve (SSC from now on). The main reasoning is that under no-arbitrage and market completeness at some level, the price of a contingency claim must equal the payoff of a replicating portfolio built under a self-financing trading strategy (s.f.t.s.).

To build such s.f.t.s., though, one needs to keep trading, thus, incurring in liquidity costs. As they affect the portfolio value, then under a No-Free Lunch with Vanishing Risk (NFLVR) economy, such extra-costs will also affect the price of contingency claims that, naturally, will differ from the classical Geometric Brownian Motion given by the Black-Merton-Scholes Model (BMS Model from now on) (Black and Scholes, 1973, Merton, 1973).

The classical Geometric Brownian Motion (GBM) process from the BMS Model can be represented in its a stochastic differential equation (SDE) as:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (2.1)$$

where S_t is the price at t , μ is a constant drift, σ is the constant volatility, and dW_t is the differential form of the Brownian motion $W_t \sim \mathcal{N}(0, t)$. Formally, S has both the time t and the sample path $\omega \in \Omega$ as arguments, but here it is written as S_t for briefness.

The SDE in (2.1) has a solution given by:

$$S_t = S_0 \exp\{\mu t + \sigma W_t\} \quad (2.2)$$

A second argument, ΔX , representing the traded quantity of an asset is added to the stochastic process S_t . This gives $S(t, \Delta X)$ as the actual prices accounted into a s.f.t.s. portfolio and the difference between $S(t, \Delta X)$ and $S(t, 0)$ (or S_t) is taken as an additional cost due to trading.

A s.f.t.s. portfolio with positions on stocks and money under such additional costs coming from each trading is given by:

$$Y_t = Y_0 + X_0 S(0, X_0) + \int_0^t X_{u-} dS(u, 0) - X_t S(t, 0) - \sum_{0 \leq u \leq t} \Delta X_u [S(u, \Delta X_u) - S(u, 0)] - \int_0^t \frac{\partial S}{\partial x}(u, 0) d[X, X]_u^C, \quad (2.3)$$

where X is the stock holding, $S(t, 0)$ is the GBM (no trading occurs), Y is the money market account and the two last terms of the RHS are additional costs. They are considered to be a form of Liquidity Costs through the existence of the portfolio. Those two terms are always positive and, then, always decrease the portfolio value:

$$L_T = \sum_{0 \leq u \leq t} \Delta X_u [S(u, \Delta X_u) - S(u, 0)] - \int_0^t \frac{\partial S}{\partial x}(u, 0) d[X, X]_u^C \quad (2.4)$$

Each price S_t differs from trading prices due to Liquidity Costs through time. Such deviation is expressed by the term $S(u, \Delta X_u)$ that represents the market price $S(t, 0)$ when affected by ΔX traded units. This deviation is defined on CJP Model as the Stochastic Supply Curve.

Although the prices accounted into the s.f.t.s. are different from the conventional GBM, agents remain price takers. When there is no trading, then $S(t, \Delta X = 0) = S(t, 0)$, meaning that the prices still follows the typical GBM or any “true-price dynamics”. This is possible under the assumption of the existence an upward Stochastic Supply Curve that is the very core of CJP Model. Its behavior is given by [Cetin et al. \(2004, p.328\)](#) as follows:

- (i) $S(t, \Delta X)$ is $\mathcal{F}_t$ -measurable and non-negative.
- (ii) $\Delta X \mapsto S(t, \Delta X, \omega)$ is a.e. in ω a non-decreasing function in ΔX a.s. (i.e. $\Delta X \leq \Delta Y \Rightarrow S(t, \Delta X, \omega) \leq S(t, \Delta Y, \omega)$ a.s. $\mathbb{P}$ and a.e. in ω).
- (iii) S is $\mathcal{C}^2$ in its second argument.
- (iv) $S(\cdot, 0)$ is a semi-martingale and $\partial S / \partial \Delta X$ has finite quadratic variation.

The SSC reacts differently depending on the trading side: an order at the selling side would mean $\Delta X < 0$, moving the prices down and, on the converse, as an order at the buying side would mean $\Delta X > 0$ would move the prices up.

As a concrete example, [Cetin et al. \(2006, p.315-316\)](#) presents:

$$S(t, \Delta X) = e^{g(t, D_t, \Delta X)} S(t, 0), \quad (2.5)$$

where $S(t, \Delta X) = f(t, D_t, \Delta X)$, D_t is an n -dimensional, $\mathcal{F}_t$ -measurable semi-martingale, and $f : \mathbb{R}^{n+2} \rightarrow \mathbb{R}^+$ is Borel measurable. The vector stochastic process D_t generates uncertainty in the economy.

The SSC relies not only on the traded quantity ΔX , but may also depend on the time the trade occurs and how liquid is the asset at that particular state and time. If the SSC is stationary, that is, independent of $t \in \mathbb{R}^+$, then another possible form is:

$$S(t, \Delta X) = e^{\alpha \Delta X} S(t, 0) \quad (2.6)$$

So, liquidity risk arises when there is a trade amount big enough to move the prices through the limit book order of the underlying asset. This effect has its intensity set by the α parameter and market conditions at a particular time.

2.2 Almgren-Chriss Framework: Optimal Execution and Permanent Market Impact

Almgren-Chriss framework ([Almgren and Chriss, 1999, 2001](#)) deals with the problem of how to unwind a position over a time window from a large trader perspective.

On these contexts, one may want to split the original quantity into small pieces to be executed gradually. But while this is performed there is a risk of price changes due to market dynamics.

Thus, to deal with Market Liquidity Risk one faces “the fundamental trade-off between execution costs on the one hand, and price risk on the other hand: a fast execution of shares leads to high execution costs, while a slow execution exposes the trader to adverse market fluctuations. This trade-off, which is central in the literature on optimal execution, leads to nontrivial optimal execution strategies” ([Guéant, 2016, p.40](#)).

The above trade-off between execution costs and price risk is not exclusive to Almgren-Chriss framework, but they are “unanimously considered, and rightly so, as the pioneers of the early literature on optimal execution” (Guéant, 2016, p.40).

The formulation in Almgren-Chriss framework is how to find an execution algorithm that schedules the trading happening in a period $t \in [0, T] \subset \mathbb{R}^+ \cup \{0\}$. For this purpose, the optimization problem is defined in terms of maximizing some quantity in function of the trader’s cash account that, by definition, varies according to the rate of trading, which is the “control” of a dynamic optimization problem in Calculus of Variations. The problem has a non-trivial nature as the trader is large enough not only to incur in relevant liquidity costs¹ and there is a lasting price movement that makes the asset price go up/down as a function of the bought/sold quantity.

Consider a single-stock portfolio at time t with $q_t \in \mathbb{R}$ shares. When $q_t > 0$ the portfolio is long position in q_t shares while, if $q_t < 0$ then the trader has a short position in the stock of $-q_t$ shares. The main role of the Almgren-Chriss framework is to solve the problem of unwinding this portfolio over a time window $[0, T]$, where the trader’s position is given by the process $(q_t)_{t \in [0, T]}$.

The dynamics of $(q_t)_{t \in [0, T]}$ is given by:

$$dq_t = v_t dt, \quad (2.7)$$

where $(v_t)_{t \in [0, T]}$, the instantaneous trading volume or trading velocity, is in the space of the progressively measurable control processes $\mathbb{H}^0(\mathbb{R}, (\mathcal{F}_t)_t)$ and satisfies the unwinding constraint $\int_0^T v_t dt = -q_0$ with $\int_0^T |v_t| dt \in L^\infty(\Omega)$, the set of almost surely bounded random variables.

The set of admissible controls, $\mathcal{A}$, is denoted by

$$\mathcal{A} = \left\{ (v_t)_{t \in [0, T]} \in \mathbb{H}^0(\mathbb{R}, (\mathcal{F}_t)_t), \int_0^T v_t dt = -q_0, \int_0^T |v_t| dt \in L^\infty(\Omega) \right\}. \quad (2.8)$$

As the meaning of $(v_t)_{t \in [0, T]}$ was explained, the framework proposes that the stock mid-price, $(S_t)_{t \in [0, T]}$, can be modeled by a diffusion random component and a permanent component of market impact with linear dependence:

$$dS_t = \sigma dW_t + k v_t dt, \quad (2.9)$$

when σ is the arithmetic volatility of the stock, dW_t the differential form of the Brownian motion W_t , k is a nonnegative parameter modeling the magnitude

¹Indeed, in CJP Model sense, since the trading price comes from a transient impact.

of the permanent market impact. If v_t models the trading volume process, the trader itself is moving the prices under linear market-impact. This is what greatly differs from an economy from the classical BMS and [Cetin et al. \(2004\)](#) since the stochastic process regarding the stock price is no longer the classical GBM.

Since the main interest of Almgren-Chriss is on Optimal Execution, there is no s.f.t.s. but the trader's cash account also incurs on liquidity costs and differs from the stock price perceived as the "true price".

$$\tilde{S}_t = S_t + g\left(\frac{v_t}{V_t}\right), \quad (2.10)$$

where g is an increasing function satisfying $g(0) = 0$ and, as usual, if the trader buys (sells), then the trading price accounted into the trader's cash account is higher (lower) than S_t . The variable $(V_t)_{t \in [0, T]}$ is the market volume process and is assumed to be deterministic, continuous, positive, and bounded process. Here $\tilde{S}_t$ takes the role of the SSC on CJP Model of modeling the liquidity costs.

Actually "liquidity costs" is a term from CJP model, as in Almgren-Chriss literature it is referred as "execution costs" (as in [Guéant \(2016, p.43\)](#)), and the phenomena of $\tilde{S}_t \neq S_t$ due to $|g\left(\frac{v_t}{V_t}\right)| > 0$ is referred as "realized impact" ([Almgren et al., 2005](#)) or "temporary impact" ([Almgren and Chriss, 2001, p.8-9](#)). In this work "liquidity costs" will be preferred.

As noted by ([Guéant, 2016, p.43](#)), the assumptions on $(V_t)_{t \in [0, T]}$ seems odd because volume cannot be exactly predicted, but it corresponds to the way many algorithms work in practice, with static volume curves computed using historical data.

Denoting $\rho = v_t/V_t$, the liquidity cost function L is expressed by:

$$L(\rho) = \rho g(\rho) \quad (2.11)$$

The assumptions on the function $L: \mathbb{R} \mapsto \mathbb{R}$ are the following:

- (H1) No fixed cost (i.e. $L(0) = 0 \Leftrightarrow \tilde{S} = S_t$)
- (H2) L is strictly convex, increasing in $\mathbb{R}^+$ and decreasing in $\mathbb{R}^-$,
- (H3) L is asymptotically super-linear (i.e. $\lim_{|\rho| \rightarrow +\infty} \frac{L(\rho)}{|\rho|} = +\infty$).

The functional form of $L(\rho)$ is usually assumed to be strictly convex power function like $L(\rho) = \eta|\rho|^{1+\phi} + \psi|\rho|$ with $\phi, \psi > 0$. The additional term $\psi|\rho|$ models proportional costs such as the bid-ask spread, trading fees, and/or stamp duty. For instance, the initial Almgren-Chriss models employed $L(\rho) = \eta\rho^2$.

With liquidity costs, the trader's cash account process is given by:

$$\begin{aligned} dX_t &= -v_t \left(S_t + g \left(\frac{v_t}{V_t} \right) \right) dt \\ &= -v_t S_t dt - V_t L(\rho_t) dt. \end{aligned} \quad (2.12)$$

The goal of the Almgren-Chriss framework is to find an optimal strategy $(v_t)_t \in \mathcal{A}$ to liquidate the portfolio subject to a maximization criterion. Classically, [Almgren and Chriss \(1999, 2001\)](#), [Almgren and Li \(2016\)](#) chooses the Mean-Variance Criterion, that being:

$$\sup_{v_t \in \mathcal{A}_{ad}} \mathbb{E}[X_T] - \frac{\gamma}{2} \mathbb{V}[X_T], \quad (2.13)$$

where $\mathbb{V}[X_T]$ is the variance of the “terminal” cash-process X_T .

[Guéant \(2016, p.47\)](#) gives a CARA (Constant Absolute Risk Aversion) utility function²:

$$\sup_{v_t \in \mathcal{A}_{ad}} \mathbb{E}[-\exp(-\gamma X_T)], \quad (2.14)$$

where $\gamma > 0$ is a positive constant known as the absolute risk aversion coefficient of the trader. Almgren-Chriss's optimization problems and will be discussed in detail on Chapter 6.

As stated by [Guéant \(2016, p.40\)](#), the market impact model relying on the linear market impact is far from realistic. Market prices are obviously resilient and stock prices recover from the permanent impact with an exponential decay.

Additionally, the calibration of k in [Almgren et al. \(2005\)](#), for instance, employed a highly proprietary dataset from Citigroup that linked each order, the “pieces”, with the main position to be executed, here the target q_T . Therefore, working in empirical evidences in favor of this design would be challenging if not impossible due absence of proper data.

In a recent interview ([Risk, 2017](#)), Robert Almgren himself admitted that he no longer employs his classical approach in his day-to-day business at Quantitative Brokers. But Almgren-Chriss' still an influential and useful framework since it incorporates both Stochastic and Calculus of Variations in the most insightful manner.

As seeking for empirical evidences of whether Market Liquidity Risk is of im-

²Probably because CARA utility function has the property to be indifferent from the agent's initial wealth, avoiding Funding Liquidity discussions.

portance on day-to-day reality of trading, the Stochastic Supply Curve is chosen as the main hypothesis for estimation. This is realized by an microstructural approach building the LOB for fixed times, which is discussed on next chapter.

Chapter 3

The Limit Order Book

The Limit Order Book (LOB from now on) is considered to be the “ultimate microscopic level of description of financial markets” (Bouchaud et al., 2002, p.2).

Even though it is a static picture, a snapshot of market conditions at a fixed time, it is far from being stationary and well-behaved: complicated global phenomena emerge as result of interactions between heterogeneous agents (Gould et al., 2013, p.1709).

Market conditions are path-dependent and building a LOB through time relies heavily on data consistency and studying its conditional behavior is complex. As explained by Gould et al. (2013, p.1714) this is so because its state space is huge: if there are P different choices for a price in a given LOB, then the state space of the current depth¹ profile alone expressed in a lot size ν is $\mathbb{Z}^P$.

A LOB organizes prices and quantities for an order-driven market and, as one the main points in this work, when there is Market Liquidity Risk it may change and change deeply, rapidly and drastically in such a way that it would greatly differ from the its previous states in time.

Market Liquidity Risk is both of microstructural and its dynamic natures that can be studied by building several LOB for several fixed times². A rich, detailed, and high-quality historic data from LOBs provides a suitable testing ground for several theories, which gives motive for the efforts of the next section.

¹Volume/quantity for a bid/ask price.

²A remarkable interactive example of implementation that show the LOB evolution by second for a Bitcoin market is one from (Stubbings, 2016)

3.1 Building the LOB for Brazilian Equity Market

Building a LOB for Brazilian Equity Market was based on B3's High Frequency Data (B3, 2017b) downloaded and organized with assistance of GetHFData package in R from Perlin and Ramos (2016) and some customizations on its open source code.

A proper dataset of orders would suffice to build a LOB, but this work also relied on trades datasets for the sake of data consistency. By combining these two datasets is possible to organize an "Event-Based" data architecture prone to deliver LOBs at several fixed-times.

All those procedures are discussed in this section.

3.1.1 B3's High-Frequency Data

By the very definition of Market Liquidity Risk, high-frequency data (HFD from now on) is a *conditio sine qua non* to study how it happens at microstructural level.

Data employed in this work is from B3's ³ ftp ⁴ from trading days of 2017.

B3's HFD is, at least until now⁵, publicly available in raw format for order-driven markets of equities, options and other derivatives⁶ on indexes, foreign currencies and commodities, including venues for fractional⁷ transactions.

Despite its availability, the task of acquiring working HFD is not limited to download huge files. Transforming them into working, tidy and usable HFD demands time and considerable computational effort⁸, but recent developments presented a very convenient way: the "GetHFData" package implementation by Marcelo Perlin and Henrique Ramos from UFRGS (Perlin and Ramos, 2016), now

³B3 stands for "Brasil/Bolsa/Balcão" is the resulting company from the merge between BM&F BOVESPA and Central de Custódia e de Liquidação Financeira de Títulos approved on March 22nd 2017 by the Competition Authority of Brazil (Conselho de Administração de Defesa Econômica - CADE).

⁴Namely B3 (2017b) .

⁵B3 has been changing discontinuing several free services and standards for the common public, being the most controversial the ending of the "Boletim Diário Informativo" - BDIN. Indeed this is a characteristic of Brazilian Markets and imposes several difficulties in modeling.

⁶A very interesting characteristic of Brazilian Markets is that several products such as forwards may be found in order-driven organized venues.

⁷For trading quantities summing less than a lot.

⁸"As an example, the trading records of November 3rd 2015 from Bovespa's equity market, relatively small, is stored as a single compacted zip file with approximately 30 MB. When unpacked, the result is a text file with 310 MB of content." (Perlin and Ramos, 2016, p.445).

in its fifth release (November 27th 2017, the first one was in October 2016). GetHFDData package in R language offers several functions to download, unzip and aggregate HFD by intraday intervals.

Even though “GetHFDData” is a great advance that will certainly facilitate research on market microstructure, some customization was employed on the open source code of the package if one wants to, say, optimize for computational performance, get both selling and buying members and the initiating order of a trade, etc. .

Adjustments were needed before the direct usage of raw data. One must be aware of several business logic rules to be applied when treating the raw data, those being: trading hours change throughout the year (based Daylight Saving Times and tracking USA Markets) and even in so-called holidays⁹ and what are the expected behavior of calls in the opening and closing of a trading day¹⁰.

3.1.2 Handling B3’s HFD and Event-Based Data Architecture

The procedure in this work uses both Orders and Trades HFD extracted from B3 (2017b) using GetHFDData R package Perlin and Ramos (2016).

Theoretically¹¹, a LOB could be constructed only by using Orders HFD in B3 (2017b). But there are disadvantages since it would require even more time consuming and computational efforts to do so by requiring a tick-by-tick matching algorithm on each side of the LOB. Incorporating Trades HFD also adds more data consistency into the procedure.

Some modifications were necessary on the open-source code¹². from Perlin and Ramos (2016) with respect to Trades HFD and Orders HFD:

1. The package processes the raw data presenting information based only on the trade-initiating or aggressor side, but building of the LOB requires information on both sides;
2. The problem with presenting trades based on the aggressor side is not a problem itself, but if when it makes sense. At the begging (ending) of the trading day there is an auction to define the opening (closing) price, thus

⁹Specifically on Ash Wednesday.

¹⁰On these parts of an usual day the trades are settled at prices established by B3, thus making the concept of “trade initiating order” (also known as the “aggressor order”) simply unsound.

¹¹E.g.: less noisy data and improvements in data consistency and documentation for the public.

¹²It can be found at Perlin and Ramos (2017)

the concept of trade-initiating/aggressor side is not of proper use for those kind of entries. In fact, this fact is represented in the raw data as both sides are marked as being the “passive side” of the trade. This was changed on the original code;

3. The package also dropped the Trade Number, applied to assign the last trading price for each line of the Events dataset;
4. Two extra columns serving as time indexes were added.

(Perlin and Ramos, 2016), resulting in the following structure:

Figure 3.1: A Subset from Trades HFD data frame for MULT3 on 20th

Date	Ticker	#Trade	ID_Buy	ID2_Buy	Agg Buy	ID_Sell	ID2_Sell	Agg Sell	Tradetime	Buy Mem.	Sell Mem.	TradeSign	Price	Qty
20/03/2017	MULT3	11660	84779574308	1224794	2	84779578021	1224795	1	12:00:31.172	45	8	-1	65,76	200
20/03/2017	MULT3	42670	84780088382	3295649	1	84780088351	3295650	2	16:48:08.404	13	8	1	66,25	200
20/03/2017	MULT3	42680	84780088382	3295651	1	84780088226	3295652	2	16:48:08.404	13	45	1	66,25	200
20/03/2017	MULT3	42690	84780088382	3295653	1	84780088357	3295654	2	16:48:08.404	13	45	1	66,25	300
20/03/2017	MULT3	44270	84780107003	3371162	2	84780103968	3371163	2	17:09:14.607	16	147	N/A	66,2	300
20/03/2017	MULT3	44260	84780107003	3371160	2	84780107415	3371161	2	17:09:14.607	16	3	N/A	66,2	100
20/03/2017	MULT3	44250	84780107003	3371158	2	84780107406	3371159	2	17:09:14.607	16	3	N/A	66,2	300
20/03/2017	MULT3	44240	84780107003	3371156	2	84780107371	3371157	2	17:09:14.607	16	3	N/A	66,2	200
20/03/2017	MULT3	44170	84780107003	3371142	2	84780106646	3371143	2	17:09:14.607	16	16	N/A	66,2	1900
20/03/2017	MULT3	44180	84780107003	3371144	2	84780102898	3371145	2	17:09:14.607	16	40	N/A	66,2	300
20/03/2017	MULT3	44190	84780107003	3371146	2	84780103496	3371147	2	17:09:14.607	16	93	N/A	66,2	300
20/03/2017	MULT3	44200	84780107003	3371148	2	84780102627	3371149	2	17:09:14.607	16	93	N/A	66,2	100
20/03/2017	MULT3	44210	84780107003	3371150	2	84780104988	3371151	2	17:09:14.607	16	72	N/A	66,2	300
20/03/2017	MULT3	44220	84780107003	3371152	2	84780106735	3371153	2	17:09:14.607	16	72	N/A	66,2	100
20/03/2017	MULT3	44230	84780107003	3371154	2	84780104338	3371155	2	17:09:14.607	16	3	N/A	66,2	100

The dataset presents trading day, ticker, a primary ID and a secondary ID (matching with Orders HFD) for both ask and bid sides and their respective members, the aggressor side, trading price and traded quantity at a given time up to the milliseconds.

When put into a proper format, Trades HFD is remarkably informative as all its entries also matches the Orders HFD for both selling and buying sides.

Orders HFD gives the realized trajectory of an order during the trading day, identifying them with a unique primary ID and a secondary ID based on the execution type and order status.

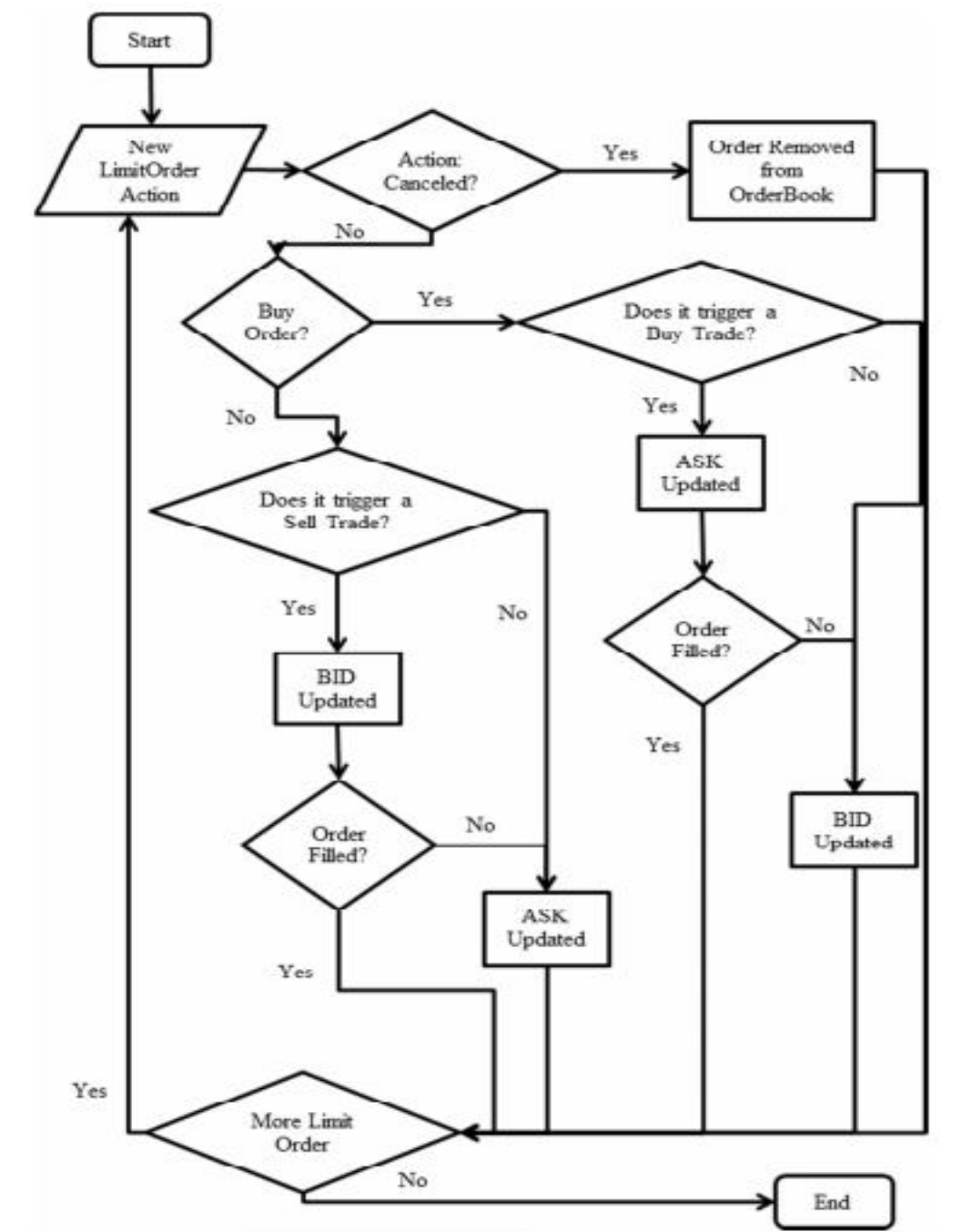
Figure 3.2: A Subset from Orders HFD data frame for MULT3 on 20th

Date	Ticker	Side	ID	ID_2	ExecutionType	Time	Price	Total Q	Trad Q	OrderStat	Agg	Mem
20/03/2017	MULT3	Buy	84779574308	1209368	New	11:59:10.542000	65,58	100	0	New	0	45
20/03/2017	MULT3	Buy	84779574308	1210389	Update	11:59:15.930000	65,59	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1210481	Update	11:59:15.944000	65,62	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1210513	Update	11:59:15.949000	65,66	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1211270	Update	11:59:17.380000	65,67	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1211278	Update	11:59:17.385000	65,68	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1213814	Update	11:59:28.540000	65,69	100	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1216637	Update	11:59:51.449000	65,69	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1217134	Update	11:59:53.180000	65,7	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1217618	Update	11:59:53.395000	65,71	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1217655	Update	11:59:53.419000	65,72	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1218288	Update	11:59:54.963000	65,73	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1218297	Update	11:59:54.967000	65,74	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1219091	Update	12:00:00.369000	65,75	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1219128	Update	12:00:00.407000	65,74	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1219234	Update	12:00:00.557000	65,75	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1219640	Update	12:00:01.673000	65,76	200	0	Replaced	0	45
20/03/2017	MULT3	Buy	84779574308	1224794	Trade	12:00:31.172000	65,76	200	200	Filled	2	45
20/03/2017	MULT3	Buy	84780088382	3295648	New	16:48:08.404000	66,25	13400	0	New	0	13
20/03/2017	MULT3	Buy	84780088382	3295649	Trade	16:48:08.404000	66,25	13400	200	Partially F	1	13
20/03/2017	MULT3	Buy	84780088382	3295651	Trade	16:48:08.404000	66,25	13400	400	Partially F	1	13
20/03/2017	MULT3	Buy	84780088382	3295653	Trade	16:48:08.404000	66,25	13400	700	Partially F	1	13
20/03/2017	MULT3	Buy	84780088382	3295655	Cancel	16:48:08.404000	66,25	13400	700	Canceled	0	13
20/03/2017	MULT3	Buy	84780107003	3360369	New	17:03:38.839000	66,21	4000	0	New	0	16
20/03/2017	MULT3	Buy	84780107003	3371142	Trade	17:09:14.607000	66,21	4000	1900	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371144	Trade	17:09:14.607000	66,21	4000	2200	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371146	Trade	17:09:14.607000	66,21	4000	2500	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371148	Trade	17:09:14.607000	66,21	4000	2600	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371150	Trade	17:09:14.607000	66,21	4000	2900	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371152	Trade	17:09:14.607000	66,21	4000	3000	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371154	Trade	17:09:14.607000	66,21	4000	3100	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371156	Trade	17:09:14.607000	66,21	4000	3300	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371158	Trade	17:09:14.607000	66,21	4000	3600	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371160	Trade	17:09:14.607000	66,21	4000	3700	Partially F	2	16
20/03/2017	MULT3	Buy	84780107003	3371162	Trade	17:09:14.607000	66,21	4000	4000	Filled	2	16

The idea is that an order frequently do not quit the LOB right after it appears: it will only vanish by completely fulfilling its trading quantity or by canceling. As it remains in the LOB it may be updated several time and/or it may be slowly traded until the total trading quantity is met.

Árdal (2013)¹³ presents a flowchart that discusses the effects on the LOB by execution type of order that is very similar to the approach here discussed:

Figure 3.3: Flowchart from (Árdal, 2013, p.12)



¹³B3 (2017b) standard and market though differs in nature from Árdal (2013) thesis subject, namely the bonds issue by the Housing Financing Fund of Iceland, a market place with very few participants.

Basically a LOB is an accumulated position, or a balance, that has taken into account several orders arriving, changing or vanishing through time. Thus, organizing the LOB is task that relies on an “Event Based” perspective.

Every entry for every order in Orders HFD have different effects that had to be accounted when building the LOB from several times.

To this purpose, a Event Based Data Architecture was conceived by taking into account “Four Layers”: Trades, Orders, Events and Depth.

The core of this procedure is the “Events” layer that is built based on entries from Trades and Orders layers and on the following assumptions and guiding principles:

1. Every order must have a “New” execution type entry as its first appearance in the LOB ¹⁴, generating a first entry in the Events layer;
2. Every “Trade” execution type entry must have an ask and bid sides (two orders involved) and a trading price. Those are given by the Trades HFD and each of its entries that generate two entries in the Events layer, one for each side of the LOB;
3. Every “Update” execution type entry must have a previous associated entry to modify, thus making an entry for ripping of the previous state and another for its substitution in the Events layer;
4. Every “Cancel” execution type entry must have a counterpart, generating only one entry in the Events layer;
5. “Reentry” execution type entries are associated with orders with hidden quantities but do not change the state of the LOB and are discarded.

But HFD from B3 (2017b), as expected from any standard HFD publicly available, has inconsistencies and, for instance, non-trivial noisy entries that needs thoughtful handling depending on the strategy one chooses to build an functional LOB.

Some of the problems regarding the assumptions listed above with the corresponding treatment:

1. Orders without “New” execution type entries. “Artificial” entries were created at the very beginning of the Orders HFD;
2. Trades with counterpart found in Orders HFD: they are disregarded as Trades HFD was employed in this procedure;

¹⁴An examples of exchange in which every order expires at the end of trading day is the Unfortunately this is not so for B3, but orders trespassing a trading day are very unusual.

3. More than one “New” execution type entry for an primary Order ID: the earliest entry was considered, discarding the latest ones.

A final challenging problem related to the nature and meaning of some orders was addressed: some orders had “zero prices”. A first guess would then indicate that they were market orders but they behave in very different way in regard to priority criteria of a common LOB: most of them were issued at the beginning of the trading day just to be finally settled at its end, frequently at closing calls period.

Since the core depths of the LOB are the most useful information it was decided to take these “zero price” orders into account in the Events layer, but discarding their depths when building the LOBs.

When Orders and Trades layers are merged they result in the Events layer.

Finally, the Depth layer summarizes the changes in quantities of each trading side and price level during the trading day.

From such data architecture then one simply fixes a given time for the LOB and take the sums of the varying quantities from the Depth layer, grouping by price level and order side. Still some noisy results were found, like trespassing bid/ask price level or negative depths. Those were filtered off as they simply violate the very definition of a LOB.

From Orders and Trades layers, the final two steps are represented in ASCII SQL standard below:

```
SELECT
    SIDE, PRICE_LEVEL, SUM(DELTA_DEPTH) AS VOLUME
FROM
    (SELECT
        SIDE, PRICE_LEVEL, SUM(DELTA_DEPTH)
        FROM
            (SELECT
                TIME, SIDE, PRICE_LEVEL, SUM(QUANTITY) AS DELTA_DEPTH
            FROM
                EVENTS
            GROUP BY
                TIME, SIDE, PRICE_LEVEL
            ) AS DEPTH
    )
WHERE
    TIME <= FIXED_TIME_FOR_LOB
GROUP BY
    SIDE, PRICE_LEVEL
```

Chapter 4

Estimating the Stochastic Supply Curve

Proving the actual existence of the SSC on CJP Model for a certain market or ticker, that being empirical evidences supporting its assumptions, is essential because it is the solely source of liquidity costs.

There are two approaches to calibrate the parameters of CJP Model: [Cetin et al. \(2006\)](#) and [Blais and Protter \(2010\)](#). The first is the pioneering work on the matter and uses simple “Trades & Quotes” (TAQ) data and the latter takes a more natural but complex microstructural approach by relying on several LOB data through time.

Both alternatives are presented in the next section, with more attention to [Blais and Protter \(2010\)](#) because, naturally, it can be applied in Brazilian Equity Market thanks to the LOB data acquired on Chapter 3.

4.1 First Approach: TAQ Calibration

The estimation of α presented in [Cetin et al. \(2006, p.502-511\)](#) proposes several functional forms for regressions using historical data of five commonly traded stocks at the NYSE and with options listed on the CBOE, that being Reebok (RBK), IBM, General Eletric (GE), FDX Technologies (FDX) and Barnes & Nobles (BKS). The regressions were run, specially in the form given by [\(4.1\)](#), for each one 1011 trading days from Jan 3rd, 1995 to December 31st, 1998. Then α estimates of each day were averaged and their robustness evaluated.

The functional form for a Least Squares estimation is found from a discrete form of [\(2.6\)](#) over a partition $0 \leq \tau_0, \tau_1, \dots, \tau_i, \tau_{i+1} \leq T$:

$$S(\tau_{i+1}, \Delta X_{\tau_{i+1}}) = \exp\{\alpha \Delta_{\tau_{i+1}}\} S(\tau_{i+1}, 0).$$

But $S(\tau_{i+1}, 0)$ is the classical GBM, the “true price” unaffected by any trades, with know closed form solution. Then:

$$\begin{aligned} S(\tau_{i+1}, \Delta X_{\tau_{i+1}}) &= \exp\{\alpha \Delta X_{\tau_{i+1}}\} S(\tau_{i+1}, 0) \\ &= \exp\{\alpha \Delta X_{\tau_{i+1}}\} \exp\{\tau_{i+1}\mu + \sigma W_{\tau_{i+1}}\}. \end{aligned}$$

Dividing $S(\tau_{i+1}, \Delta X_{\tau_{i+1}})$ by $S(\tau_i, \Delta X_{\tau_i})$:

$$\begin{aligned} \left(\frac{S(\tau_{i+1}, \Delta X_{\tau_{i+1}})}{S(\tau_i, \Delta X_{\tau_i})} \right) &= \frac{\exp\{\alpha \Delta X_{\tau_{i+1}}\} \exp\{\tau_{i+1}\mu + \sigma W_{\tau_{i+1}}\}}{\exp\{\alpha \Delta X_{\tau_i}\} \exp\{\tau_i\mu + \sigma W_{\tau_i}\}} \\ &= \exp\{\alpha(\Delta X_{\tau_{i+1}} - \Delta X_{\tau_i}) + \mu(\tau_{i+1} - \tau_i) + \sigma(W_{\tau_{i+1}} - W_{\tau_i})\} \end{aligned}$$

where the error $\epsilon_{\tau_{i+1}, \tau_i}$ equals $\epsilon\sqrt{\tau_{i+1} - \tau_i}$ with $\epsilon \sim \mathcal{N}(0, 1)$.

Finally, by taking the log-return:

$$\ln \left(\frac{S(\tau_{i+1}, \Delta X_{\tau_{i+1}})}{S(\tau_i, \Delta X_{\tau_i})} \right) = \alpha(\Delta X_{\tau_{i+1}} - \Delta X_{\tau_i}) + \mu(\tau_{i+1} - \tau_i) + \sigma \epsilon_{\tau_{i+1}, \tau_i}, \quad (4.1)$$

It is worth noting some theoretical and practical aspects of such approach:

1. The so called true price, $S(t, 0)$, is unobservable (Cetin et al., 2006, p.502). By taking the log-return this term is omitted.
2. If $\alpha = 0$, then 4.1 to the log-return of the classical GBM.
3. It is necessary to identify which side initiated the trading (if the volume is of buying or selling side). In Cetin et al. (2006), Lee and Ready (1991) algorithm was employed.
4. The main concern is on small quantities driven by frequent hedging of options (as mentioned before, CJP Model has super-replication as its main problem). This makes sense since there is no permanent market impact moving the prices on CJP Economy.

A Least Squares estimation on (4.1) in Cetin et al. (2006) gives 0.0001, 0.0005, 0.001 and 0.002 as reasonable values of α for IBM, FDX, BKS, GE and RBK.

A controversial finding is that according to [Cetin et al. \(2006, p.509\)](#) the evolution of α *"is not a serious concern"* when calibrating liquidity costs, which is very surprising.

The basic idea of the [Lee and Ready \(1991\)](#) algorithm is that if the price increases then the trade must be buyer initiated, and if it decreases, is seller initiated. It was indeed instrumental because the trading initiating side on common TAQ or even tick by tick data is hardly recorded.

But it happens that for CJP Model this procedure seems somehow tautological. As pointed by [Blais and Protter \(2010, p.826\)](#), this model falls into a circular proof since it is using an algorithm that delivers inputs (i.e.: the aggressor side of the trade) by assuming the existing of a mechanism that is indeed the exactly behavior of the Stochastic Supply Curve that one wants to estimate.

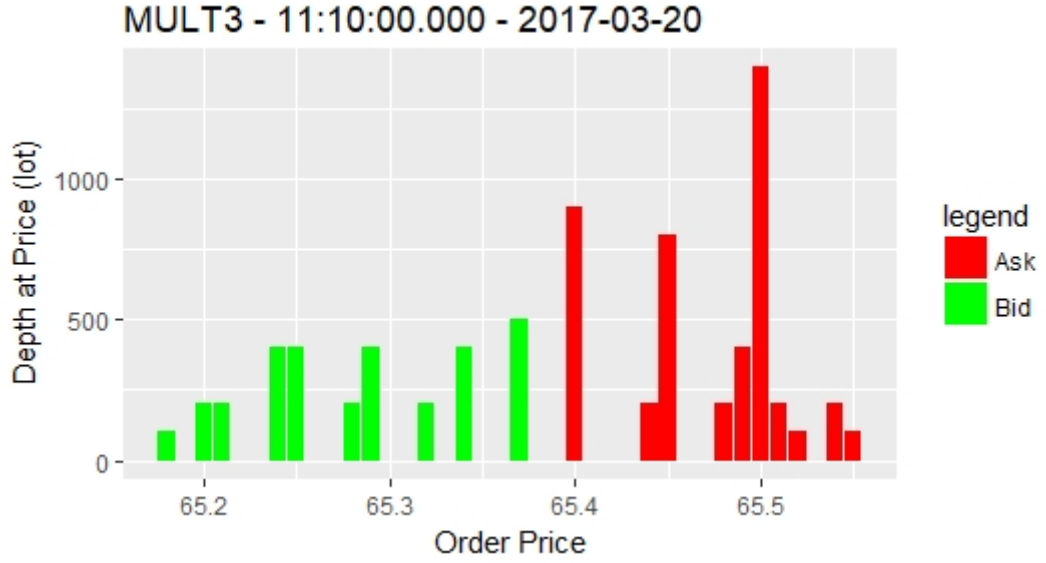
4.2 The Limit Order Book Approach

[Blais and Protter \(2010\)](#) propose a different approach that is fairly more complex than its previous counterpart since it has microstrutural fundamentals as it relies on LOB data¹ in several points in time.

The idea is to organize the best bids and asks and, then, look at its functional form against v_t with the following reasoning: an ask entry generates a potential buyer-initiated trade and a bid entry generates a potential seller-initiated traded. Then from a common LOB such kind of information is available as it is visually verifiable from Figure 4.1:

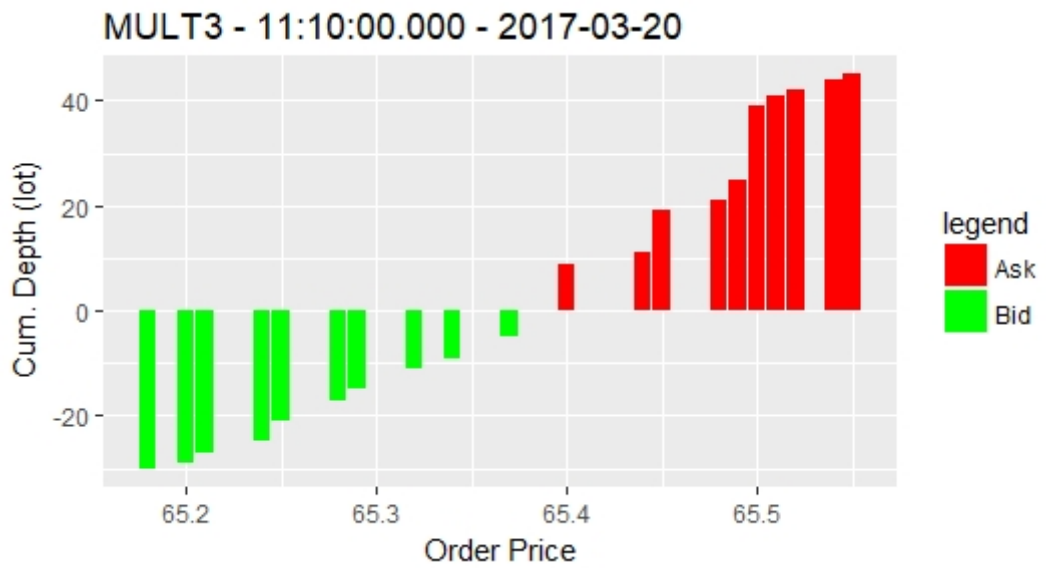
¹LOB data from Morgan Stanley ([Blais and Protter, 2010, p.827](#))

Figure 4.1: LOB of MULT3 from March 20th - 11h10



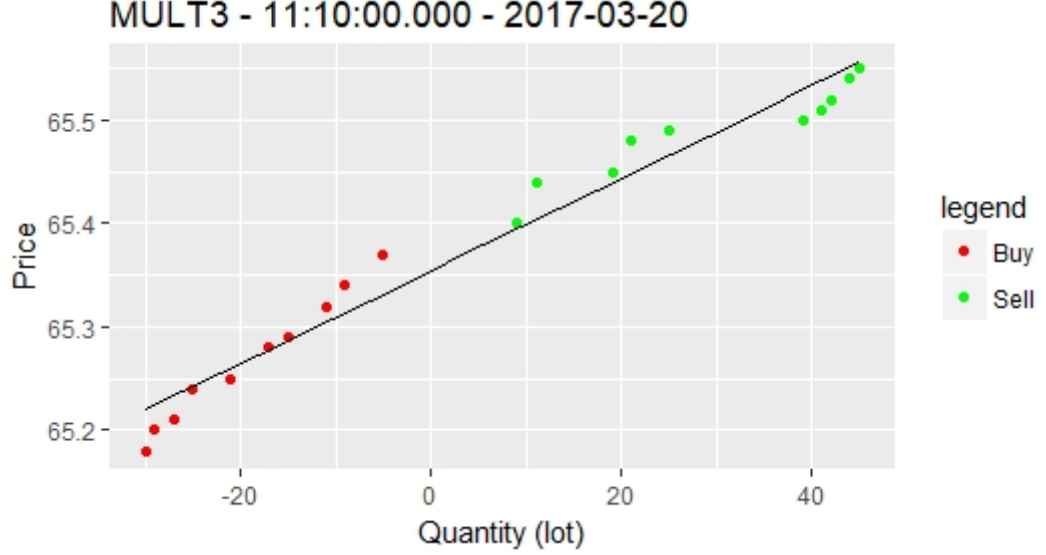
Let $(P_i, \sum_{i=1}^k)$ be the k -th level of depth at price P_i from the bid side of a LOB as in Figure 4.1. Each level of depth on the bid side provides liquidity for a given buying order of size v_B and, on the converse, a $(-P_j, \sum_{j=1}^m)$ is a point providing liquidity for a selling order. Since the volume is monotone over the depth of each side of the LOB, the Stochastic Supply Curve is expected to have an upward slope. Firstly, the cumulative quantity by side and price level is the first step to build such as in Figure 4.2:

Figure 4.2: Cumulative LOB Depths of MULT3 from March 20th - 11h10



From an usual Cumulative LOB Depth it is very clear that the Stochastic Supply Curve appears very naturally².

Figure 4.3: SSC of MULT3 from March 20th - 11h10



As in Figure 4.3, Blais and Protter (2010, p.827) takes the 10 best asks and bids for each regression applied through time with the intention to focus on the more central part of the LOB Depth³.

For several LOBs through time they propose a simple linear supply curve in the form of:

$$S(t, \Delta X) = \alpha \Delta X + S(t, 0). \quad (4.2)$$

The functional form in (4.2) is assigning a stationary independent and identically distribution to α , which makes usual parametric statistical inference a straightforward way to reject that $S(t, v) = S(t, 0), \forall v_t \in \mathbb{R} \times \mathbb{R}^+$, thus confirming statistically that the SSC is non-trivial (Blais and Protter, 2010, p.828).

In regression of n supply curves, $\{\hat{S}_i(\Delta X) = \hat{\alpha}_i \Delta X + b_i\}_{i=1}^n$, they assumed

$$(\alpha - \hat{\alpha}_i) \sim \mathcal{N}(0, \sigma^2). \quad (4.3)$$

²Since any remaining point violating monotonicity is, by definition, a noisy entry.

³Which makes sense and also has to do with the fact that CJP focuses on “small investors” (Çetin et al., 2010).

Thus, with $Z = \frac{\hat{\alpha}_i - \alpha}{(\sigma_{\hat{\alpha}_i - \alpha})^{\frac{1}{2}} \sqrt{n}} \sim \mathcal{N}(0, 1)$ they tested $H_0 : M = 0$ against $H_1 : M \neq 0$. [Blais and Protter \(2010, p.828\)](#) found very significant evidences of $\alpha > 0$ parameter for order book data for the week of July 14th 2003 to July 18th 2003.

In contrast to [Cetin et al. \(2006\)](#), [Blais and Protter \(2010, p.829\)](#) take into consideration the stochastic behavior of $\alpha_t, \forall t \in \mathbb{R}^+$, finding evidence that α_t changes through time with small variance and can be thought as a differentiable function or even as a mean reverting stochastic process ([Blais and Protter, 2010, p.829](#)). Then, they also suggest:

$$S(t, \Delta X) = \alpha_t \Delta X + S(t, 0). \quad (4.4)$$

It is worth noting that $S(t, 0)$ is modeled explicitly, differently from [Cetin et al. \(2006\)](#). In the framework, $S(t, 0)$ is also an estimated parameter,

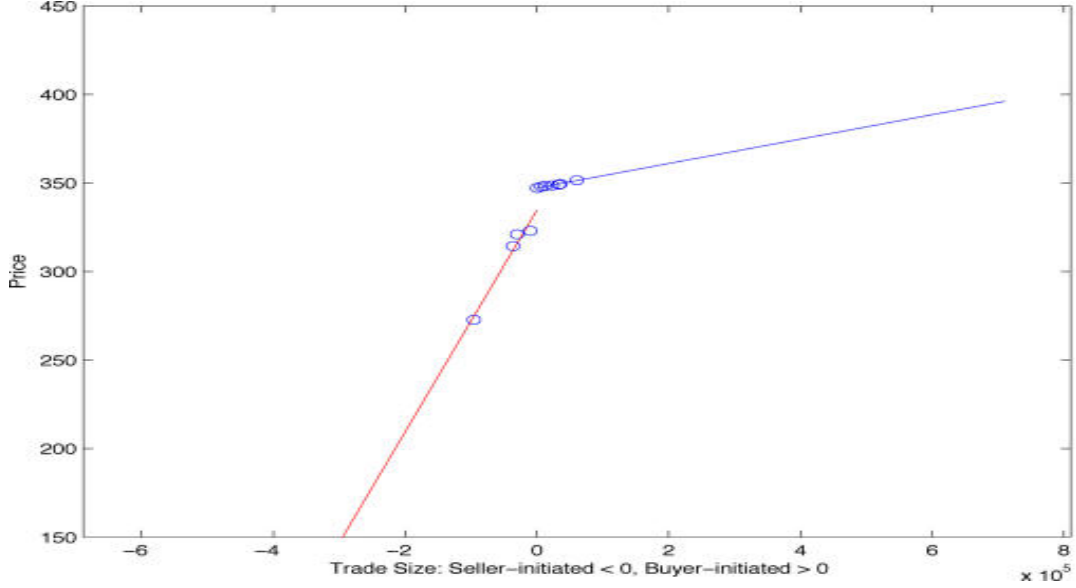
Usually $S(t, 0)$ is represented by a proxy like the mid-price ([Avellaneda and Stoikov, 2008](#), [Guéant, 2016](#)) or it is avoided as in ([Cetin et al., 2006](#)) (i.e. the take the log-return as Brownian motion increments are independent). The best way to deal with the unobservable stock price process remains an open question and interesting topic for subsequent research.

Finally, for an illiquid stock, [Blais and Protter \(2010, p.832\)](#) suggest the so-called “Jump linear SSC”:

$$S(t, \Delta X) = [\alpha_t^- \Delta X + S(0, t)^-] \mathbb{1}_{\{\Delta X < 0\}} + [\alpha_t^+ \Delta X + S(0, t)^+] \mathbb{1}_{\{\Delta X \geq 0\}} \quad (4.5)$$

And give as an example:

Figure 4.4: Allied Domecq: Jump linear supply curve (Blais and Protter, 2010, p.832)



4.2.1 The LOB Approach: Literature Review

Árdal (2013, p.11) applies SSC framework for three bonds issued by the Housing Financing Fund in Iceland using HFD provided by the Icelandic Stock Exchange covering the period of July 8th 2004 until February 5th 2010.

The author employed common least squares regression testing for the following functional forms assuming stationary α parameter (Árdal, 2013, p.8):

$$S_1(t, \Delta X) = S(t, 0)[\alpha_1 \cdot \Delta X + 1] + \epsilon_{1,t} \quad (4.6)$$

$$S_2(t, \Delta X) = \exp S(t, 0)[\alpha_2 \cdot \Delta X] + \epsilon_{2,t} \quad (4.7)$$

$$S_3(t, \Delta X) = \left[\alpha_3 \cdot \text{sign}(\Delta X) \cdot \sqrt{|\Delta X|} \right] S_3(t, 0) + \epsilon_{4,t} \quad (4.8)$$

$$S_4(t, \Delta X) = [\alpha_4 \cdot \text{sign}(\Delta X) \cdot \ln(1 + |\Delta X|)] S(t, 0) + \epsilon_{4,t}. \quad (4.9)$$

The LOB was examined only five times each trading day and regressions were run limited to at least $n = 3$ bid and ask levels (Árdal, 2013, p.15) to prevent for extreme cases too far from the core of the SSC ⁴.

Also segregated the analysis into “Before Crisis”, “During the Crisis”, “Weak Market Making” and “Normal Market Marking” time periods Árdal (2013, p.14-

⁴Making these bonds much less liquid than equities. For instance, Blais and Protter (2010) employed $n = 10$.

16). The “Before Crisis” period goes from July 8th 2004 to October 10th 2008. During the peak of the crisis, market makers abandoned their posts in the bond market and liquidity dried up during the so called “During Crisis” period lasted from October 11th 2008 to December 15th 2008. Some months into the crisis, the House Financing Fund and newly established banks signed a market making agreement looser than previous pre-crisis conditions, marking a period of “Weak Market Making” from December 16th 2008 to June 30th 2009. Finally, in July 1st 2009 a new normal market making agreement was signed and liquidity returned to a “Normal Market Making” period.

He finds statistically significant evidence of non-trivial SSC by same inferential procedures (Árdal, 2013, p.29) (Árdal, 2013, p.19-23) and different α values for each period: greater liquidity for “Before Crisis” and “Normal Market Making” periods and lesser liquidity in “During the Crisis” and “Weak Market Making”. Finally, functional form regressions also varied with time periods: less liquid periods with SSC resembling 4.4 and exactly (4.5).

Other works involving estimating the SSC with empirical findings were not found anywhere else.

Nevertheless Blais and Protter (2010) approach to the SSC has a scarce literature, the same cannot be said with respect to the general acceptance of its empirical findings.

Jarrow et al. (2012, p.1340) explicitly makes use of (4.4) form to represent liquidity costs. The linear impact form of the actual trading prices was also found in (Klöß, 2013, p.15).

The assumption of constant density outside the bid-ask spread in (Roch and Soner, 2013, p.7) and (Cohen and Szpruch, 2012, p.218) also refers to (Blais and Protter, 2010) work⁵ for corroboration.

On the very extreme opposite, Gu and Steffensen (2015, p.12) recalls the nature of α_t as expressed in (4.4). They incorporate α_t insight to represent time-varying liquidity affecting time-varying volatility in a optimal execution problem under dynamic mean-variance criterion.

Originally, CJP Model main concern was to discuss how option hedging and its pricing under the replicating portfolio argument are affected by liquidity costs, but restricted to not so large positions⁶.

But it seems that the flexible and natural way that SSC represents microstructural elements make it convenient for several model designing, a popularity that made also Blais and Protter (2010) a common “battle-horse”, sometimes in the most surprising and controversial styles.

⁵But for very liquid assets. This is odd in view of (Blais and Protter, 2010, p.829) and (4.4).

⁶Which is even explicitly stated in Çetin et al. (2010) title.

And this is so despite its “fairly old” data⁷. As Jarrow and Protter (2015, p.131) states in a work 10 years later: based on Chi-X Europe knowledge, they affirm that behavior of the SSC on its center for highly liquid stocks is still linear, even with the advent of HFT.

Whether or not all those “well-suited” assumptions relying on interpretations of (Blais and Protter, 2010) had survived the “test of time”, next section tries to shed some light into Brazilian Markets.

4.3 SSC Estimation on Brazilian Markets

The building of the LOB in Chapter 3 enables the full implementation of Blais and Protter (2010).

The ticker and date from Chapter 3, MULT3’s⁸, is again used in this example.

For fixed times on the trading day of March 20th 2017⁹ the LOBs were built in 1 minute interval basis from 10h00 AM to 16h55 PM¹⁰ converted in lot units, delivering a total of 416 LOBs. From this total, several LOBs did not present enough depth to consider 10 bids and ask, totalizing $n = 20$. The alternative was to consider the minimum on the available number of bid or ask, avoiding to run an unbalanced SSC.

The respective 416 SSCs can be found in Annex ¹¹.

At first sight, the SSC give significant evidence of non-trivial Liquidity Costs, which is a marking reassurance on the validity of the main feature of CJP Model.

A dynamic minute-by-minute exposition of the SSC makes clear the stochastic nature of Liquidity Costs exactly as it was very much mentioned on the literature.

Both dynamic features of functional form and stochastic behavior of α -parameter are two main topics that naturally arises and, due to econometric aspects of estimation techniques, are very much relate.

One of those, for instance, is not explicit in both Blais and Protter (2010) and Árdal (2013): by assumption it is mandatory that $\alpha, S(t, 0) \in \mathbb{R}^+$. So, the classical Least Squares may not be at very soul of CJP Model¹².

⁷Data is from 2003/2004.

⁸MULT3 is the ticker of common shares from Multiplan SA, a real estate company founded in 1975.

⁹For instance, an expiration day for equity options of physical settlement.

¹⁰The typical trading hour is from 10h00 AM to 17h00 PM but a little later and earlier of opening and ending times of the trading hours some opening and closing calls are still being settle and registered outside auction calls. They are disregarded to avoid abnormal LOB estimates. For example, see 3.1 for trades without an “aggressor side”.

¹¹Actually, there are 420 plots. The last four were included to stress the fact that indeed the LOBs will have uncommon behavior at the very end of a trading day

¹²Recalling that for Blais and Protter (2010, p.828) and (4.3) the normality assumption on

In addition, if one wants to consistently keep track of α through time then it would be desirable to employ the same functional specification for all times. But as presented for several SSC observations, the SSC presents different types of non-linearities hardly incorporated by an one-variable least-squares approach.

For a start, prospecting the possibility of $S(t, 0)$ to be a reasonable representation of a classical GBM or “marginal/true price”, estimations were run based on (2.5) and its log-level form taken:

$$\begin{aligned} S(t, \Delta X_i) &= S(t, 0) \exp(\alpha_t \Delta X_i + \epsilon_i) \Rightarrow \\ \log(S(t, \Delta X_i)) &= \log(S(t, 0)) + \alpha_t \Delta X_i + \epsilon_i. \end{aligned} \quad (4.10)$$

If $S(t, 0)$ is a log-normal GBM than $\log(S(t, \Delta X))$ is Gaussian. This model, though, does not address $\alpha > 0$ condition.

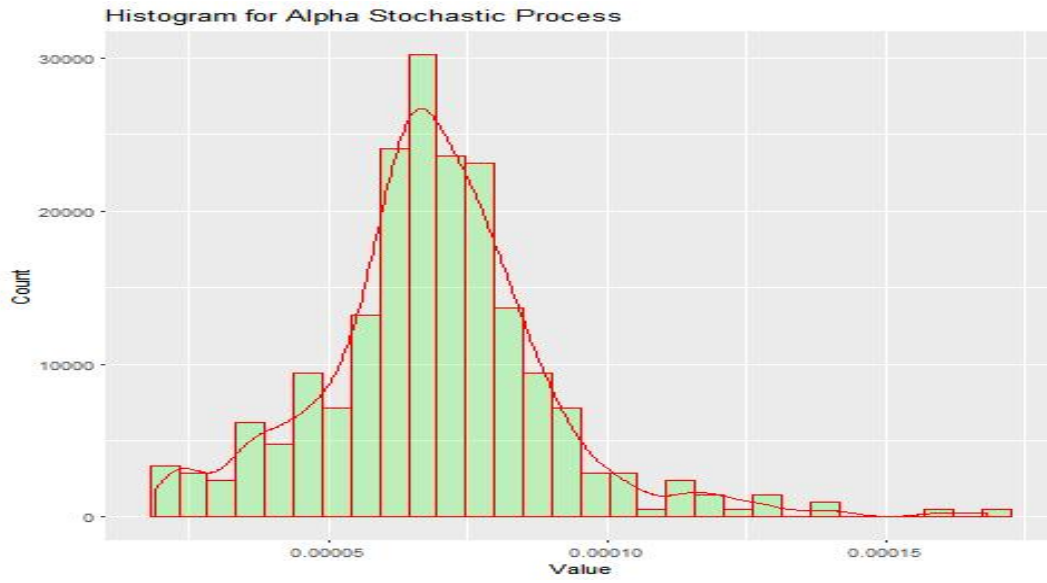
Parametric interpretation of log-level assumes a convenient form in this approach. Taking the derivative of Y with respect to ΔX and then solving for α_t one obtains directly:

$$\begin{aligned} \frac{\partial}{\partial \Delta X} S(t, \Delta X_i) &= \frac{\partial}{\partial \Delta X} S(t, \Delta X) \exp(\alpha_t \Delta X_i + \epsilon_i) \\ &= \alpha_t \overbrace{S(t, \Delta X) \exp(\alpha_t \Delta X)}^{S(t, \Delta X_i)} \\ \alpha_t &= \frac{\partial}{\partial \Delta X} S(t, \Delta X_i) \frac{1}{S(t, \Delta X_i)}. \end{aligned} \quad (4.11)$$

This means that a change in one ΔX unit, for instance, shares in lot units, will delivers a percentage change in $\alpha_t\%$.

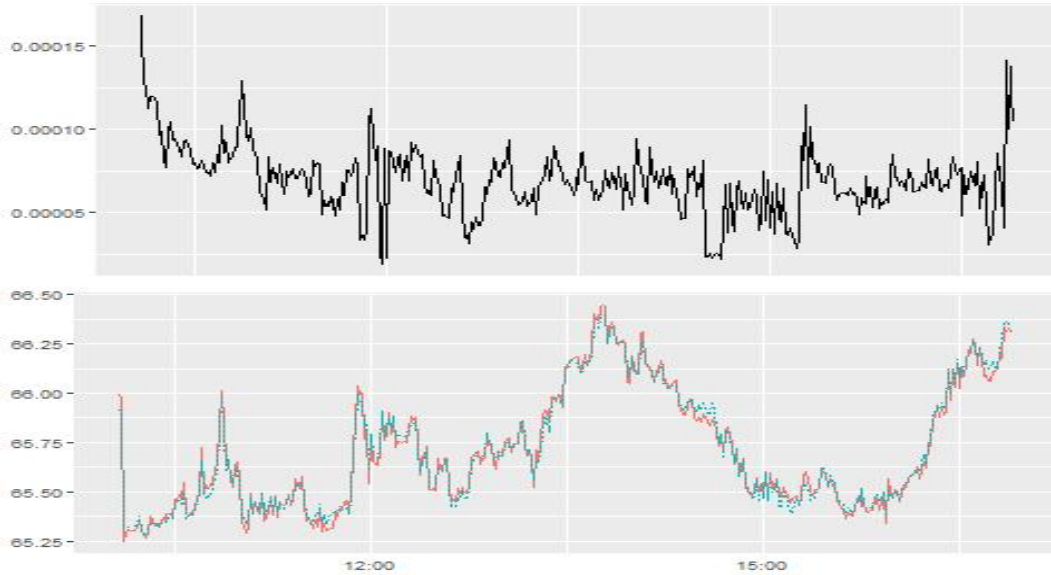
$(\alpha - \hat{\alpha}_i)$ is acceptable for stationary (i.e. time-independent) estimates in the form of (4.2).

Figure 4.5: Histogram of $\hat{\alpha}_t$ from 10h05 AM to 16h50 AM - MULT 3



Estimates for $\hat{\alpha}_t$ (orange fitted line) and $\hat{S}(t, 0)$ (orange horizontal dotted line) with the mid-price (blue horizontal dotted line) were plotted for the period from 10h05 AM to 16h50 AM and presented below¹³:

Figure 4.6: Estimates of α_t from 10h05 AM to 16h50 AM - MULT 3



¹³Both estimates were too extreme for the first 5 and last five minutes of the original dataset and were dropped.

Objectively, fitted curves for model 4.10 resembled an OLS level-level case and could not incorporate the shape-changing feature of the SSCs. It is a challenge to find an adequate functional form for a regression in this way, considering that the specification must comply with the assumptions of CJP Model.

Nevertheless, there are some interesting findings worth mentioning:

1. Most models incorporate the mid-price as a proxy for the true stochastic processes for the GBM, a variable that $\hat{S}(t, 0)$ tracked very closely;
2. It is visible is the fact that although $\hat{S}(t, 0)$ is a representation for the “true prices”, the actual trading prices are indeed varying with quantities, which makes the mid-price an unfortunate representation for realistic portfolio or cash processes;
3. Values of α were very close from those found in the TAQ approach from Cetin et al. (2006);

Wide bid-ask spreads events that lasted for few minutes were spotted on the LOBs, being the most surprising those happening at the very beginning and ending of the trading day, somehow in contrary with the stylized fact that an U-shape curve for the Market Trading Volume¹⁴ would mean higher liquidity.

4. Also, extreme observations in LOBs through successive periods of this particular trading day, although presenting a huge volume at that some prices, shows that there is a level of illiquidity from the asset as the LOB got empty of usual agents and it was visualizing the orders and constant updates, particularly on bid prices, of the market market. If the only source of liquidity close to competitive prices is the market maker of MULT3, then at that particular time there was a lack of liquidity.
5. The stochastic nature of α_t is well represented by plotting the estimates given by 4.10 over the trading day. As suggested by Blais and Protter (2010, p.829), it would not be an “heroic hypothesis” to assume α_t as some kind of mean-reverting stochastic process.

This behavior on LOB core to its extremes also happens with BBDC3¹⁵ for April 13th 2017 and it is visible on a 15 to 15 minutes basis and sometimes takes minutes to liquidity return to the core of the LOB (as is visible in the grid plot 08 for March 20th 11h45 to 11h59 for MULT3).

¹⁴The V_t deterministic, continuous and bound process in Almgren-Chriss’ framework.

¹⁵Banco Bradesco common shares in Figure A.29 on the Annex.

Additionally, the fact that α_t , a liquidity parameter measuring Liquidity Cost, is being thought as a mean-reverting stochastic process with properties of its own is of great interest and contrasts greatly with the simplistic treatment relegated to the "instantaneous/realized/executed" costs. This will be discussed taking [Almgren and Chriss \(1999, 2001\)](#) and [Guéant \(2016\)](#) as benchmarks for the portfolio process.

Questions about how and when a more general class of equities on other trading days are affected by Market Liquidity Risk that could be modeled by a Stochastic Liquidity Cost Process arrives. As a general and common cause of liquidity, Chapter 5 discusses Market Liquidity e Funding Liquidity in context of option hedging, arguing that an option exercised that demands "physical delivery" act as a special type of funding bind.

Chapter 5

Stochastic Supply Curve under Pressure

HFD extracted from B3 (2017b) as reported on Chapter 3 and SSC implementation for a Brazilian equity on Chapter 4 opens several possibilities. But until now the focus was on MULT3 in March 20th.

Here some possibilities regarding comparative analysis are explored between tickers and time periods. Market Liquidity to be greater or of serious concern. In the spirit of Árdal (2013), hypothesis involves change on functional forms of SSC and size of Liquidity Costs measured by the SSC.

Chapter 1 briefly mentioned the close link between Market Liquidity and Funding Liquidity described by Levy (2015). Funding Liquidity Risk, although normally meaning cash scarcity, is essentially a liability to be claimed by a counterpart. It can be, for instance, a contractual obligation to deliver a given underlying asset due to a equity forward or equity call option, thus making it a liability of “physical settlement”¹.

A common practice is to Δ -Hedge the contingency claim in which one is short. Due to the very nature of dynamic hedging and portfolio replication, Market Liquidity Risk is serious concern not only by Liquidity Costs but also in face of a imminent execution when close to moneyness and exercise dates, that is a situation when the spot price is close to the strike price and close to the expiration date. This nuances are measured by the Greek Gamma, the second-derivative of the option price with the respect to price that measures the sensitivity of Delta when the spot price change.

Working as an analogy to the “stricto sensu” notion of Funding Liquidity Risk, a short position in call options a few trading days before the expiration would

¹Of course, financial products are not tangible. This term in opposition to a settlement by money/cash disbursement.

be a pressuring factor to acquire the underlying asset for physical settlement, amplifying Market Liquidity Risk.

5.1 Dynamic Hedging under Market Liquidity Risk

According to the Black-Scholes model, dynamic hedging is a pretty straightforward task that involves tracking optimal weights of a balanced portfolio built on the money-market account that yields a risk-free rate and the underlying asset. Through time until option maturity the portfolio value will replicate the equity option's payoff. The trading strategy that must be followed, i.e. “weights” of the portfolio, is then given by the Greek “Delta” (Δ), which is the first derivative of BSM Pricing Formula with respect the price of the underlying asset.

In a frictionless market as in the Black-Merton-Scholes Economy one can trade the underlying asset at any time and quantity without affecting the market prices or incurring costs due to the change of prices on the LOB. It means that the hedger adjusts the replicating portfolio and after the order turns in to a trade he remains Δ -hedged. In reality it is not that simple and several difficulties, namely Market Liquidity Risk, may be of relevance.

5.1.1 The Saw-Tooth Pattern

The slang “Gamma Week” is fairly known among traders; it refers to the week of the last trading days before the expiration date. On these days the Greek Gamma has a tendency to have an absolute growth² that may be even greater when the spot price of the underlying is oscillating around the strike.

In such conditions the optimal replicating portfolio may change rapidly and Δ -hedging may become very costly.

A well-documented and relatively recent example is given by [Lehalle et al. \(2012\)](#): on July 19th of 2012 several American blue-chip stocks presented a uncommon behavior named “Saw-Tooth Pattern”.

²Positive if one is short in call options and negative if in put options.

Figure 5.1: Saw-Tooth Pattern - COKE - Graph from Guéant (2016, p.172)

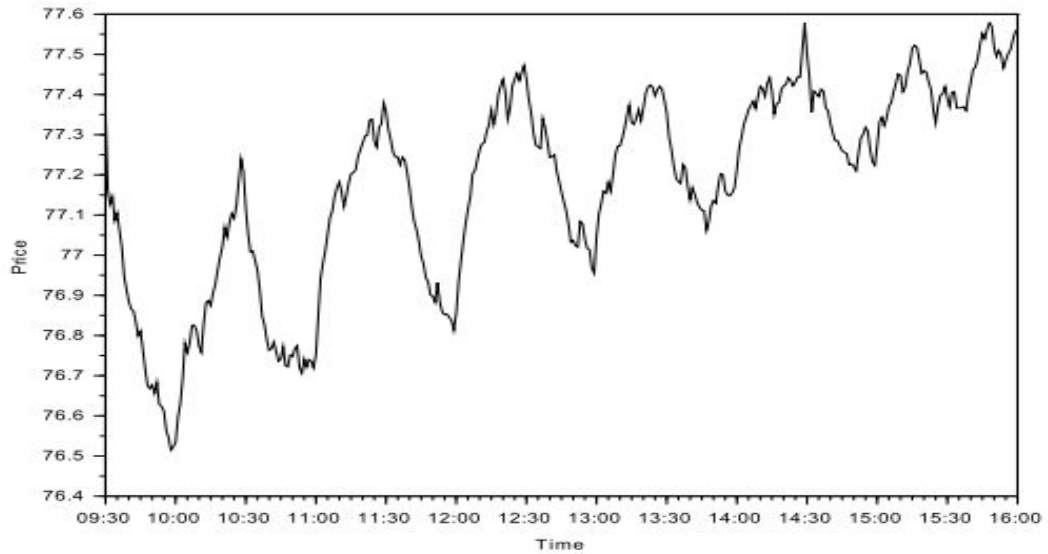
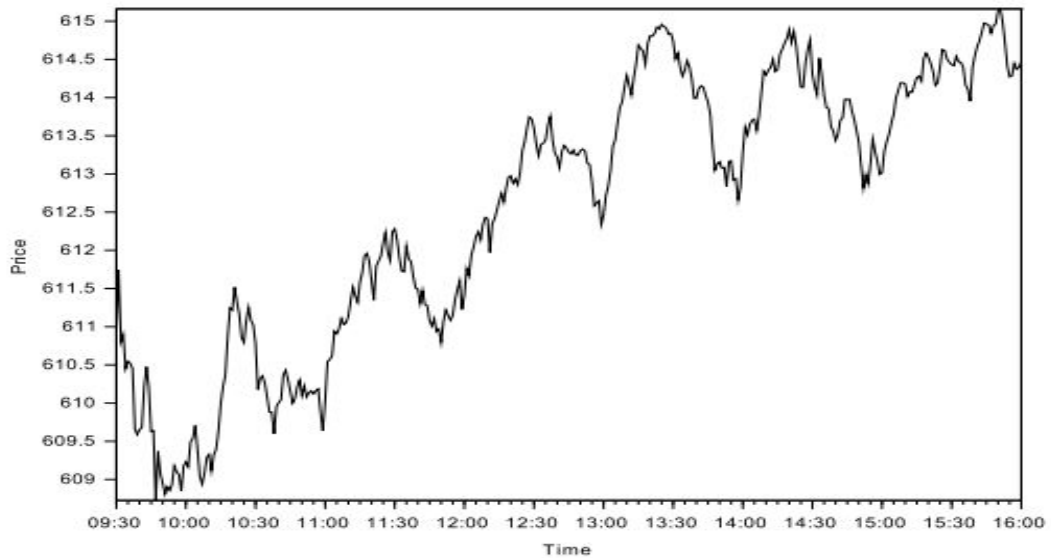


Figure 5.2: Saw-Tooth Pattern - APPL - Graph from Guéant (2016, p.172)



Figures 5.1 and 5.2 show that the stock prices behaved in a uncommon way, awkwardly resembling a “saw-tooth”: they reached local minimum every hour and then a local maximum the first half-hour of each hour (Guéant, 2016, p.171). This also happened with Mc-Donald’s and IBM.

Lehalle et al. (2012) analyzed in detail that specific day and pointed out that 19th was one trading day before the 3rd Friday of July, an expiration day for options in the American Market.

Then, such patterns were certainly due to a very large trader who was blindly Δ -Hedging a large options position³. At every half-hour a trading decision was made in respect to the state of the replicating portfolio.

As Guéant (2016, p.171-p.174) details, the most plausible history is the following:

1. At 10h00 the Δ -hedging strategy consisted in buying shares based on an execution strategy (for instance a VWAP algorithm) over 30 minutes;
2. At 10h30 the price had increased due to Permanent Market Impact. Since the portfolio was to replicate a negative Γ position (or for having a long *Gamna* position) the price increased Δ ;
3. Then, to remain Δ -hedged, a sell order was executed over the next half-hour;
4. Again because of Market Impact, the stock price went down until 11h00;
5. Similarly to step 1, but one hour later, the trader had to buy shares to, again, remain Δ -hedged;
6. But then such buying orders pushed up the prices until 11h30, changing Δ and forcing the trader to sell from the next half hour and son on.

The usage of an execution algorithm disregarding market price impacts for sure has a main role on the process, but certainly being short on a ATM/ITM option and trading on the very last day of the “Gamma Week” certainly made the situation much worse. Of course, one could not expect that this idiosyncratic Saw-Tooth Pattern will ever repeat itself again since its causes and how to prevent such uncommon behavior, as its costly consequences, are fairly know. But the one main stressful event that made arise such phenomena, that is how a trader has to keep Δ -hedging in a hurry facing an uncertain exercise, never ceases.

On Brazilian markets the option issuer has two choices when the position is ITM or very to close to ITM moneyness: trading the underlying asset and just settling the contract by physical delivery or block the exercise by buying a contract with the same features of the issued option until the last trading day before the expiration date.

Thus, instead of replicating on the following trading days of expiration, if options on critical days are being bought at abnormal prices, far above a reasonable referential price, than it may a symptom of several kinds of “illnesses”: “ill-covered”⁴ position short on ITM options (if even existent), a highly risk aversion for Δ -hedging or, last but not least, a pronounced unwillingness to Δ -hedging

³A negative Γ -position as it seems.

⁴A portfolio far below a desirable Δ .

since there is an anticipation that the liquidity for the underlying asset will definitely vanish, which is indeed the main point of this work.

The Permanent Market Impact affected negatively its portfolio performance in two ways: by moving the prices it was necessary to *Delta*-Hedge constantly; the movements were upward when it was necessary to buy and downward when it was necessary to sell.

In those terms, Market Liquidity Risk was surely present. But in regard to liquidity risks/execution costs/temporary it is not clear how it affects Δ -Hedging as those costs affects only the current prices accounted into the portfolio: this is addressed by simulating a CJP Economy in the next subsection.

5.1.2 Simulation in a CJP Economy

On the first CJP Model in (Cetin et al., 2004), the SSC is the source of liquidity costs that could be avoided by continuous trading strategies of finite variation, making the market approximately complete⁵ (in a $\mathcal{L}^2$ -sense) if there exists a unique equivalent martingale measure for $S(t, 0)$ (Gökay and Soner, 2012, p.318). On this setup the classical Black-Scholes formula indeed still giving the option price but the Δ -Hedging optimal strategy is somehow different and relies on averaging the classical dynamic hedging strategy as studied in Cetin et al. (2006).

In reality, though, traders face both discrete-time trading and a finite variation strategy with liquidity costs vanishing to zero is for sure never attainable due to taxes and fees that make frequent trading a cost-ineffective approach.

Following the classical Δ -Hedging in a discrete-time setup, some Monte Carlo simulations for scenarios of dynamic-hedging an European call option were run under a constant SSC as the source of liquidity costs in the usual form of:

$$S(t, \Delta X) = e^{g(t, D_t, \Delta X)} S(t, 0), \tag{2.5}$$

with $t = 0$ and $D_t = 0$: $S(t, \Delta X) = e^{\alpha \Delta X} S(t, 0)$.

A discrete-time Dynamic Δ -Hedging Strategy for a European call option consists in managing a portfolio with Δ -Shares and a cash balance (lend or borrowed at a risk-free rate) through time.

The economy is the same as given in CJP model, which makes the definitions of the s.f.t.s., liquidity cost and the properties of the Stochastic Supply Curve the

⁵“Weaker” conditions to fulfill a “weaker” Second Fundamental Theorem of Finance.

same too.

Recalling the definitions in Chapter 2 for the s.f.t.s.:

$$Y_t = Y_0 + X_0 S(0, X_0) + \int_0^t X_u dS(u, 0) - X_t S(t, 0) - \sum_{0 \leq u \leq t} \Delta X_u [S(u, \Delta X_u) - S(u, 0)] - \int_0^t \frac{\partial S}{\partial x}(u, 0) d[X, X] u^C, \quad (2.3)$$

The last terms of the equation (2.3) represent liquidity costs over maturity time:

$$L_T = \sum_{0 \leq u \leq t} \Delta X_u [S(u, \Delta X_u) - S(u, 0)] - \int_0^t \frac{\partial S}{\partial x}(u, 0) d[X, X] u^C. \quad (2.4)$$

Considering a partition $0 = t_0 \leq t_1 \leq \dots \leq t_n = T$ with $\Delta t = t_i - t_{i-1}$, $\forall i = 1, \dots, n$, the discrete time version of (2.4) would be:

$$L_T = \sum_{i=1}^n \Delta X_i [S(t_i, \Delta X_i) - S(t_i, 0)] + X_0 [S(0, X_0) - S(0, 0)], \quad (5.1)$$

where $\Delta X_i = X_{t_i} - X_{t_{i-1}}$. Finally, given C_0 as the price of a derivative at time 0 of a hedging strategy, the hedging error is defined as:

$$\sum_{i=0}^{n-1} X_{t_i} (S_{t_{i+1}} - S_{t_i}) - C + C_0 - L_T, \quad (5.2)$$

Where L_T is the total liquidity cost until maturity time, C_0 is the call premium received by the issuer at t_0 , C is the final pay-off of the option and $\sum_{i=0}^{n-1} X_{t_i} (S_{t_{i+1}} - S_{t_i})$ is the selling and buying of stocks just before the exercise of the option. There is no FV updating nor Cash Balance accounting because $r_f = 0$.

The following simulation results were execute in R with specifications given by [Ku et al. \(2012\)](#): risk-free rate $r_f = 0.00$; $\sigma = 0.2$; $K = 100$; $T = 1$ year; Weekly and daily revision, $N_w = 52$ and $N_d = 300$ respectively; and Liquidity parameter α of 0, 0.0001, 0.0005, 0.001 and 0.002, and Spot price of $S_0 = 100$. It was not specified on how many trials the simulations were run, then the following simulations are settled in $B = 50.000$, fixing R's random seed as "011092" for $N = 52$ and $N = 300$.

The economy is the same as given in CJP model, which makes the definitions of the s.f.t.s., liquidity cost and the properties of the Stochastic Supply Curve the

same too.

For start, a Base Case scenario with no liquidity cost was ran with weekly ($N = 52$) and daily ($N = 300$) to verify the soundness and validity of the present implementation. Table 5.1 presents some results about Hedging Error. Those are compared with the theoretical standard deviation of Hedging Error given by Derman and Miller (2016, p.111-112) as:

$$\sigma_{HE} \approx \sqrt{\frac{\pi}{4}} \frac{\sigma}{\sqrt{n}} \frac{\partial C}{\partial \sigma}. \quad (5.3)$$

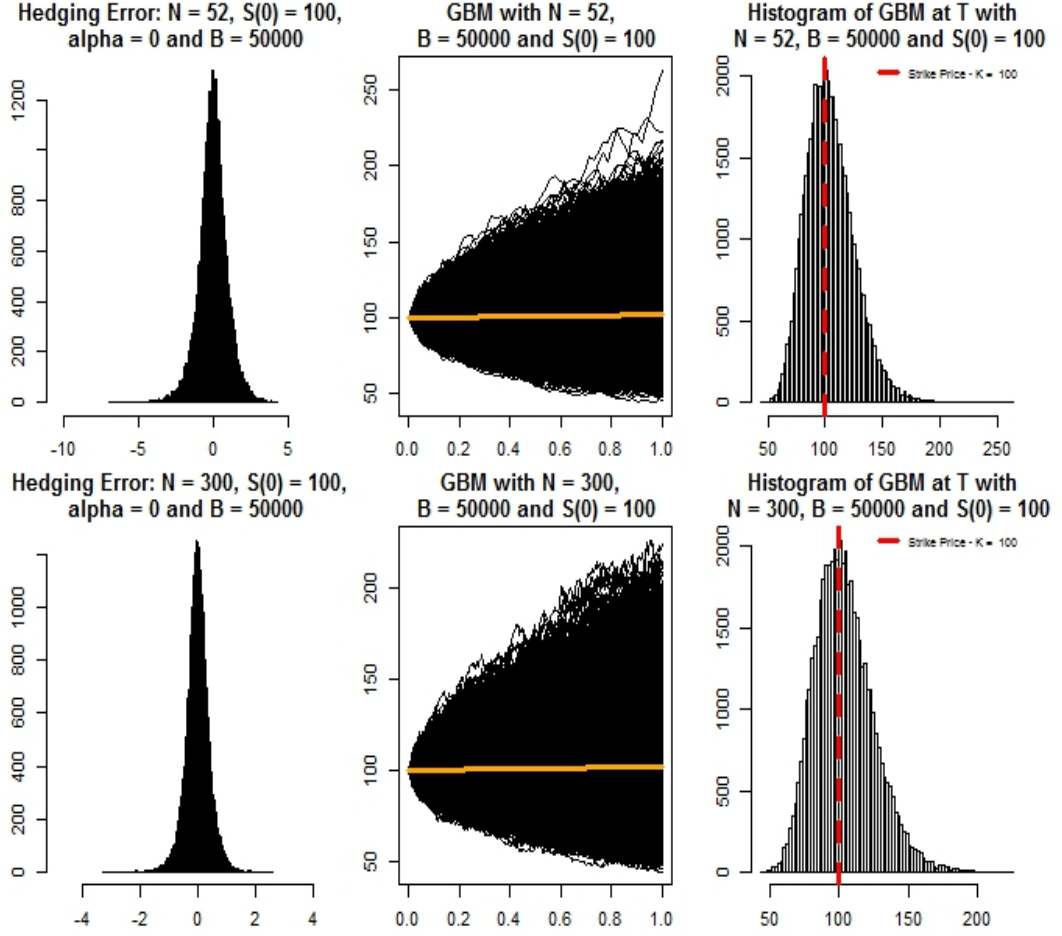
The equation states that the Hedging Error standard deviation must be proportional to Kappa (also called “Vega”) and diminishing with hedging frequency. From Table 5.1 it is possible to see that both simulation of the Base Case scenario as the Hedging Error goes to zero and the set of empirical σ_{HE} match (5.3).

Table 5.1: Hedging Errors without Liquidity Risk and $N = 52$ and $N = 300$

Hedging Error	$N = 52$	$N = 300$
Mean	-0.003912	0.000693
Empirical σ	0.956679	0.399802
Theoretical σ	0.97569	0.406212
Maximum	4.325	2.59
Minimum	-6.9635	-3.2425
Median	0.0072	0.0009

The Figure 5.3 gives a histogram of Hedging Error, a plot with simulated Brownian paths and a histogram of GBM at $t = T$ for both weekly and daily revisions simulations.

Figure 5.3: Histogram of Hedging Errors, Paths of GBM and Histogram of $S(T)$



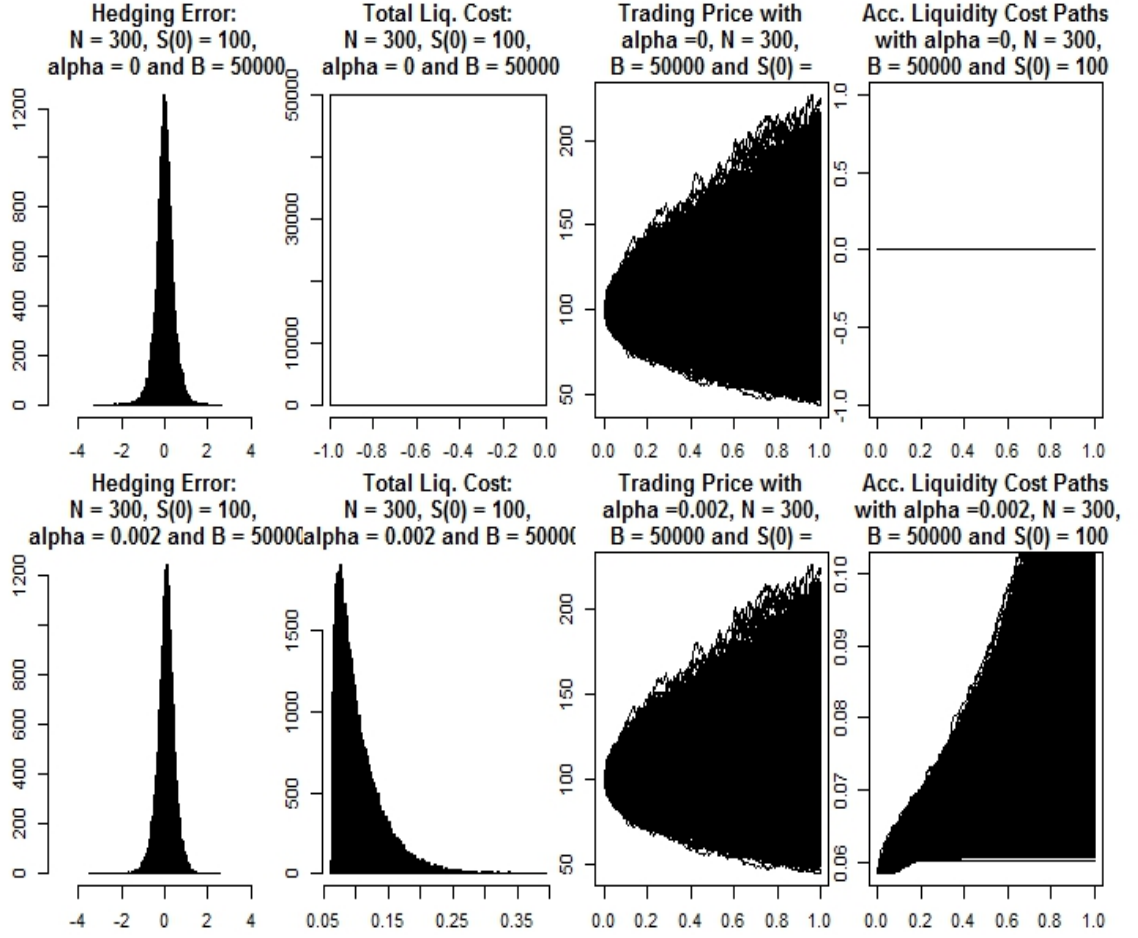
With $N = 52$ it was found that 54.414% of the trials ended ITM, matching $\frac{\partial C}{\partial S}(t_0, S_0) = \Delta_0 = 0.53983$, and with $N = 300$ it was found that 54.046% ended ITM. The Delta paths for all trials can be appreciate in Figure [A.30](#) at the Appendix.

From all these results, the fundamental scheme employed here is working as it should be for a conventional discrete time Δ -Hedging.

When introducing the SSC on the trading price there will be a marked difference, as L_T (5.1) is never zero.

From the SSC on (2.6) it is possible to obtain "friction" prices affected by trading of Δ -shares at each time. Figure [5.4](#) summarizes simulations with $N = 300$ and $\alpha = 0$ and 0.002.

Figure 5.4: Trading Prices with $B = 50000$, $\alpha = 0.0000$ and 0.002 for $N = 300$



When $\alpha = 0$ there are no liquidity costs as depicted on 5.4. With $\alpha > 0$, the total liquidity costs have a characteristic χ^2 distribution because L_T in (5.1) is never below zero.

Table 5.2 shows what is not much visible in Figure 5.4: Liquidity Costs have an effect on hedging error and also deliver a slightly higher variance than (5.3).

Table 5.2: Hedging Errors with $B = 10000$, $\alpha = 0$ and 0.002 , $r_f = 0.00$ for $N = 52$ and 300

Hedging Error	N = 52 & $\alpha = 0$	N = 52 & $\alpha = 0.002$	N = 300 & $\alpha = 0.000$	N = 300 & $\alpha = 0.002$
Mean	-0.003912	-0.111951	0.000693	-0.107053
Empirical σ	0.956679	0.964018	0.399802	0.405303
Theoretical σ	0.97569	0.97569	0.406212	0.406212
Maximum	4.325005	4.229693	2.589966	2.398281
Minimum	-6.963524	-7.234885	-3.24251	-3.638936
Median	0.007184	-0.089124	0.000886	-0.093122

It is expected that as $n \rightarrow +\infty$, the liquidity costs diminishes and the s.f.t.s. approaches a continuous hedge strategy of finite variation. This trend appears somehow in Table 5.2.

This is tested with a series of simulations with $B = 5.000$ or 10.000 , making $n \rightarrow +\infty$ through a sequence with $N = 21, 52, 84, 300, 1000$, and 10000 .

When $B = 5.000$, $r_f = 0.00$ but $\alpha = 0.002$, it seems that what is stated by Cetin et al. (2006, p.512-516) is somehow converging, but not monotonically, to -0.063 . This can be better assessed in Table 5.3 below and Figure A.32 at the Annex.

Table 5.3: Hedging Errors with $r_f = 0.00$, $\alpha = 0$ and $B = 5,000$

Hedging Errors	N=21	N=52	N=84	N=3000
Mean	-0.055155	-0.064433	-0.069845	-0.062026
Empirical σ	1.497193	0.965676	0.754204	0.42007
Theoretical σ	1.535338	0.97569	0.767669	0.406212
Maximum	4.844636	3.629177	3.328904	1.942721
Minimum	-8.271941	-4.797408	-3.807625	-2.515207
Median	-0.02142	-0.048161	-0.051305	-0.054352

The Call Premium to do the Hedging Errors accounting comes from Ku et al. (2012, p.139) and found at the Annex on Table A.1.

But this is so because $r_f = 0.00$, which prevents L_0 to grow when updated to FV. Then, one more scenario were run, settled with $\alpha = 0.002$, and $r_f = 0.05$ to amplify the impacts from liquidity costs.

Firstly, on scenarios without liquidity risk ($\alpha = 0$) even when $r_f = 0.05$ and no matter what N is employed: hedging errors is asymptotically diminishing and standard deviation also matches (5.3) as shown in Table 5.4 and Figure A.31 at the Annex.

Table 5.4: Hedging Errors with $r_f = 0.05$, $\alpha = 0$ and $B = 10,000$

Hedging Error	N = 21	N = 52	N = 84	N = 300
Mean	-0.012586	0.005392	-0.003863	0.001498
Empirical σ	1.452816	0.929982	0.731176	0.392699
Theoretical σ	1.451359	0.922323	0.725679	0.383993
Maximum	6.545	4.4456	3.3028	2.0179
Minimum	-8.9911	-7.0206	-3.7396	-2.55

Then for $\alpha = 0.002$ and $r_f = 0.05$ the accounting of Hedging Errors employed the usual BS Price for the Call Premium received by the issuer of the contingency claim. The Hedging Errors are summarized in 5.5 and the Histograms of Hedging Errors are presented in Figure A.33 at the Annex.

Table 5.5: Hedging Errors with $B = 10000$, several N and $\alpha = 0.002$, $r_f = 0.05$

Hedging Error	N = 21	N = 52	N = 84	N = 300	N = 1,000	N = 10,000
Mean	-0.14526	-0.12709	-0.13529	-0.13033	-0.13093	-0.132707
Empirical σ	1.46172	0.93727	0.73776	0.39775	0.22288	0.079801
Theoretical σ	1.45136	0.92232	0.72568	0.38399	0.21032	0.06651
Maximum	6.40093	4.25571	3.12108	1.8051	0.74452	0.108921
Minimum	-9.30596	-7.38504	-4.07491	-2.86282	-2.87302	-0.744795
Median	-0.1051	-0.10687	-0.11706	-0.11691	-0.1179	-0.118768

The persistent expectation of Hedging Errors turned to be around 0.13, which shows indeed that liquidity costs are stronger when $r_f > 0$ even when $n \rightarrow +\infty$.

5.2 Equity Options on Brazilian Market

The only exchange presenting order-driven markets for equity options and equities is also B3. As any country, Brazil has some unique characteristics and own business logic/rules relevant for modeling.

This section discusses some relevant topics on Brazilian equity options and describes some information and publicly available sources.

5.2.1 Equity Options

When issued in Brazil, a standard equity option⁶, European or American, is part of series arranged by market makers with approval of B3. When approved, the

⁶Traded on the exchange instead being an OTC instrument, which may be a highly customized contract. Since the strike price changes when there are earning associated with the underlying asset, it is not a “plain vanilla option”.

equity option series has its ticker code roughly named after its features as follows:

- First 4 letters stand for the code of the underlying asset, specifically stocks;
- A fifth letter to specify the month of expiry: A to L for call options and M to X on put options. This naming is ambiguous since there are some option series with maturities over one year;
- Last two or three digits⁷, helps to differentiate the series of the option for a expiration expiration month and sometimes the letter 'E' at end of the contract may indicate if it is an European contract⁸.

A ticker code is unique until its expiration date but only by reading it is not sufficient to retrieve all of its features over time. A ticker code does not indicate if the equity option has its underlying on a ordinary or preferential share neither if it is an European or American contract. Also, the same ticker code may refer to totally different options depending on the time it exists. For instance, by September 30th 2015 ABEVA20 was a series of European Options with ordinary shares as underlying and strike price of BRL 19.94 while on October 20th 2016 it was an American option on ordinary shares as underlying and strike price of BRL 20.24.

Strike prices change through time and keep track of those changes is not an easy task. When stock earnings⁹ or events like stock splitting occur they affect option strikes but B3 automatically adjusts them without changing the ticker codes.

Another problem is that even the underlying asset may change. An example would be the trend of getting rid of preferred shares, unifying different equity tickers under an unique common stock. Since B3 was only Bovespa it released a new standard named “Novo Mercado”, a non-obligatory certification for listed company that complies with so called “Best Corporate Governance practices”. One imposition to become a Novo Mercado company is being exclusively listed in the stock exchange by ordinary shares. For this reason, plenty of public companies are issuing deals proposing to share holders an exchange between preferential and ordinary stocks and also releasing the so-called “units”¹⁰.

⁷On December 4th of 2017 B3's the series will be a three-digit number (B3, 2017f)

⁸Normally when data is reporting exercises. Unfortunately, it is not the case for HFD from B3 (2017b).

⁹Normally dividends or interests on net equity, although events like asset-liability management of debentures (private debt) emissions may considerably change the strike price as happened with MPXE11 (MPX Energia) (B3, 2011).

¹⁰Units is an account unity that groups a certain number of preferential and ordinary shares and is also listed on B3 Stock Exchange. This procedure may suit an intermediary state that also seeks to extinguishing preferential stocks. Normally they are coded with a "11" for the last two digits.

Only in 2017 those changes in equity and sometimes for the underlying asset of equity options happened to Vale (B3, 2017g), Sanepar¹¹ (B3, 2017d), Eletropaulo (B3, 2017e), and Suzano (B3, 2017h). When such switch of asset position changes over the whole market the underlying of contingency claims on equities also changes from preferred to common stocks, a considerable modification on these contracts.

5.2.2 Equity Options Exercise

As mentioned previously, market liquidity risk may occur close to the expiration day like the emblematic example of the “Sawtooth Pattern” on July 19th 2012, which was a last trading day before expiration of options in USA and on the Gamma Week. Thus, how and when an option is exercised are decisive factors on underlying assets in those contexts.

In Brazil, the expiry date is the third Monday of a given month if it is a trading day, otherwise it is the subsequent trading day. The so called Gamma Week is different from USA since it happens on the previous trading week. But the week is an intense one: Brazilian ETF options expiration dates are on the second Wednesday of the contract month, that being the very same Gamma Week of equity options.

After the trading day it was bought, an American option can be exercised at any moment of on any trading day before its expiration date but on shorter time window on its expiration date, namely 10h00-13h00¹². An European option can only be exercised on the expiration date following the same hours of the American option. Normally an option is exercised when it has a positive payoff, although OTM options may also be exercised if the underlying price is close enough to the strike price depending on authorization from B3.

Equity options are of “physical settlement”: the trader must deliver the underlying asset when the contract is exercised¹³. Basically, then the equity option is exercised or may be close to be exercised, the trader has two choices:

1. A closing purchase transaction of the same options’ series, thus netting her position. The purchasing for this end, though, only starts to count on the trading day after the purchase¹⁴ and it is allowed until the last trading day before the expiration. Actually, before April 24th 2017 the offsetting of an

¹¹Common and preferred stocks were converted to an “Unit”, namely SAPR11.

¹²Schedule may change on Daylight Saving Time

¹³Or pay the holder for a stock at the strike price if it is a put option.

¹⁴This means that if a trader of an American option is exercised on a given trading day, purchasing the options’ series will not offset the current exercised position at that very moment.

exercised position by buying the issued contract on the same day was back then allowed (B3, 2015);

2. When dealing with indexes, the trader pays the options' payoff or, when the contract is of "physical" settlement as in equity options, deliver the underlying asset.

When there is a exercise, the contract is settled when the contracting part of the options contract receives the stock.

5.2.3 Equity Options Features

Acquiring data on features of a certain equity options demands the constant tracking of several events, variables and rules of business logica and, as mentioned previously, these informations are not easily obtained from B3 (2017b) or ticker codes.

The end of BDIN reported in the Chapter 4¹⁵ indeed made the data treatment far more challenging, as the former standard presented far better documentation.

Then it becomes necessary to use of another two different kinds of data about equity options:

- "COTAHISTAYYYY.txt" (B3, 2017a) are files holding closing information of several trading days. With the proper treatment they provide strike prices¹⁶, expiration date and the type of underlying stock (common or preferred)¹⁷. But does not include options' volatility, referential prizes. It is worth note that since it is a dynamic file where it reports closing data for a whole trading day, one must be aware about the possibility that a ticker is absent of a given day because it was not traded;
- "PEYYYYMMDD.ex" (B3, 2017c) presents the equity options referential prizes (EORP from now on) files that report static information about the available equity options series in activity but it is only available from December 2016 to nowadays. Unfortunately it lacks information on the underlying asset, but reports options' volatility, referential premiums and, most importantly, the adjusted strike prices.

¹⁵Not without negative reactions (Reclame, 2017).

¹⁶As discussed earlier, the strike prices may change dramatically through maturity.

¹⁷For equity options there is a column of the dataset that gives the first four digits of the ticker code and the second tells the type of the stock. But this information will identify underlying asset with certainty if the company has only one common share and only one preferred share.

The approach here proposed for basic information on equity options features is not ideal because it does not guarantees perfect matching between option contracts and underlying assets¹⁸ and also there are inconsistencies between EORP and historical quotations datasets¹⁹. If proper documentation were publicly available²⁰, For instance, “BVBG028”²¹ would be a far more accurate way²² to identify ticker’s features in general.

Nevertheless, this procedure presented several valid observations and give a general view with respected with equity options features, thus making it acceptable for exploratory data analysis.

5.2.4 Offsetting Market Liquidity Risk: Option Prices

As convenient as an American call option might be it also was the corresponding offsetting a not so long time ago: by just buying the exact same options series one was allowed to prevent the usual delivery of underlying assets not always available or in trader’s portfolio holdings.

But since April 24th 2017 acquiring an equity options in a given trading day aiming to offset the corresponding position was no more allowed²³ (B3, 2015). For European options the impact was objectively none because there is only a specific time window on a trading day for the exercise²⁴, but it this was of essence for obvious reasons. There was no choice but to deliver the underlying asset.

To assess events where there may be relevant SSC a dataset relating American call options exercises, equity prices, options prices. Then after a first selection equity options referential premiums were compared with trading prices.

As mentioned, options with changes in underlying, equity ticker with more than one common share emission or more than one preferred share emission in 2017 are all filtered²⁵.

After applying all proper treatment there are exactly 5386 entries for American call options exercises. They were split in exercises before and after B3 (2015).

¹⁸All equity options switching the underlying asset were excluded as far as all equities with more than two preferred shares (because identification by “PN/ON” symbol is ambiguous).

¹⁹For example, “BBDCl15”, “BBDCK13”, “VALEJ6”, “VALEJ4” and “VALEI2” were examples of equity options series from 2017 with their types unmatched between COTAHIST and EORP data.

²⁰At least in B3 (2018) it was missing.

²¹Financial instruments file, a daily 150 MB “.xml” file

²²Curiously enough, “COTAHIST” files are still being update to current trading days and, not surprisingly, they still make references to BDI (namely, column BDI holds the same codes as the former market report). The new BVBG files, of course, have totally different rules.

²³The official communication date from July 2015, two years earlier.

²⁴Namely the expiration date from 10h00 to 13h00.

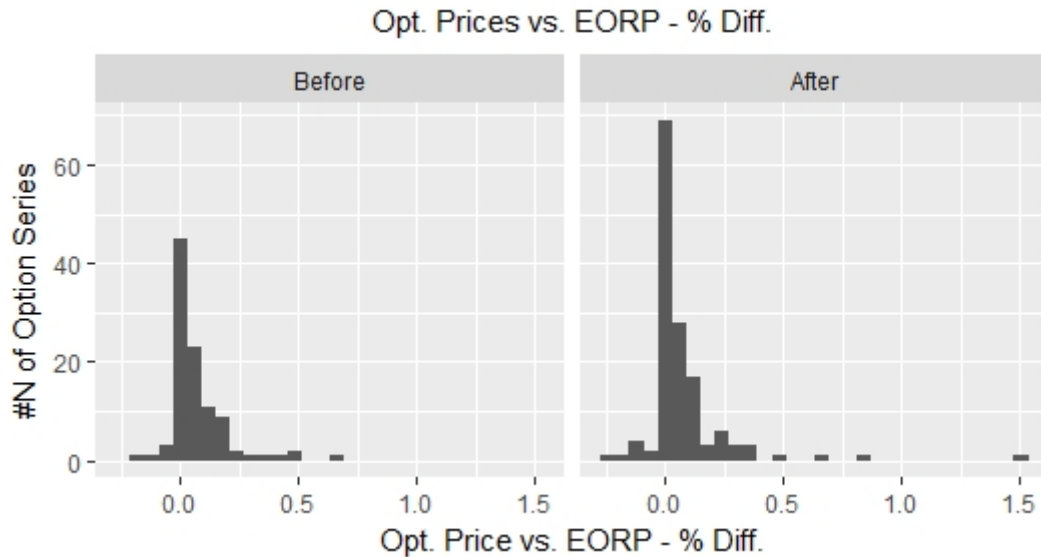
²⁵Lists with all those ticker are in the Annex.

The events before exercise rules would, in theory, present higher prices on call options since offsetting was possible at that time. Dates of exercises apart from the expiration were examined with an odd finding: from 1818 entries on exercises there are only 218 happening before the expiration.

Almost the same happened with exercise data after the changes: from 3568 exercises entries only 360 happened before the expiration date.

As it seems, exercises of American options in Brazil (at least for no known trivial reason at all) are more likely to be exercised at the expiration date, same as Europeans.

Figure 5.5: Best Bid Option Prices vs. EORP



Expressed in percentage quantities, the differences remain: traders are indeed willing to pay more for an option worthing more than its referential price in execution days, consistently more in execution days. Differences between trading prices on options and their referential premiums were expected to change due to changes in offsetting rules, but both appeared identical in regard to their distributions.

As a marking example with MULT3, same ticker as in Chapter 04, on an expiration day of January 2017, it was spotted the same bid-ask spread with market market volume on the extremes of the LOB on the middle of the day:

Figure 5.6: MULT3 - Expiration Date - 12h30-12h44

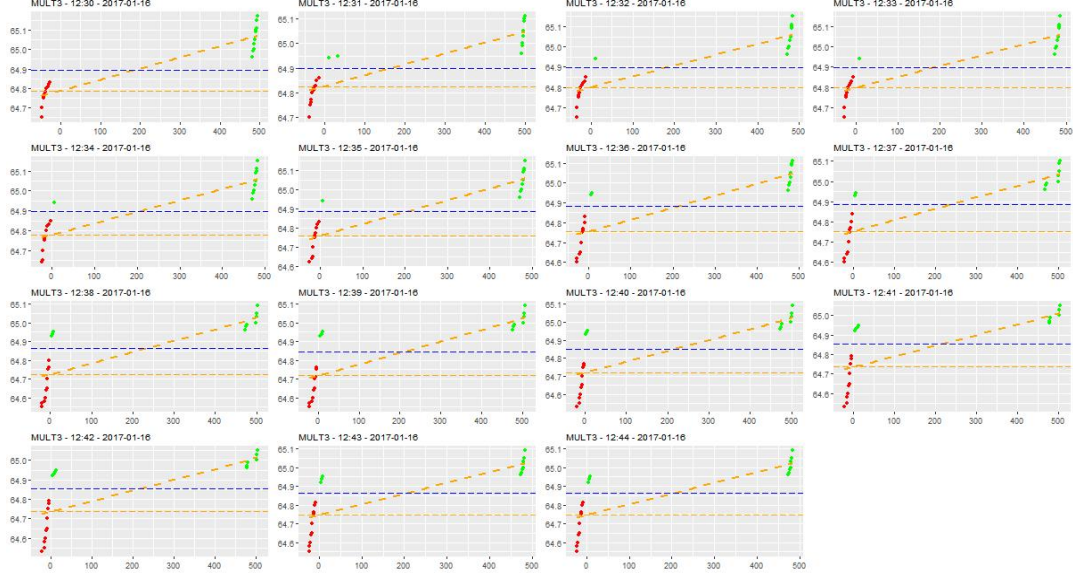
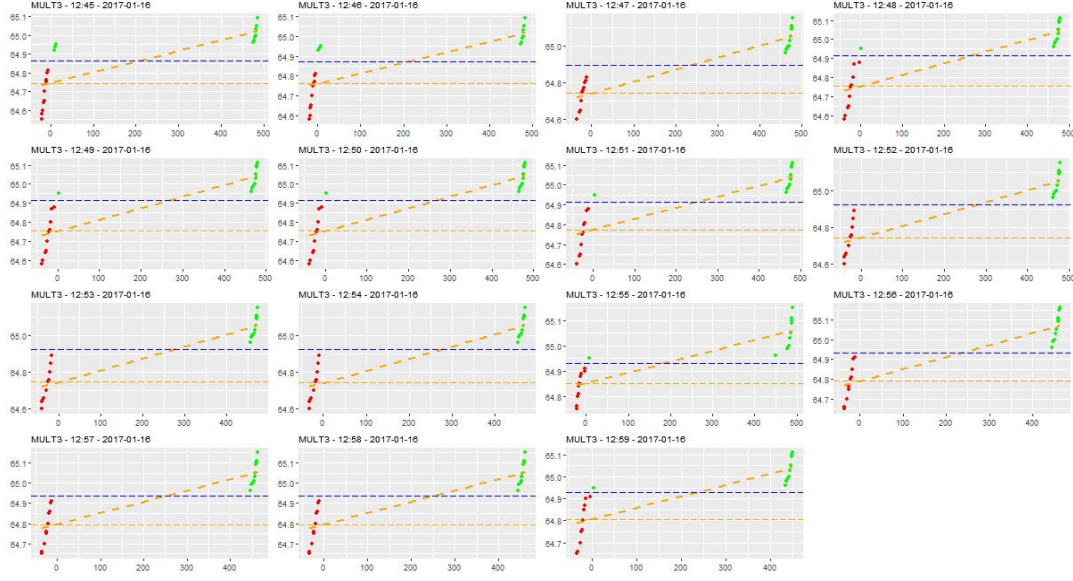


Figure 5.7: MULT3 - Expiration Date - 12h45-12h59



This was indeed no surprise since the time window for exercise of equity options is from 10h00 to 13h00.

Without the intention to be exhaustive, as building and running through the LOB time-to-time is physically and time constraining, here some hypothesis in the same terms as the “Saw-Tooth” were explored.

With the possibility of Liquidity Costs representations behaving in such a way, a discussion on how simplistic assumption on its nature could affect some remarkable models on portfolio process is taken in Chapter 06.

Chapter 6

On Execution Strategies

This final chapter aims to give context and explore possibilities in the context of Optimal Execution Strategies by applying some of the empirical findings on the previous chapters.

To do so, this study takes Almgren-Chriss's portfolio process as a working benchmark.

Although different in almost every aspect, CJP Model and Almgren-Chriss framework have Market Liquidity Risk as main subject. Also, the Liquidity Cost component, appearing as an “execution cost ” function is, of course, present as Almgren-Chriss Execution Strategy solely operates on market-orders.

At first sight, the execution cost is an unimportant component under stylized facts of non-linear cost impact and to be calibrated by power-rules that, for the sake of simplicity, assumes $\eta|\rho|^{1+\phi}$ convex form with respect to an averaging process of the trading rate on deterministic, continuous, positive and bounded process such as the market volume process $(V_T)_{t \in [0, T]}$ (Guéant, 2016, p.43). And this is even so when Almgren-Chriss framework is broadly known for the permanent market impact.

As misleading as deterministic Liquidity Costs, such assumptions are understatements as there is no Optimal Execution Strategy without a suitable execution cost function as it will be detailed on this last chapter.

6.1 The Optimal Execution Problem

Optimal Execution problems are generally given in a setup where a trader wants to unwind single-stock portfolio position. At time $t \in [0, T]$ the quantity of shares is denoted by q_t and it varies between times $t_1, t_2 \in [0, T]$ with $t_2 > t_1$ by

the following rules:

$$\begin{aligned} dq_t &= v_t dt \\ \int_{t_1}^{t_2} dq_t &= \int_{t_1}^{t_2} v_t dt \\ q_{t_2} - q_{t_1} &= \int_{t_1}^{t_2} v_t dt, \end{aligned} \tag{6.1}$$

recalling that $(v_t)_{t \in [0, T]}$, the instantaneous trading volume or trading velocity, is in the space of the progressively measurable control processes $\mathbb{H}^0$ and satisfies the unwinding constraint $\int_0^T v_t dt = -q_0$ with $\int_0^T |v_t| dt \in L^\infty(\Omega)$, the set of almost surely bounded random variables.

The set of admissible controls, $\mathcal{A}$, is denoted by

$$\mathcal{A} = \left\{ (v_t)_{t \in [0, T]} \in \mathbb{H}^0(\mathbb{R}, (\mathcal{F}_t)_t), \int_0^T v_t dt = -q_0, \int_0^T |v_t| dt \in L^\infty(\Omega) \right\}. \tag{6.2}$$

If one aims to totally unwind liquidate a long-position:

$$\begin{aligned} q_T - q_0 &= \int_0^T v_t dt \\ -q_0 &= \int_0^T v_t dt. \end{aligned} \tag{6.3}$$

The converse, acquiring $(q_T - q_0)$ stocks through time-window $[0, t]$, results in $q_T - q_0 = \int_0^T v_t dt$.

The cash-balance process, dX_t , of the portfolio is a result of successive trading given by $(v_t)_{t \in [0, T]}$ at actual trading prices that are given by $\tilde{S}^1$, which is a function of $S(t, 0)$, the “true” or “marginal” (Cetin et al., 2004) price ², the trading volume v_t and generic function $g(\cdot)$:

$$dX_t = -v_t \left[\tilde{S}(S(t, 0), v_t, g(\cdot)) \right] dt \tag{6.4}$$

at the end of the period $[0, T]$ the cash-balance process is:

¹Note that the subscript t is omitted: $\tilde{S}$ is to be considered as a mapping.

²As mentioned in Chapter 4, sometimes represented as the mid-price. Actually there is no need to be the mid-price as long as the choice is robust in regard to the extreme volatility of the bid-ask spread (Guéant, 2016, p.42).

$$\begin{aligned} \int_0^T dX_t &= - \int_0^T v_t \left[\tilde{S}(S(t, 0), v_t, g(\cdot)) \right] dt \\ X_T &= X_0 - \int_0^T v_t \left[\tilde{S}(S(t, 0), v_t, g(\cdot)) \right] dt. \end{aligned} \quad (6.5)$$

The success of a trading strategy is firstly based on its benchmark. This could be an utility function, VaR-based, the classical Mean-Variance or Mean Quadratic Criteria that could even be dynamically adaptive (Gatheral and Schied, 2013, Guéant, 2016, Lorenz and Almgren, 2011, Tse et al., 2013).

Those variations addresses the way a functional is obtained from (6.5), but not the central question: what is the nature of $\tilde{S}$?

$\tilde{S}$ what is consensual is that is definitely a function of $S(t, 0)$ and v_t , but how this relationship works to properly model liquidity costs is not: CJP Model bets on SSC Curve, as for Almgren-Chriss framework it is given a fairly simplistic $L(\cdot)$ because the focus is on kv_t component.

From the original Almgren-Chriss setup, dS_t followed (2.9) and liquidity costs were represented then by (2.11), an average form of the positive, deterministic, continuous, bounded process $(V_t)_{t \in [0, T]}$ with respect to $(v_t)_{t \in [0, T]}$. Controversial or not, each assumption have a practical sense with respect to mathematical properties that make possible or ease the task of solving the problem with tools of Calculus of Variations.

6.1.1 Almgren-Chriss Portfolio Process

This subsection details the exposition of Guéant (2016, p.39-64) focusing on the portfolio process of Almgren-Chriss' framework.

$$dS_t = \sigma dW_t + kv_t dt, \quad (6.6)$$

As for $g(\cdot)$ it is expressed as in (2.11), thus:

$$g\left(\frac{v_t}{V_t}\right) = \rho L(\rho) \quad (6.7)$$

The trader's cash account at the end of the execution process (i.e. $t = T$) is obtained by integrating the cash process (6.5):

$$\begin{aligned}
\int_0^T dX_T &= - \int_0^T v_t \left(S_t + g \left(\frac{v_t}{V_t} \right) \right) dt \\
X_T &= X_0 - \int_0^T v_t g \left(\frac{v_t}{V_t} \right) dt - \int_0^T v_t S_t dt \\
X_T &= X_0 - \int_0^T V_t L(\rho) dt - \int_0^T v_t S_t dt
\end{aligned} \tag{6.8}$$

Integrating $\int_0^T v_t S_t dt$ by parts and by using (6.1) and (6.6):

$$\begin{aligned}
\int_0^T v_t S_t dt &= (S_T q_T - S_0 q_0) - \int_0^T q_t dS_t \\
&= S_0 q_0 - \int_0^T q_t (k v_t dt + \sigma dW_t) \\
&= S_0 q_0 - k \int_0^T q_t v_t dt - \sigma \int_0^T q_t dW_t,
\end{aligned} \tag{6.9}$$

which gives:

$$X_T = X_0 + q_0 S_0 - k \int_0^T q_t v_t dt + \sigma \int_0^T q_t dW_t - \int_0^T V_t L(\rho) dt \tag{6.10}$$

By (6.1) and with another integration by parts on $\int_0^T q_t v_t dt$:

$$\begin{aligned}
\int_0^T q_t v_t dt &= (q_T^2 - q_0^2) - \int_0^T q_t v_t dt \\
\int_0^T q_t v_t dt &= -\frac{1}{2} q_0^2
\end{aligned} \tag{6.11}$$

Then, by using (6.11) on (6.10):

$$X_T = X_0 + q_0 S_0 - \frac{k}{2} q_0^2 + \sigma \int_0^T q_t dW_t - \int_0^T V_t L(\rho) dt \tag{6.12}$$

Since $\mathbb{E} \left[\sigma \int_0^T q_t dW_t \right]$ is the expectation of an Ito's Integral that goes to zero it has mean

$$\begin{aligned}
\mathbb{E}[X_T] &= X_0 + q_0 S_0 - \frac{k}{2} q_0^2 + \cancel{\sigma \int_0^T q_t dW_t} - \int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt \\
\mathbb{E}[X_T] &= X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt,
\end{aligned} \tag{6.13}$$

and variance

$$\begin{aligned}
\mathbb{V}[X_T] &= \mathbb{E}[X_T^2] - \mathbb{E}[X_T]^2 \\
\mathbb{V}[X_T] &= \mathbb{E} \left[\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 + \sigma \int_0^T q_t dW_t - \int_0^T V_t L(\rho) dt \right)^2 \right] \\
&\quad - \mathbb{E} \left[\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt \right)^2 \right]
\end{aligned}$$

The second term of the right-hand side, $\mathbb{E}[X_T]^2$, is whole deterministic while the first term, $\mathbb{E}[X_T^2]$, will present identical deterministic parts with additional terms being products between deterministic parts and the Ito's Integral $\sigma \int_0^T q_t dW_t$ (denoted as $\zeta(dW_t)$) and the squared Ito's Integral itself.

$$\begin{aligned}
\mathbb{V}[X_T] &= \mathbb{E} \left[\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 + \sigma \int_0^T q_t dW_t - \int_0^T V_t L(\rho) dt \right)^2 \right] \\
&\quad - \left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt \right)^2 \\
\mathbb{V}[X_T] &= \mathbb{E} \left[\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \cancel{\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt} \right)^2 \right] + \mathbb{E}[\zeta(dW_t)] + \\
&\quad \mathbb{E} \left[\left(\sigma \int_0^T q_t dW_t \right)^2 \right] \\
&\quad - \left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \cancel{\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt} \right)^2
\end{aligned}$$

The independence of dW_t over time and by the expectation of the Ito's Integral $\mathbb{E}[\zeta(dW_t)] = 0$ as the remaining term is subjected to Ito's Isometry.

$$\begin{aligned}
\mathbb{V}[X_T] &= \mathbb{E}[\zeta(dW_t)] + \mathbb{E}\left[\left(\sigma \int_0^T q_t dW_t\right)^2\right] \\
\mathbb{V}[X_T] &= \sigma^2 \int_0^T q_t^2 dt.
\end{aligned} \tag{6.14}$$

Which finally gives

$$X_T \sim \mathcal{N}\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - \int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt, \sigma^2 \int_0^T q_t^2 dt\right). \tag{6.15}$$

The variance of the terminal cash process is an increasing function of $\sigma > 0$, the stock volatility, and it depends on the strategy through the term $\int_0^T q_t^2 dt$. Formally, by taking the derivative with respect to time:

$$\frac{d}{dt} \mathbb{V}[X_T] = \frac{d}{dt} \left(\sigma^2 \int_0^T q_t^2 dt \right),$$

and from (6.2), (6.1) and the integral on the compact $[0, T]$ it is possible to reach:

$$\begin{aligned}
\frac{d}{dt} \mathbb{V}[X_T] &= \sigma^2 \int_0^T \frac{d}{dt} q_t^2 dt \\
&= 2\sigma^2 \int_0^T q_t v_t dt.
\end{aligned} \tag{6.16}$$

The integral on the LHS was already solved in (6.11), which makes

$$\begin{aligned}
\frac{d}{dt} \mathbb{V}[X_T] &= 2\sigma^2 \int_0^T q_t v_t dt \\
&= 2\sigma^2 \frac{-1}{2} q_0^2 \\
&= -\sigma^2 q_0^2
\end{aligned} \tag{6.17}$$

This result seems counter-intuitive, but it has the following meaning: this present example models a portfolio that is unwinding a long-position in $q_0 > 0$ shares aiming for $q_T = -q_0$ and, as far as Almgren-Chriss's Execution Strategy is schedules a trading curve such that $t \mapsto q$ is non-increasing, it is clear that the faster the execution, smaller the cash process' variance.

The expectation of the cash balance at the end of the execution schedule,

$\mathbb{E}[X_T]$, incorporates both permanent market impact and liquidity costs:

$$\mathbb{E}[X_T] = \overbrace{X_0 + q_0 S_0}^{MtMvalue} - \overbrace{\frac{k}{2} q_0^2}^{perm.m.i.} - \overbrace{\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt}^{LiquidityCosts}. \quad (6.18)$$

The first term is the Mark-to-Market value of the portfolio at time $t = 0$. The second term corresponds to costs coming from the permanent market impact and, surprisingly, is not dependent on v_t , thus, being unavoidable.

The third and final term, the Liquidity Cost, is the only one varying with the control $(v_t)_{t \in [0, T]}$. The function $L\left(\frac{v_t}{V_t}\right)$ is assumed to be convex and it is minimal when is proportional do V_t .

Without such assumption on L the whole mathematical foundations would not fit into the well-known class of the so-called “Bolza Problems” (Pennanen and Perkkio, 2014, Rockafellar, 1970, 1971), in other hand without the execution cost function L there would be nothing to vary in terms of the admissible control³ because other terms as the linear market impact are also constant with respect to $(v_t)_{t \in [0, T]}$ and does not change the portfolio performance with respect to the expectation $\mathbb{E}[X_T]$. The variance $\mathbb{V}[X_T]$ is actually the only element, remaining L under suspension, that is in terms of $q_t = v_t dt$, but since it is a penalizing component⁴ and its sole presence would make the problem of no trade-off at all.

Almgren-Chriss’ framework is well know for the permanent market impact, but for the optimization problem, the execution cost, L , is a main defining element of the cash process. The empirical findings developed in this work for the execution cost, here called Stochastic Liquidity Cost, are that it is not deterministic, its more likely that its functional representation may change through time and, finally, its liquidity parameters can be though as a stochastic process, a mean-reverting stochastic process in its on right.

³Which would certainly be a worrying situation in a Calculus of Variations problem.

⁴Remember the classical $\mathbb{E}[X_T] - \frac{\gamma}{2} \mathbb{V}[X_T]$.

Chapter 7

Conclusions

This work studied Market Liquidity Risk with a motivation to doing so on microstructural level by estimating the Stochastic Supply Curve from Çetin-Jarrow-Protter Model (Cetin et al., 2004, 2006, Çetin et al., 2010).

For this purpose, LOBs of Equity Brazilian markets were built with publicly available HFD from B3 (2017b) with help of GetHFDData from Perlin and Ramos (2016).

This was done by combining both Orders and Trades HFD , thus adding consistency and quality to the results and avoiding time-consuming and non-vectorized procedures by matching schemes/engines.

The LOBs were organized in four layers, orders, trades, events and depth, on an event-based approach. This means that each execution type and order status was properly categorized and treated according to its actual impact on the LOB and respective entries, frequently more than one for each Orders HFD entry, were generated. Fixing the relevant time horizon as the current trading day, the “Events” layer was built in such a way that it would generate a relevant depth grouped by event time, trading side, volume and bid/ask price and prone to deliver a LOB at each fixed time.

Naming those LOBs as “functional” means that, though acceptable at some degree, they are not free of some assumptions and “ad-hoc” treatment, being the case of “zero-order prices” and “ghost entries” the two most relevant and need additional effort to design a more thoughtful methodology to improve the building procedure.

In Chapter 04 the Stochastic Supply Curve literature and estimation procedures were exposed as how the previous empirical findings are interpreted and taken as ruling assumption for further modeling purposes.

The LOB approach of Blais and Protter (2010) was the working option as it was a more natural way to assess Liquidity Costs in a microstructural level. The SSC were estimated for MULT3 in March 20th for 1 minute time-interval and for

BBDC3 for 15 minute -time-interval¹ for detailed evaluation of dynamic features and time-varying nature of its functional form.

It was found that the SSC through a trading time is non-trivial and its nature is time-varying with respect to its functional time. The liquidity parameter, α_t was also found to be of dynamic nature and, in the same line as [Blais and Protter \(2010\)](#), it was found that it could be though as a mean-reverting stochastic process at its own right.

Difficulties regarding a more generic functional form or estimation procedure that would address the shape-shifting nature of the adjustment of the SSC, specially under CJP Model assumptions, is definitely a point to be improved and a theme for further research.

What motivates or at least amplifies in some given contexts a pressure in trading and therefore exposition to Market Liquidity Risk is the Funding Liquidity Risk, specially a imminent settlement of a liability, that is a funding bind.

In this line and as a suggestion to more generic contexts where Stochastic Liquidity Costs Liquidity would emerge as a liquidity commonality between equities, Chapter 5 recalled the duality between Market Liquidity and Funding Liquidity Risk arguing that “physical delivery” due to exercises on call options would serve as an analogy for funding bind.

This arguing was inspired in the “Saw-Tooth” pattern, well studied by [Lehalle et al. \(2012\)](#) and described by [Guéant \(2016\)](#). That specific trading day was the last one before expiration day for options in USA markets, therefore a critical days.

To evaluate illiquidity related to option hedging, or the anticipation of it, information on features of equity options were of need. Unfortunately, this was found to be challenging due to idiosyncratic standards of Brazilian markets.

An “indirect” way to keep track of strikes, referential prizes, tickers and options’ underlying assets was designed. Then, comparisons between Equity Options Referential Prizes ([B3, 2017c](#)) and market-to-market prices reported on exercises on American Options.

Persistent overpaying for options in exercise dates were found and, before changes of rules on offsetting options, this would sign a unwilling from traders to acquire underlying assets on the market due to illiquidity.

Without the intention of concluding and exhaustive on this suggestion on liquidity commonalities, a time-window close to the end of expiration hours was presented to point the suggestion at least worth mentioning with Chapter 05.

As Stochastic Liquidity Costs are shown to be a feasible representation of execution cost, in contrast with assumptions that would relegate this component as

¹Figure [A.29](#) also in Annex.

static and deterministic, a evaluation on relaxation of this hypothesis is exposed on a remarkable reference on Optimal Execution Costs, the Almgren-Chriss' framework (Almgren and Chriss, 1999, 2001) based on (Guéant, 2016) exposition.

Almgren-Chriss' framework main concern is in how a large trader can unwind/acquire a long/short position in stocks. The problem, though, is that the position is large enough to even affect market prices, therefore with discouraged to trade too fast. But at a slow pace, there is also the risk of exposition to price movements. The opposition between this two factors makes a trade-off that has to be solved by means of non-trivial strategies.

Detailing the main features and reasoning about Almgren-Chriss' process portfolio it was found that although the remarking feature of the model is the linear permanent market impact it is not decisive on the Optimal Execution Strategy. This is even more surprising when the execution costs, the component comparable to Liquidity Costs in CJP Model, once saw as of simplistic treatment is in fact the one variable that is varying with the control v_t in the expectation of the cash process.

The emphasis on Liquidity Costs, now emphasized here as Stochastic Liquidity Costs, was motivated by its central role as a threshold between ledger and accruals adjustments and cash generation, the very transition from mark-to-market and realized and delivered asset performance.

Although ubiquitous and intuitive in theoretical and practical contexts, Liquidity Costs are of no trivial behavior. Therefore, the discussions, developments on HFD and market data, empirical findings are in hope to motivate an equally broad consensus on the worthiness of further improvements and developments.

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Appendix A

Annex

A.1 Building the LOB

A.1.1 Trades HFD

Trades HFD is obtained by modifying a subroutine from “GetHFDData” package (Perlin and Ramos, 2016) to deal with the raw files from B3 (2017b). Original implementation of this functionality can be found at Perlin and Ramos (2017).

```
load_trades_raw <- function(fname){

col.names <- c('SessionDate', 'InstrumentSymbol', 'TradeNumber',
               'TradePrice', 'TradedQuantity',
               'Tradetime',
               'TradeIndicator',
               'BuyOrderDate',
               'SequentialBuyOrderNumber', 'SecondaryOrderID',
               'AggressorBuyOrderIndicator',
               'SellOrderDate',
               'SequentialSellOrderNumber', 'SecondaryOrderID2',
               'AggressorSellOrderIndicator',
               'CrossTradeIndicator',
               'BuyMember', 'SellMember')

my.col.types <- readr::cols(
  SessionDate = readr::col_date(format = ""),
  InstrumentSymbol = readr::col_character(),
  TradeNumber = readr::col_character(),
```



```

TradePrice = readr::col_character(),
TradedQuantity = readr::col_character(),
Tradetime = readr::col_character(),
TradeIndicator = readr::col_integer(),
BuyOrderDate = readr::col_date(format = ""),
SequentialBuyOrderNumber = readr::col_character(),
SecondaryOrderID = readr::col_character(),
AggressorBuyOrderIndicator = readr::col_integer(),
SellOrderDate = readr::col_date(format = ""),
SequentialSellOrderNumber = readr::col_character(),
SecondaryOrderID2 = readr::col_character(),
AggressorSellOrderIndicator = readr::col_integer(),
CrossTradeIndicator = readr::col_integer(),
BuyMember = readr::col_character(),
SellMember = readr::col_character()
)
# fix for "no visible binding" NOTE from CHECK
#utils::suppressForeignCheck(col.names) # DOESNT WORK
# Unfortunately this seems to be the easiest way to keep
#CHECK quiet about the variables

InstrumentSymbol <- TradeIndicator <- SessionDate <-
  TradePrice <- NULL
TradedQuantity <- Tradetime <- AggressorBuyOrderIndicator <-
  TradeDateTime <- NULL
last.price <- TradeSign <- BuyMember <- CrossTradeIndicator <-
  SellMember <- NULL

# read data (warning are from last line)
suppressWarnings(suppressMessages(
  my.df <- readr::read_csv2(file = fname,
                             skip = 1,
                             col_names = col.names,
                             progress = F,
                             col_types = my.col.types)
))

# remove cancelled trades

```

```
##### ADDS T AND T_CHAR COLUMNS.
##### THERE IS ALSO A MODIFICATION ON TRADESIGN COLUMN
##### TO ADDRESS THE FACT THAT ON
##### OPENING AND CLOSING AUCTIONS
##### THE TRADE INITIATING SIDE
##### DEFINITION IS NOT APPLICABLE.
```

```
my.df <- dplyr::filter(my.df, TradeIndicator==1) %>%
  mutate(TradeSign = if_else(AggressorBuyOrderIndicator==2 &
    AggressorSellOrderIndicator==2, 2,
    (AggressorBuyOrderIndicator==1)*1-
    (AggressorBuyOrderIndicator==2)*1)
    ) %>%
  mutate(TradeDateTime =
    lubridate::ymd_hms(paste(SessionDate, Tradetime),
    tz = "America/Sao_Paulo")) %>%
  mutate(t = as.numeric(TradeDateTime)) %>%
  mutate(t_char = as.character(t))

return(my.df)
}
```

A.1.2 Orders HFD

Orders HFD is obtained by modifying a subroutine from “GetHFData” package (Perlin and Ramos, 2016) to deal with the raw files from B3 (2017b). Original implementation of this functionality can be found at Perlin and Ramos (2017).

```
#### This functions returns a data.frame with a given LOB Side.
#### One must run it two times:
#### on OFER_VDA files and on OFER_CPA files.
#### Finally, both must be merged by an "UNION ALL" operation.

load_orders_raw <- function(fname) {
  type.matching <- 'partial'; my.assets <- ''
  # set column meanings
  #(ftp://ftp.bmf.com.br/MarketData/OFER_CPA_LAYOUT_english.txt)
  tab.order.side <- data.frame(x.in = (1:2),
                               x.out = c('Buy', 'Sell'))

  tab.execution.type <- data.frame(x.in = c(1:9, 11,12),
                                    x.out =
c('New', 'Update', 'Cancel', 'Trade', 'Reentry',
'New_Stop_Price', 'Rejected', 'Removed',
'Stop_Price_Triggered', 'Expired', 'Eliminated')
)

  tab.order.status <-
    data.frame(x.in = c('0', '1', '2', '4', '5', '8', 'C'),
              x.out =
c('New', 'Partially_Filled', 'Filled', 'Canceled', 3
'Replaced', 'Rejected', 'Expired')
)

  tab.aggressor.indicator <- data.frame(x.in = 0:2,
                                         x.out = c('Neutral', 'Aggressor', 'Passive'))

  # import data
  col.names <- c('SessionDate', 'InstrumentSymbol', 'OrderSide',
'SequentialOrderNumber', 'SecondaryOrderID', 'ExecutionType',
'PriorityTime', 'PriorityIndicator', 'OrderPrice',
```

```
'TotalQuantity', 'TradedQuantity', 'OrderDate',
'OrderDatetime', 'OrderStatus', 'AgressorIndicator', 'Member'
)
```

```
my.col.types <- readr::cols(
  SessionDate = readr::col_date(format = ""),
  InstrumentSymbol = readr::col_character(),
  OrderSide = readr::col_integer(),
  SequentialOrderNumber = readr::col_double(),
  SecondaryOrderID = readr::col_integer(),
  ExecutionType = readr::col_integer(),
  PriorityTime = readr::col_guess(),
  PriorityIndicator = readr::col_character(),
  OrderPrice = readr::col_character(),
  TotalQuantity = readr::col_double(),
  TradedQuantity = readr::col_double(),
  OrderDate = readr::col_skip(),
  OrderDatetime = readr::col_datetime(format = ""),
  OrderStatus = readr::col_character(),
  AgressorIndicator = readr::col_integer(),
  Member = readr::col_integer()
)
```

```
# fix for "no visible binding" NOTE from CHECK
#utils::suppressForeignCheck(col.names) # DOESNT WORK
# Unfortunately this seems to be the easiest
# way to keep CHECK quiet about the variables
```

```
SessionDate <- InstrumentSymbol <- OrderSide <- NULL
SequentialOrderNumber <- SecondaryOrderID <-
ExecutionType <- NULL
PriorityTime <- PriorityIndicator <- OrderPrice <-
TotalQuantity <- NULL
TradedQuantity <- OrderDate <- OrderDatetime <-
OrderStatus <- AgressorIndicator <- Member <- NULL
```

```
# read data (warning are from last line)
suppressWarnings(suppressMessages(
  my.df <- readr::read_csv2(file = fname,
```

```

    skip = 1,
    col_names = col.names,
    progress = F,
    col_types = my.col.types,
    locale =
    readr::locale(tz = 'America/Sao_Paulo',
    decimal_mark = '.'))
  ))

# replace col values for intuitive factors
idx <- match(my.df$OrderSide, tab.order.side$x.in)
my.df$OrderSide <- as.factor(tab.order.side$x.out[idx])

idx <- match(my.df$ExecutionType, tab.execution.type$x.in)
my.df$ExecutionType <- as.factor(tab.execution.type$x.out[idx])

idx <- match(my.df$OrderStatus, tab.order.status$x.in)
my.df$OrderStatus <- as.factor(tab.order.status$x.out[idx])

if(nrow(my.df)<10){
  return(data.frame())
}

cat(paste('_', Imported_', nrow(my.df), 'lines', ',
length(unique(my.df$InstrumentSymbol)),
'unique_tickers'))
cat(paste('\n----> Processing file '))

# filter for NA data
my.df <- stats::na.omit(my.df)

# if no my.assets are supplied, import everything

if (is.null(my.assets)){
  my.assets <- unique(my.df$InstrumentSymbol)
}

if (type.matching == 'exact') {
  # exact matching

```

```

idx <- my.df$InstrumentSymbol %in% my.assets

} else if (type.matching == 'partial'){
  # partial match my.assets
  my.pattern <- paste0('(',paste0(my.assets ,
    collapse = '|'),')')
  idx <- grepl(x = my.df$InstrumentSymbol, pattern = my.pattern)
}

my.df <- my.df[idx, ]

# if no data is returned, output empty dataframe
if (nrow(my.df)==0){
  cat(' _ _Found _no _data _for _my._assets._')
  return(data.frame())
}

# Set objet classes correct
options(digits.secs=3) # get correct milseconds
my.df$OrderPrice <- as.numeric(my.df$OrderPrice)
my.df$PriorityDateTime <- lubridate::ymd_hms(
  paste(my.df$SessionDate,my.df$PriorityTime),
  tz = "America/Sao_Paulo")

my.df <- my.df %>%
  mutate(t = as.numeric(PriorityDateTime)) %>%
  mutate(t_char = as.character(t))

cat(paste0(' _ _Found _', nrow(my.df), '_ _lines _for _',
  length(unique(my.df$InstrumentSymbol)),
  '_ _selected _tickers _'))
return(my.df)

# remove day times outside period from first.time and last.time

#unique.dates <- unique(my.df$SessionDate)

#first.time <-
# as.POSIXct(paste0(unique.dates[1], ' ', first.time, ' BRT'))

```

```

#last.time <-
#  as.POSIXct(paste0(unique.dates[1], ' ', last.time, ' BRT'))

#idx <- ( my.df$OrderDatetime >= first.time)&
( my.df$OrderDatetime <= last.time)

#my.df <- my.df[idx, ]
}

```

A.1.3 LOB Code

The code for the LOB comes from this dissertation and its permissive free software license is under MIT License. They can be found at [Hossaka \(2018\)](#).

```
rm(list=ls());gc()

library('GetHFDData')
library('obAnalytics')
library('tictoc')
library('readr')
library('lubridate')
library('stringr')
library('ggplot2')
library('dplyr')
library("openxlsx")
options(digits.secs = 3)
max.cols <- Sys.getenv("COLUMNS")
options(width=if(max.cols != "") max.cols else 80,
        scipen=999, digits.secs=3)

tic('Building LOB')
vticker <- c('MULT3')
vYYYYMMDD <- c('20170320')

tic("Loading Trades HFD")
if(vticker == vec.vticker[1]){ZZZ_df.all_trades <-
  readRDS(### Path for Trades HFD in rds format ###)

  df.CPA <-
    readRDS(
      ### Path for Orders HFD on the
      ### bid side in rds format###)
  df.VDA <-
    readRDS(
      ### Path for Trades HFD on the
      ### ask side in rds format ###)

  ZZZ_df.all_orders <-
```



```

df.CPA %>% bind_rows(df.VDA);
try(rm(df.CPA)); try(rm(df.VDA))}

ZZZ_df.orders <- ZZZ_df.all_orders %>%
  filter(InstrumentSymbol == vticker) %>%
  mutate(t = as.numeric(PriorityDateTime)) %>%
  mutate(t_char = as.character(t))

ZZZ_df.trades <-
  ZZZ_df.all_trades %>% filter(InstrumentSymbol == vticker)

df.t <- ZZZ_df.trades %>%
  mutate(TradeNumber =
    as.numeric(TradeNumber),
    ID_Buy =
    as.numeric(SequentialBuyOrderNumber),
    ID2_Buy =
    as.numeric(SecondaryOrderID),
    aggind_buy =
    as.numeric(AggressorBuyOrderIndicator),
    ID_Sell =
    as.numeric(SequentialSellOrderNumber),
    ID2_Sell =
    as.numeric(SecondaryOrderID2),
    aggind_sell =
    as.numeric(AggressorSellOrderIndicator),
    TradePrice =
    as.numeric(TradePrice),
    TradedQuantity =
    as.numeric(TradedQuantity),
    BuyMember =
    as.numeric(BuyMember),
    SellMember =
    as.numeric(SellMember)) %>%
  dplyr::select(1:3, 23:25, 26:28, 21:22, 20, 6, 17:19, 4:5) %>%
  mutate(Delta = TradedQuantity)
toc()

df.o <- ZZZ_df.orders %>%

```

```

    arrange(t, SequentialOrderNumber, SecondaryOrderID)

vlot <- min(df.o$TotalQuantity[df.o$TotalQuantity >
                                0],
           df.t$TradedQuantity[df.t$TradedQuantity > 0])

df.o <- df.o %>% filter(ExecutionType != 'Reentry') %>%
  arrange(PriorityDateTime,
           SequentialOrderNumber, SecondaryOrderID) %>%
  mutate(Adj = 0, Effect = 0, Level = 0,
         Side = "", EventID = 1:n()) %>%
  as.data.frame()

vmintime <- min(df.o$PriorityDateTime)

df.ghost <- df.o %>% group_by(SequentialOrderNumber) %>%
  filter(min_rank(EventID)==1) %>%
  filter(ExecutionType != 'New') %>% mutate(Adj = 1)

df.o$Adj[df.ghost$EventID] <- 1

df.ghost <- df.ghost %>% select(-EventID) %>% as.data.frame()

###Filtering criterion is just to get an empty dataframe.
df.set_ghost_trades_bid <-
  df.set_ghost_trades_ask <-
  df.set_ghost_others <- df.ghost %>%
  filter(ExecutionType == 'New') %>%
  as.data.frame()

#####New Orders - Bid Side.
df.ghost_trad_bid <- df.ghost %>%
  filter(ExecutionType == 'Trade') %>%
  inner_join(df.t, by = c("SequentialOrderNumber" = "ID_Buy",
                        "SecondaryOrderID" = "ID2_Buy"))

for(i in 1:nrow(df.ghost_trad_bid)){
  if(nrow(df.ghost_trad_bid)==0){break()}
  df.set_ghost_trades_bid[i,1] <- lubridate::ymd(vYYYYMMDD)

```

```

df.set_ghost_trades_bid[i,2:3] <- df.ghost_trad_bid[i,2:3]
df.set_ghost_trades_bid[i,4] <- df.ghost_trad_bid[i,4]
df.set_ghost_trades_bid[i,5] <- 0000001
df.set_ghost_trades_bid[i,6] <- 'New'
df.set_ghost_trades_bid[i,7] <-
  strftime(vmintime,format = "%H:%M:%OS")
df.set_ghost_trades_bid[i,8:9] <- df.ghost_trad_bid[i,8:9]

df.set_ghost_trades_bid[i,10] <- df.ghost_trad_bid[i,10]
df.set_ghost_trades_bid[i,11] <-
  df.ghost_trad_bid[i,11]-df.ghost_trad_bid[i,38]

df.set_ghost_trades_bid[i,12] <- vmintime
df.set_ghost_trades_bid[i,13] <- 'New'
df.set_ghost_trades_bid[i,14] <- 0
df.set_ghost_trades_bid[i,15] <- df.ghost_trad_bid[i,15]
df.set_ghost_trades_bid[i,16] <- vmintime
df.set_ghost_trades_bid[i,17] <- as.numeric(vmintime)
df.set_ghost_trades_bid[i,18] <-
  as.character(as.numeric(vmintime))
df.set_ghost_trades_bid[i,19] <- 2
####Effect
df.set_ghost_trades_bid[i,20] <-
  df.set_ghost_trades_bid[i,10] -
  df.set_ghost_trades_bid[i,11]
####Level
df.set_ghost_trades_bid[i,21] <- df.ghost_trad_bid[i,9]
####Side
df.set_ghost_trades_bid[i,22] <- df.ghost_trad_bid[i,3]
}

#####New Orders - Ask Side.
df.ghost_trad_ask <- df.ghost %>%
  filter(ExecutionType == 'Trade') %>%
  inner_join(df.t, by =
    c("SequentialOrderNumber" = "ID_Sell",
      "SecondaryOrderID" = "ID2_Sell"))

for(i in 1:nrow(df.ghost_trad_ask)){

```

```

  if(nrow(df.ghost_trad_ask)==0){break()}
  df.set_ghost_trades_ask[i,1] <- lubridate::ymd(vYYYYMMDD)
  df.set_ghost_trades_ask[i,2:3] <- df.ghost_trad_ask[i,2:3]
  df.set_ghost_trades_ask[i,4] <- df.ghost_trad_ask[i,4]
  df.set_ghost_trades_ask[i,5] <- 0000001
  df.set_ghost_trades_ask[i,6] <- 'New'
  df.set_ghost_trades_ask[i,7] <-
    strftime(vmintime,format = "%H:%M:%OS")
  df.set_ghost_trades_ask[i,8:9] <- df.ghost_trad_ask[i,8:9]

  df.set_ghost_trades_ask[i,10] <- df.ghost_trad_ask[i,10]
  df.set_ghost_trades_ask[i,11] <-
    df.ghost_trad_ask[i,11]-df.ghost_trad_ask[i,38]

  df.set_ghost_trades_ask[i,12] <- vmintime
  df.set_ghost_trades_ask[i,13] <- 'New'
  df.set_ghost_trades_ask[i,14] <- 0
  df.set_ghost_trades_ask[i,15] <- df.ghost_trad_ask[i,15]
  df.set_ghost_trades_ask[i,16] <- vmintime
  df.set_ghost_trades_ask[i,17] <- as.numeric(vmintime)
  df.set_ghost_trades_ask[i,18] <-
    as.character(as.numeric(vmintime))
  df.set_ghost_trades_ask[i,19] <- 2
  ####Effect
  df.set_ghost_trades_ask[i,20] <-
    df.set_ghost_trades_ask[i,10] -
    df.set_ghost_trades_ask[i,11]
  ####Level
  df.set_ghost_trades_ask[i,21] <- df.ghost_trad_ask[i,9]
  ####Side
  df.set_ghost_trades_ask[i,22] <- df.ghost_trad_ask[i,3]
}

#####Artificial "New Orders" – non-Trade entries.

df.ghost_others <- df.ghost %>% filter(ExecutionType != 'Trade')

for(i in 1:nrow(df.ghost_others)){
  if(nrow(df.ghost_others)==0){break()}

```

```

df.set_ghost_others[i,1] <- lubridate::ymd(vYYYYMMDD)
df.set_ghost_others[i,2:3] <- df.ghost_others[i,2:3]
df.set_ghost_others[i,4] <- df.ghost_others[i,4]
df.set_ghost_others[i,5] <- 0000001
df.set_ghost_others[i,6] <- 'New'
df.set_ghost_others[i,7] <-
  strftime(vmintime, format = "%H:%M:%OS")
df.set_ghost_others[i,8:9] <- df.ghost_others[i,8:9]

df.set_ghost_others[i,10:11] <- df.ghost_others[i,10:11]

df.set_ghost_others[i,12] <- vmintime
df.set_ghost_others[i,13] <- 'New'
df.set_ghost_others[i,14] <- 0
df.set_ghost_others[i,15] <- df.ghost_others[i,15]
df.set_ghost_others[i,16] <- vmintime
df.set_ghost_others[i,17] <- as.numeric(vmintime)
df.set_ghost_others[i,18] <-
  as.character(as.numeric(vmintime))
df.set_ghost_others[i,19] <- 2
###Effect
df.set_ghost_others[i,20] <- df.set_ghost_others[i,10]
if(df.ghost_others[i,6]== 'Update')
{df.set_ghost_others[i,20] <- df.set_ghost_others[i,10] <- 0}
###Level
df.set_ghost_others[i,21] <- df.set_ghost_others[i,9]
###Side
df.set_ghost_others[i,22] <- df.set_ghost_others[i,3]
}

df.set_ghost <- df.set_ghost_trades_bid %>%
  bind_rows(df.set_ghost_trades_ask) %>%
  bind_rows(df.set_ghost_others)

df.o <- df.set_ghost %>% bind_rows(df.o) %>%
  arrange(PriorityDateTime, SequentialOrderNumber,
    SecondaryOrderID) %>%
  mutate(EventID = 1:n()) %>%
  group_by(SequentialOrderNumber) %>%

```

```

mutate (OrderEvent = 1:n())

length(unique(df.o$SequentialOrderNumber))
length(unique(ZZZ_df.orders$SequentialOrderNumber))

# rm(list =
# c("df.ghost","df.ghost_others"
# ,"df.ghost_trad_ask","df.ghost_trad_bid"
# ,"df.set_ghost","df.set_ghost_others"
# ,"df.set_ghost_trades_ask","df.set_ghost_trades_bid","i")
# )

ZZZ_df.LOB_Events <- data.frame(
  A_Date = as.Date(character()), #Session date.
  A_Ticker = character(), #Ticker code.
  A_Side = character(), #Side of the order.
  A_Time = character(),

  #Time: H%M:OS% - PriorityTime in
  #Orders HFD and Tradetime in Trades HFD
  A_Type = factor(), #From Orders HFD
  A_Status = factor(), #From Orders HFD
  A_Order_ID = numeric(),
  #Sequential/ID Order from Orders HFD
  A_Order_ID2 = numeric(),
  #Secondary ID Order from Orders HFD
  A_TotalQtd = numeric(), #Total quantity
  #from Orders HFD... it may change with updates
  A_TrdQtd = numeric(),
  #Cummulative quantity from Orders... hopefully monotone
  A_Price = numeric(),
  #Order Price from Orders HFD (sometimes it is zero)
  A_LOB_Side = character(),
  #Ask/Bid or Sell/Buy
  A_LOB_Qtd = numeric(),
  #The actual quantity that the LOB is affected by an event
  A_LOB_Level = numeric(),
  #Price at which the quantity if effect is accounted

```

```

A_Entry = numeric() ,
#Some orders and definitively trades generate
# an additional registry as counterpart
A_Adj = numeric() ,
#Flag for artificial orders to correct Ghost Entries
A_t = numeric() ,
#PriorityDateTime in Orders HFD or TradeDateTime
#in Trades HFD converted in numeric format
A_t_char = character() , #t converted in character
stringsAsFactors=FALSE
)

##### BUILDING LOB #####
tic( 'Building_LOB_Parts' )

Dealing_New <- df.o %>% filter(ExecutionType == 'New'
                               & OrderEvent == 1)
df.LOB_Orders <- df.o %>% filter(ExecutionType != 'New') %>%
  bind_rows(Dealing_New)

tic();df.LOB_Orders <- df.LOB_Orders %>%
  arrange(SequentialOrderNumber ,SecondaryOrderID) %>%
  group_by(SequentialOrderNumber) %>%
  mutate( Effect = -(lag( TotalQuantity)-lag( TradedQuantity)) ,
          Level = lag( OrderPrice) , Side = lag( OrderSide));toc()

###What is being added by new orders, including artificial
###ones for dealing with "Ghost Entries".

tic( 'Dealing_New' );Dealing_New <- Dealing_New %>%
  mutate(A_Date = SessionDate ,
         A_Ticker = InstrumentSymbol ,
         A_Side = as.character( OrderSide) ,
         A_Time = PriorityTime ,
         A_Type = ExecutionType ,
         A_Status = OrderStatus ,
         A_Order_ID = SequentialOrderNumber ,
         A_Order_ID2 = SecondaryOrderID ,
         A_TotalQtd = TotalQuantity ,

```

```

    A_TrdQtd = TradedQuantity ,
    A_Price = OrderPrice ,
    A_LOB_Side = as.character( OrderSide ) ,
    A_LOB_Qtd = TotalQuantity - TradedQuantity ,
    A_LOB_Level = OrderPrice ,
    A_Entry = 0 ,
    A_Adj = Adj ,
    A_t = t ,
    A_t_char = t_char
  ) %>% mutate_if(is.factor , as.character) %>%
as.data.frame(); toc()

tic(); Dealing_Cancel_Expiry <- df.LOB_Orders %>%
  filter( ExecutionType %in% c( 'Cancel' , 'Expiry' )) %>%
  mutate( A_Date = SessionDate ,
    A_Ticker = InstrumentSymbol ,
    A_Side = as.character( OrderSide ) ,
    A_Time = PriorityTime ,
    A_Type = ExecutionType ,
    A_Status = OrderStatus ,
    A_Order_ID = SequentialOrderNumber ,
    A_Order_ID2 = SecondaryOrderID ,
    A_TotalQtd = TotalQuantity ,
    A_TrdQtd = TradedQuantity ,
    A_Price = OrderPrice ,
    A_LOB_Side = as.character( OrderSide ) ,
    A_LOB_Qtd =
      -(TotalQuantity - TradedQuantity) ,
    A_LOB_Level = OrderPrice ,
    A_Entry = 0 ,
    A_Adj = Adj ,
    A_t = t ,
    A_t_char = t_char
  ) %>% mutate_if(is.factor , as.character) %>%
as.data.frame(); toc()

###What is being actually added by an update order.
tic(); Dealing_Update_0<- df.LOB_Orders %>%
  filter( ExecutionType == 'Update' ) %>%
  mutate( A_Date = SessionDate ,

```



```

    A_Ticker = InstrumentSymbol ,
    A_Side = as.character( OrderSide ) ,
    A_Time = PriorityTime ,
    A_Type = ExecutionType ,
    A_Status = OrderStatus ,
    A_Order_ID = SequentialOrderNumber ,
    A_Order_ID2 = SecondaryOrderID ,
    A_TotalQtd = TotalQuantity ,
    A_TrdQtd = TradedQuantity ,
    A_Price = OrderPrice ,
    A_LOB_Side = as.character( OrderSide ) ,
    A_LOB_Qtd = TotalQuantity - TradedQuantity ,
    A_LOB_Level = OrderPrice ,
    A_Entry = 0 ,
    A_Adj = Adj ,
    A_t = t ,
    A_t_char = t_char
  ) %>% mutate_if( is.factor , as.character ) %>%
as.data.frame() ; toc ()
###What is being actually added by an update order.
tic (); Dealing_Update_1 <- df.LOB_Orders %>%
  filter ( ExecutionType == 'Update' ) %>%
  mutate ( A_Date = SessionDate ,
    A_Ticker = InstrumentSymbol ,
    A_Side = as.character( OrderSide ) ,
    A_Time = PriorityTime ,
    A_Type = ExecutionType ,
    A_Status = OrderStatus ,
    A_Order_ID = SequentialOrderNumber ,
    A_Order_ID2 = SecondaryOrderID ,
    A_TotalQtd = TotalQuantity ,
    A_TrdQtd = TradedQuantity ,
    A_Price = OrderPrice ,
    A_LOB_Side = as.character( OrderSide ) ,
    A_LOB_Qtd = Effect ,
    A_LOB_Level = Level ,
    A_Entry = 1 ,
    A_Adj = Adj ,
    A_t = t ,

```

```

        A_t_char = t_char
    ) %>% mutate_if(is.factor , as.character) %>%
    as.data.frame(); toc()

df.o_trades <- df.LOB_Orders %>%
    filter(ExecutionType == 'Trade')

tic(); df.Buy <- df.o_trades %>%
    inner_join(df.t , by =
                c("SequentialOrderNumber" = "ID_Buy" ,
                  "SecondaryOrderID" = "ID2_Buy" )) %>%

mutate(A_Date = SessionDate.y ,
       A_Ticker = InstrumentSymbol.y ,
       A_Side = as.character( OrderSide ) ,
       A_Time = PriorityTime ,
       A_Type = ExecutionType ,
       A_Status = OrderStatus ,
       A_TradNum = TradeNumber ,
       A_Order_ID = SequentialOrderNumber ,
       A_Order_ID2 = SecondaryOrderID ,
       A_TotalQtd = TotalQuantity ,
       A_TrdQtd = TradedQuantity.x ,
       A_Price = TradePrice ,
       A_LOB_Side = as.character( OrderSide ) ,
       A_LOB_Qtd = -1*Delta ,
       A_LOB_Level = OrderPrice ,
       A_Entry = 2 ,
       A_Adj = Adj ,
       A_t = t.x ,
       A_t_char = t_char.x ,
       T_Order_ID_Buy = SequentialOrderNumber ,
       T_Order_ID2_Buy = SecondaryOrderID ,
       T_OrderPrice_Buy = OrderPrice ,

       T_TotalQtd_Buy = TotalQuantity ,
       T_TrdQtd_Buy = TradedQuantity.x ,
       T_Tradetime = Tradetime ,
       T_Type_Buy = ExecutionType ,

```

```

T_Status_Buy = OrderStatus ,
T_agg_order_Buy =
    as.character( AgressorIndicator ) ,

T_ID_Counter = ID_Sell ,
T_ID2_Counter = ID2_Sell ,
T_TradeNumber = TradeNumber ,
T_TradeSign = TradeSign ,
T_Member = BuyMember ,
T_Member_Counter = SellMember ,
T_TrdPrice = TradePrice ,
T_TrdQtd = Delta ,
T_t = t.y ,
T_t_char = t_char.y
) %>% mutate_if( is.factor , as.character ) %>%
as.data.frame() ; toc()

tic() ; df.Sell <- df.o_trades %>%
inner_join( df.t , by =
            c( "SequentialOrderNumber" = "ID_Sell" ,
              "SecondaryOrderID" = "ID2_Sell" ) ) %>%
mutate( A_Date = SessionDate.y ,
        A_Ticker = InstrumentSymbol.y ,
        A_Side = as.character( OrderSide ) ,
        A_Time = PriorityTime ,
        A_Type = ExecutionType ,
        A_Status = OrderStatus ,
        A_TradNum = TradeNumber ,
        A_Order_ID = SequentialOrderNumber ,
        A_Order_ID2 = SecondaryOrderID ,
        A_TotalQtd = TotalQuantity ,
        A_TrdQtd = TradedQuantity.x ,
        A_Price = TradePrice ,
        A_LOB_Side = as.character( OrderSide ) ,
        A_LOB_Qtd = -1*Delta ,
        A_LOB_Level = OrderPrice ,
        A_Entry = 3 ,
        A_Adj = Adj ,
        A_t = t.x ,

```

```

A_t_char = t_char.x,

T_Order_ID_Sell = SequentialOrderNumber ,
T_Order_ID2_Sell = SecondaryOrderID ,
T_OrderPrice_Sell = OrderPrice ,
T_TotalQtd_Sell = TotalQuantity ,
T_TrdQtd_Sell = TradedQuantity.x,
T_Tradetime = Tradetime ,
T_Type_Sell = ExecutionType ,
T_Status_Sell = OrderStatus ,
T_agg_order_Sell =
    as.character(AgressorIndicator),

T_ID_Counter = ID_Buy,
T_ID2_Counter = ID2_Buy,
T_TradeNumber = TradeNumber ,
T_TradeSign = TradeSign ,
T_Member = SellMember ,
T_Member_Counter = BuyMember,
T_TrdPrice = TradePrice ,
T_TrdQtd = Delta ,
T_tl = t.y,
T_t_char = t_char.y
) %>% mutate_if(is.factor , as.character) %>%
as.data.frame(); toc()

#rm(list=c("df.o_trades"))
Dealing_Trades <- df.Buy %>%
  select(starts_with("A_")) %>% bind_rows(df.Sell) %>%
  select(starts_with("A_"))

tic("Union_All_LOB_Events"); df.LOB_Events <-
  ZZZ_df.LOB_Events %>%
  bind_rows(Dealing_Trades) %>%
  bind_rows(Dealing_New) %>%
  select(starts_with("A_")) %>%
  bind_rows(Dealing_Update_0) %>%
  select(starts_with("A_")) %>%
  bind_rows(Dealing_Update_1) %>%

```

```

select (starts_with("A_")) %>%
bind_rows(Dealing_Cancel_Expiry) %>%
select (starts_with("A_")) %>%
arrange(A_t, A_Order_ID, A_Order_ID2); toc()
colnames(df.LOB_Events) <-
  sub("A_", "", names(df.LOB_Events))

# rm(list=c("Dealing_Trades", "Dealing_New",
#           "Dealing_Update_0", "Dealing_Update_1",
#           "Dealing_Cancel_Expiry"))

df.Buy_LF <- df.Buy %>% select(ends_with("_Buy"))
df.Sell_LF <- df.Sell %>% select(ends_with("_Sell"))

df.LOB_Trades <- df.t %>% select(1:5, 7:8, 10:19) %>%
  inner_join(df.Sell_LF, by =
    c("ID_Sell" = "T_Order_ID_Sell",
      "ID2_Sell" = "T_Order_ID2_Sell")) %>%
  inner_join(df.Buy_LF, by =
    c("ID_Buy" = "T_Order_ID_Buy",
      "ID2_Buy" = "T_Order_ID2_Buy"))
toc()

tic("Summarising LOB_Depth_dataframe"); df.LOB_Depth <-
  df.LOB_Events %>%
  group_by(LOB_Side, LOB_Level, t, t_char) %>%
  summarise(delta_depth = sum(LOB_Qtd)) %>%
  arrange(LOB_Side, LOB_Level, t) %>%
  mutate(timestamp =
    as.POSIXct(t, origin = '1970-01-01')) %>%
  mutate(LOB_Time =
    strftime(timestamp, format = "%H:%M:%OS")); toc()

tic("Saving LOB_List."); LOB <- list()
LOB[["Orders"]] <- df.LOB_Orders
LOB[["Trades"]] <- df.LOB_Trades
LOB[["Events"]] <- df.LOB_Events
LOB[["Depth"]] <- df.LOB_Depth
saveRDS(LOB, ###PATH TO SAVE THE LOB WITH ALL DATASETS###); toc()

```

`toc()`

A.2 SSC for MULT3 - Chapter 04

Figure A.1: SSC of MULT3 from March 20th - 01/28

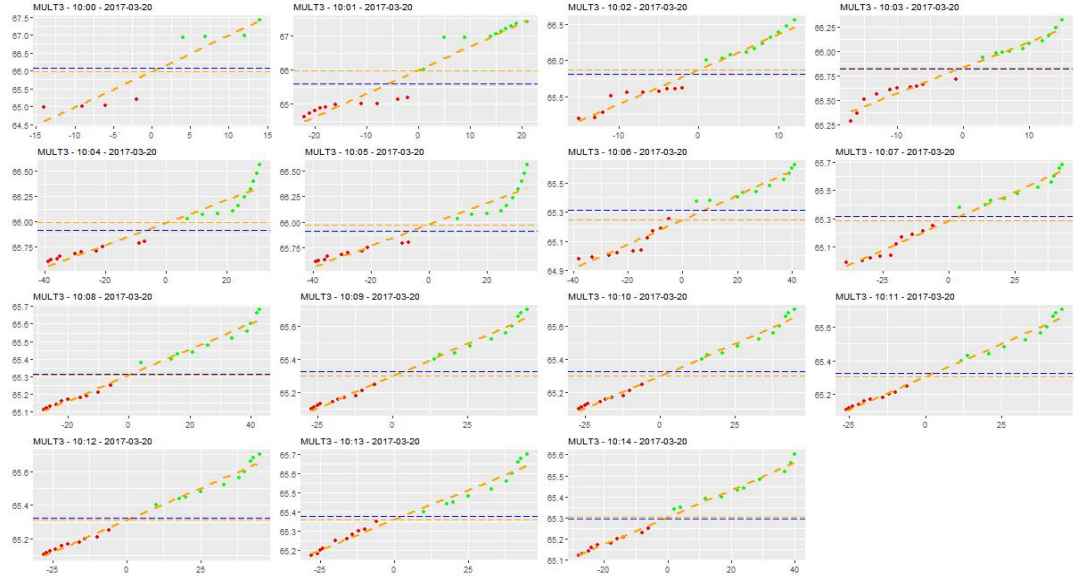


Figure A.2: SSC of MULT3 from March 20th - 02/28

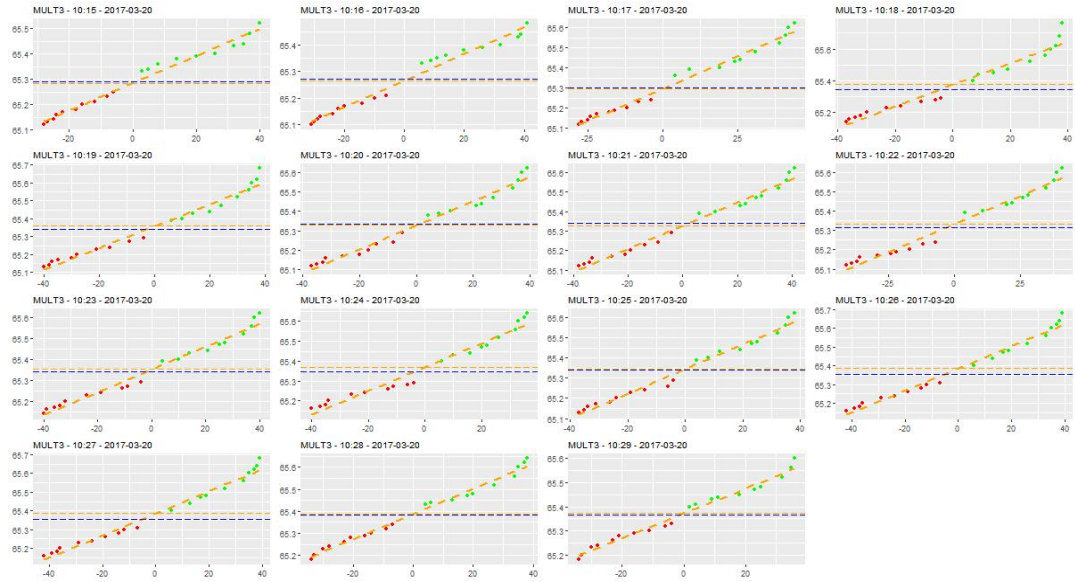


Figure A.3: SSC of MULT3 from March 20th - 03/28

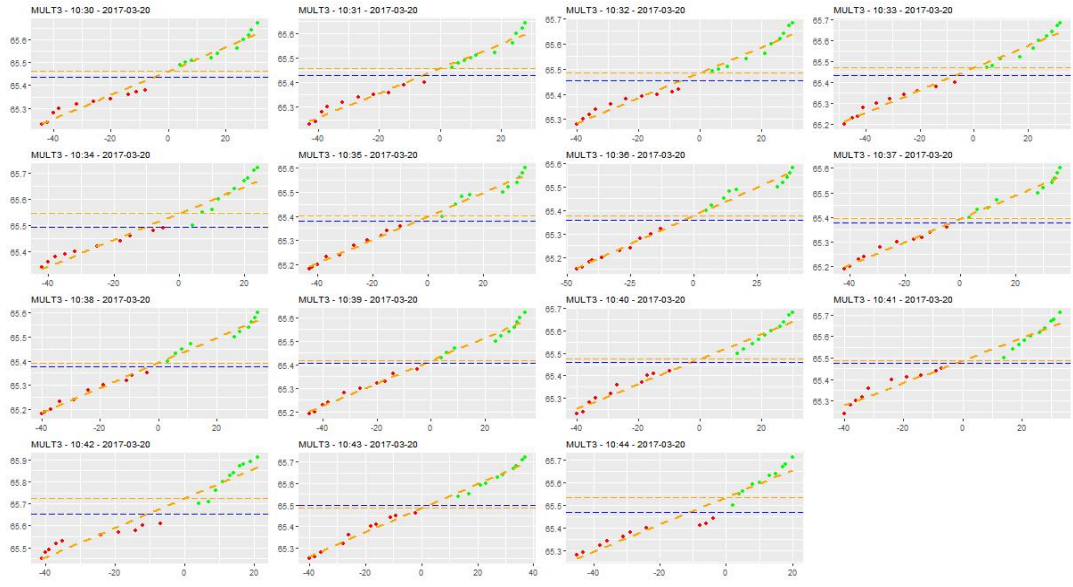


Figure A.4: SSC of MULT3 from March 20th - 04/28

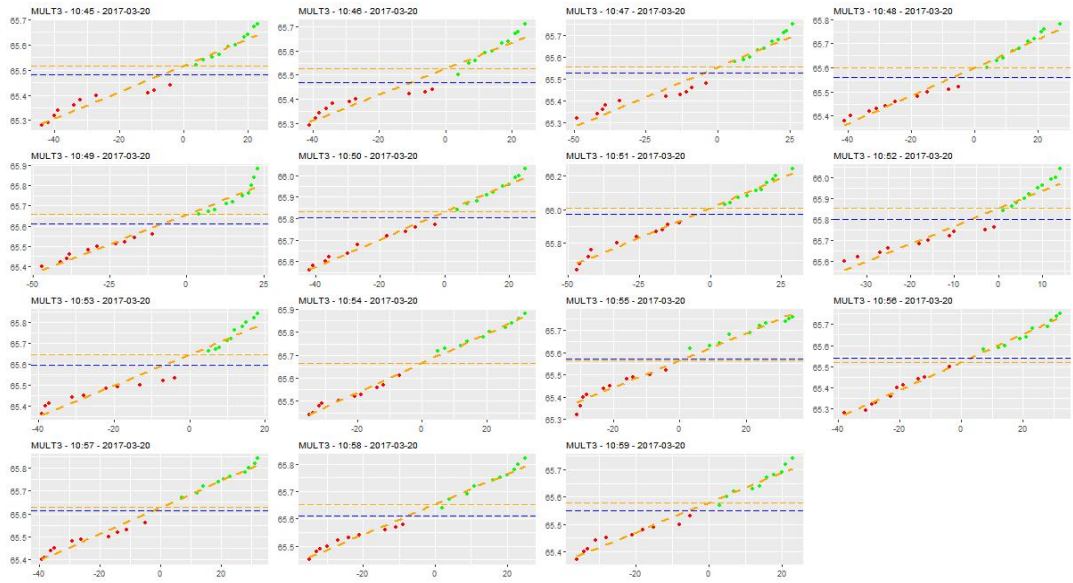


Figure A.5: SSC of MULT3 from March 20th - 05/28

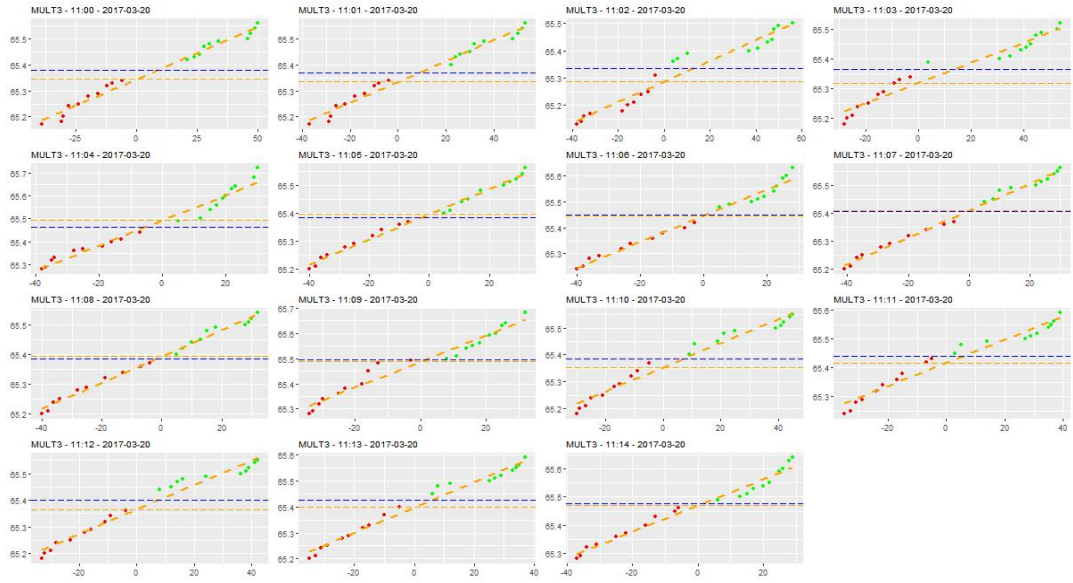


Figure A.6: SSC of MULT3 from March 20th - 06/28

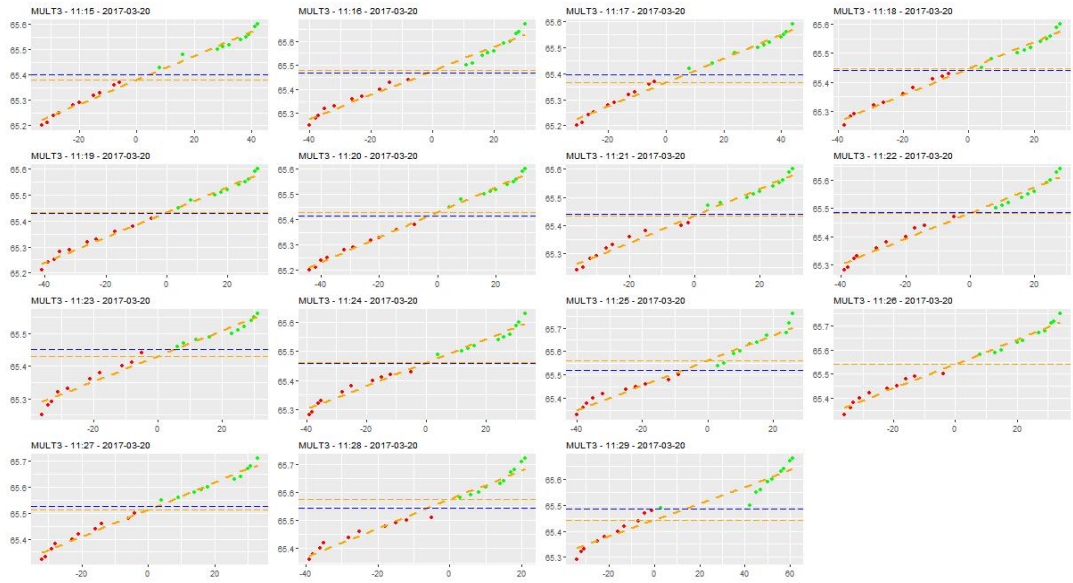


Figure A.7: SSC of MULT3 from March 20th - 07/28

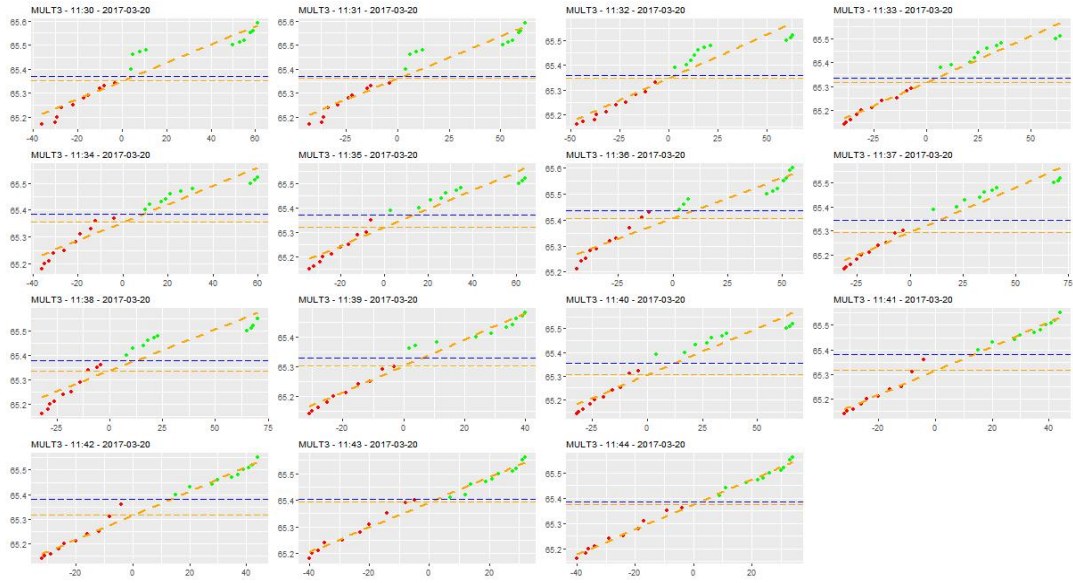


Figure A.8: SSC of MULT3 from March 20th - 08/28

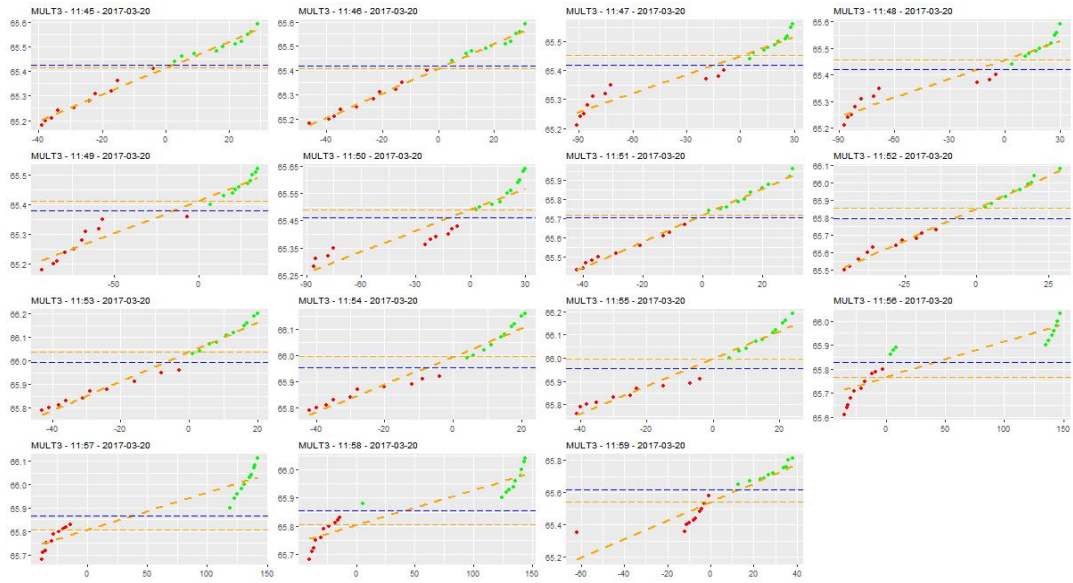


Figure A.9: SSC of MULT3 from March 20th - 09/28

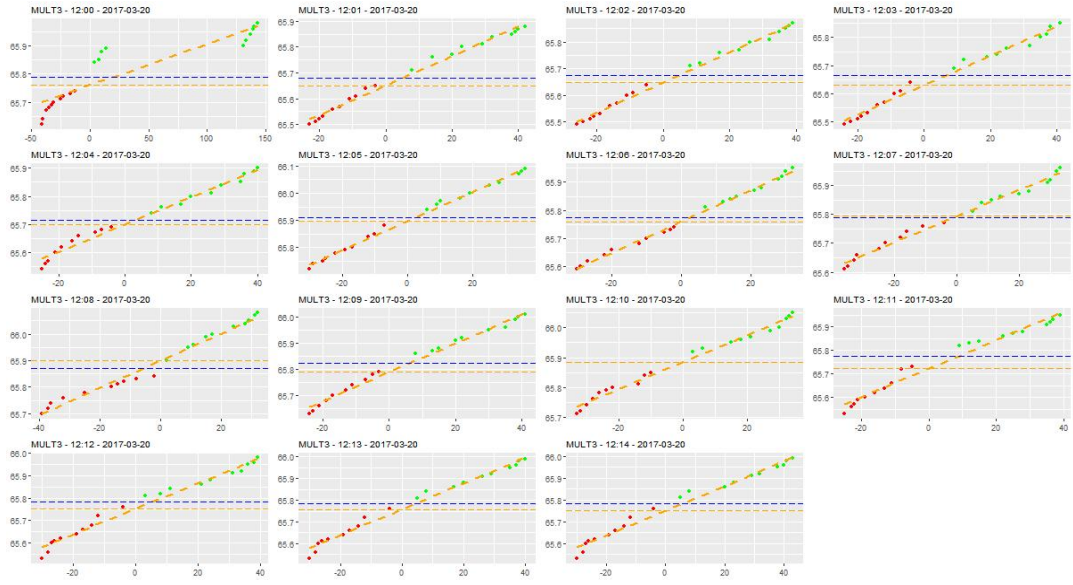


Figure A.10: SSC of MULT3 from March 20th - 10/28

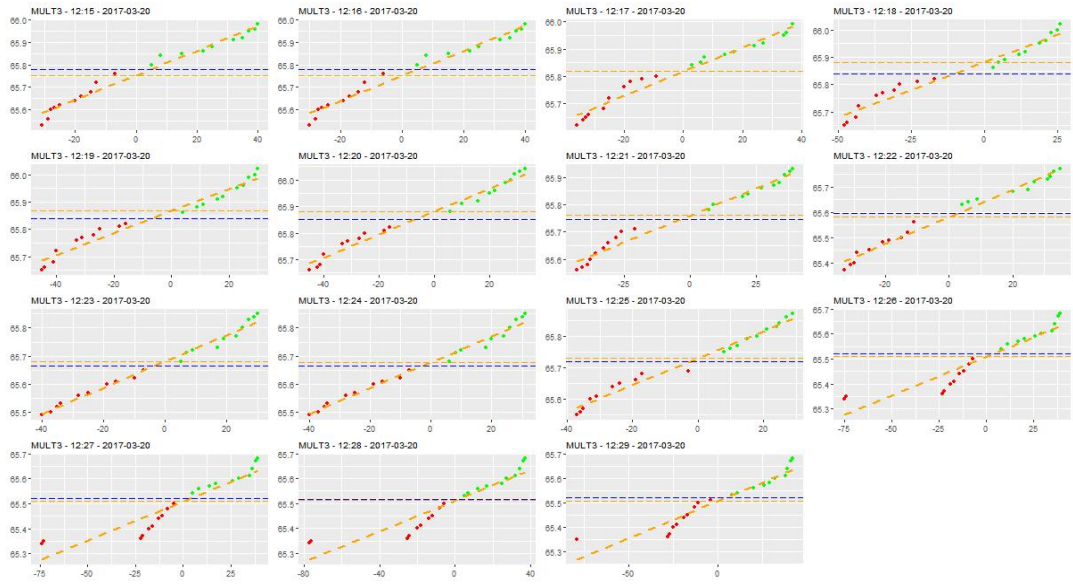


Figure A.11: SSC of MULT3 from March 20th - 11/28

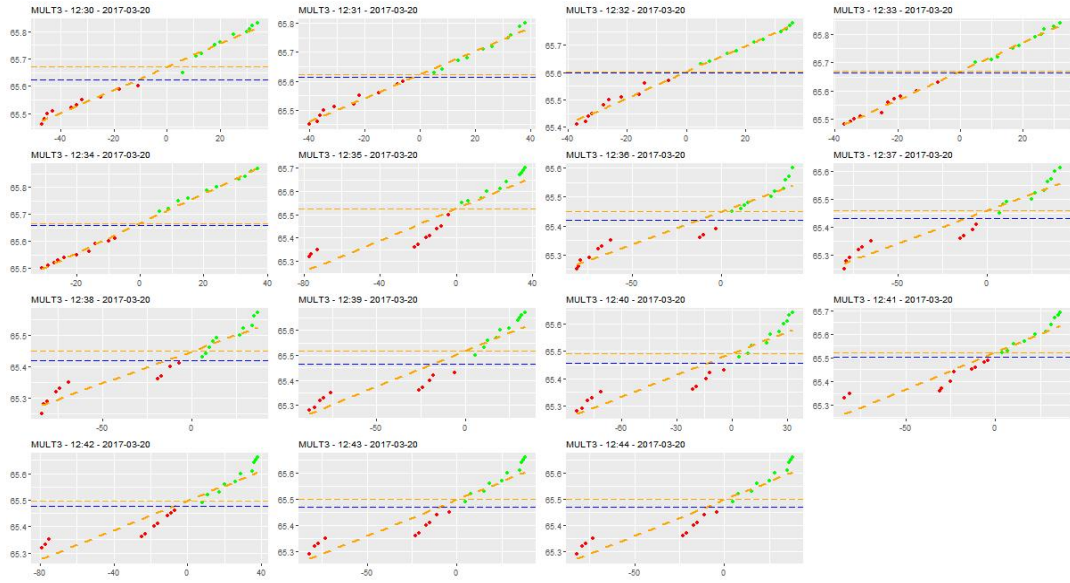


Figure A.12: SSC of MULT3 from March 20th - 12/28

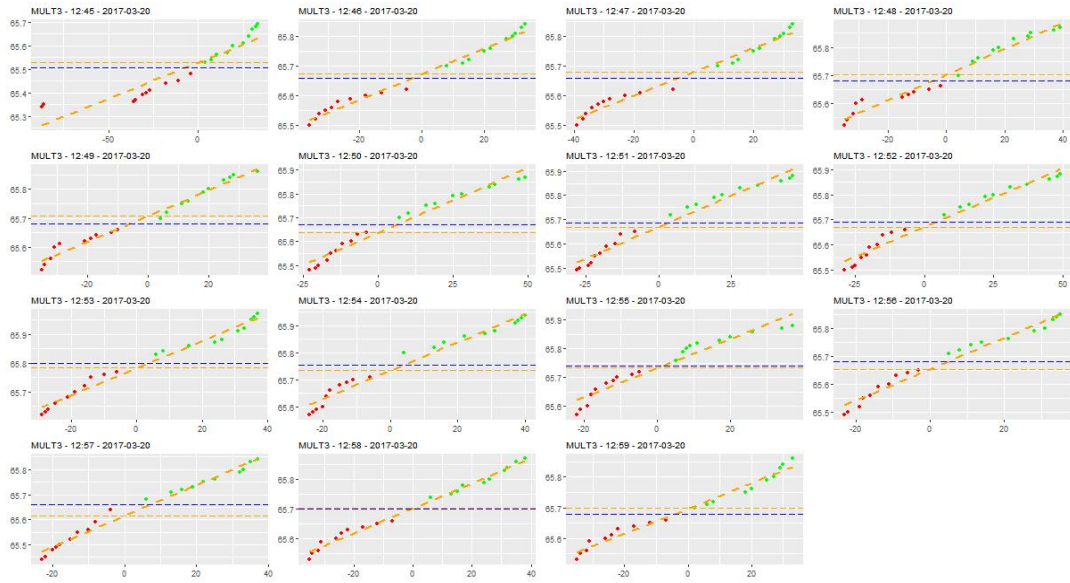


Figure A.13: SSC of MULT3 from March 20th - 13/28

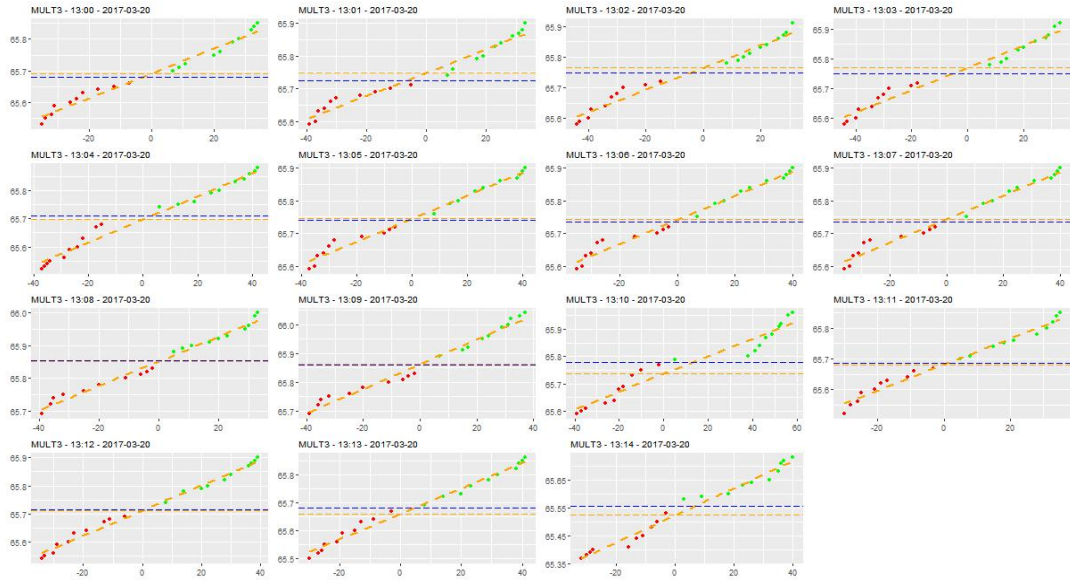


Figure A.14: SSC of MULT3 from March 20th - 14/28

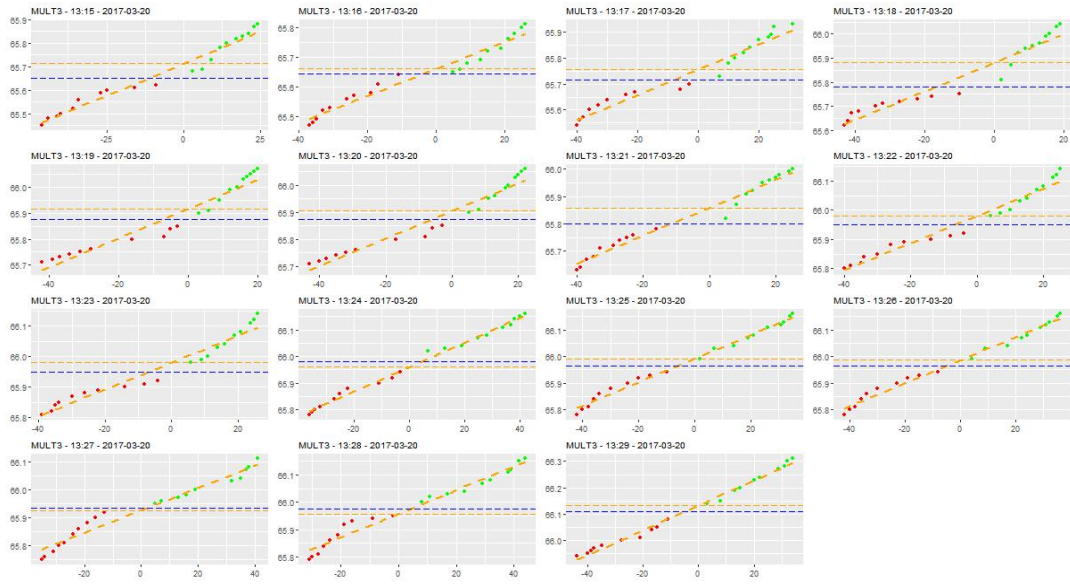


Figure A.15: SSC of MULT3 from March 20th - 15/28

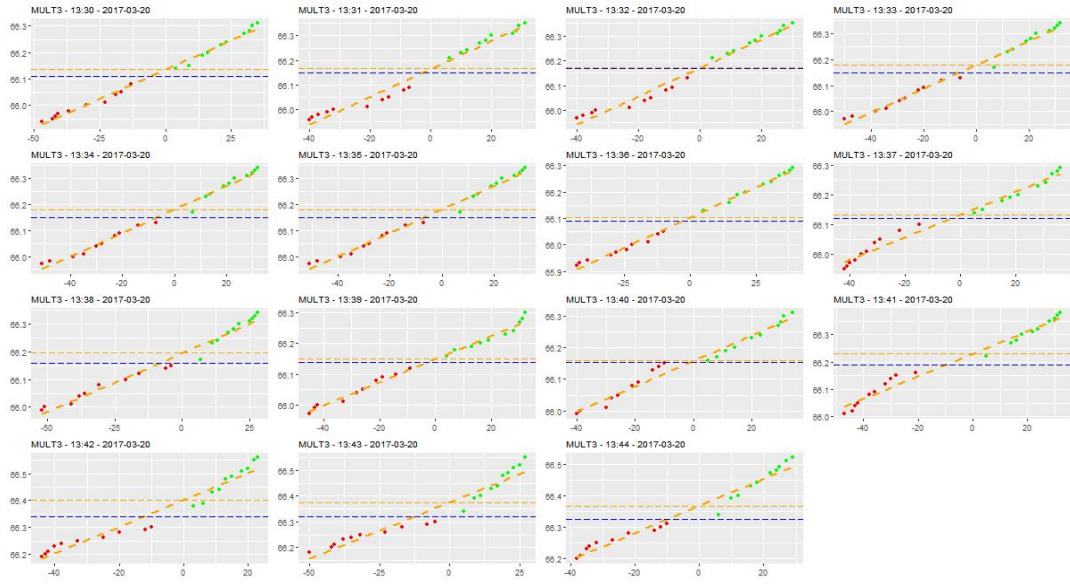


Figure A.16: SSC of MULT3 from March 20th - 16/28

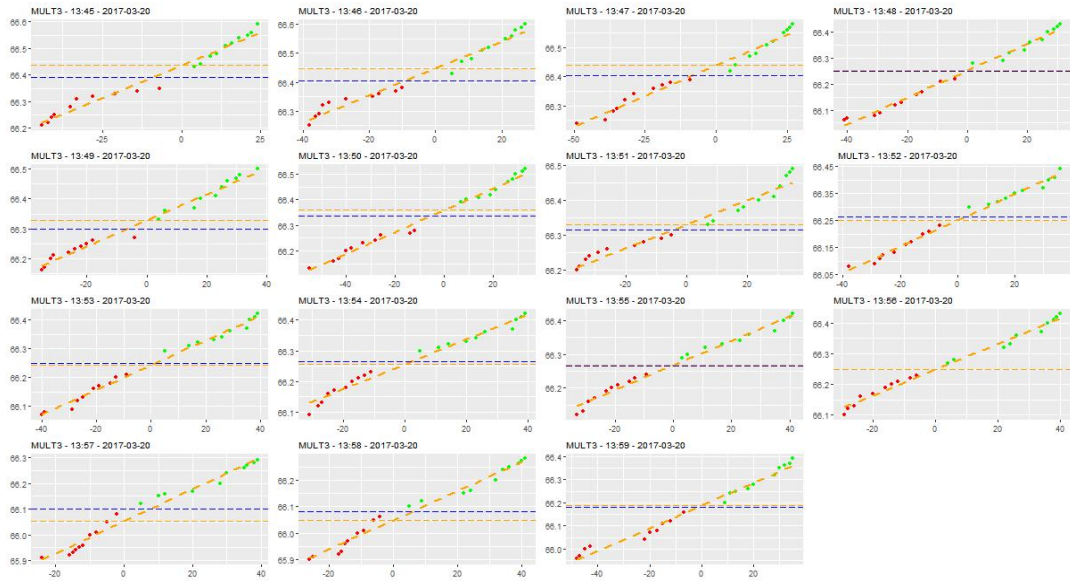


Figure A.17: SSC of MULT3 from March 20th - 17/28

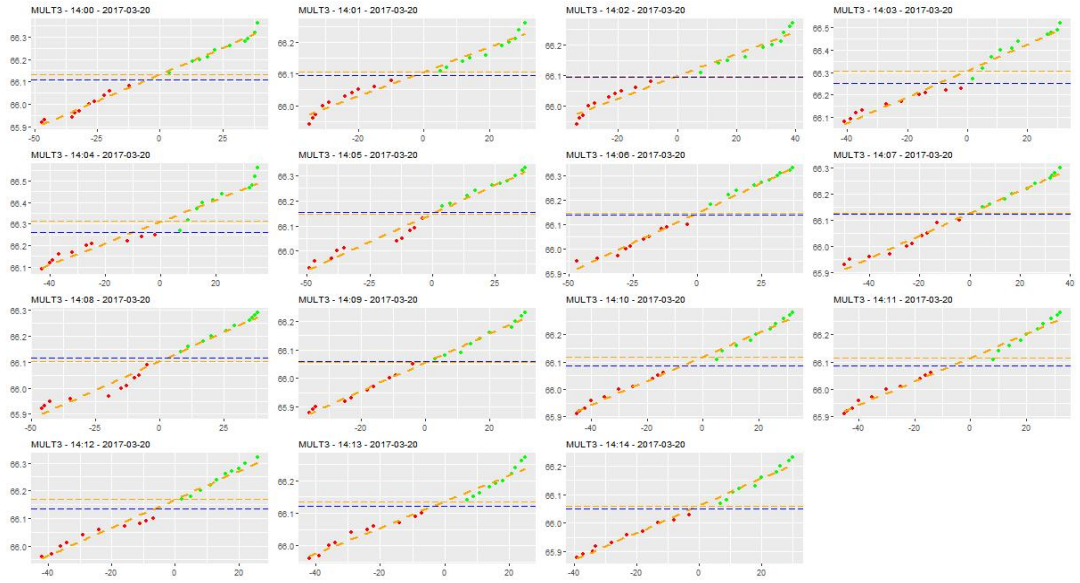


Figure A.18: SSC of MULT3 from March 20th - 18/28

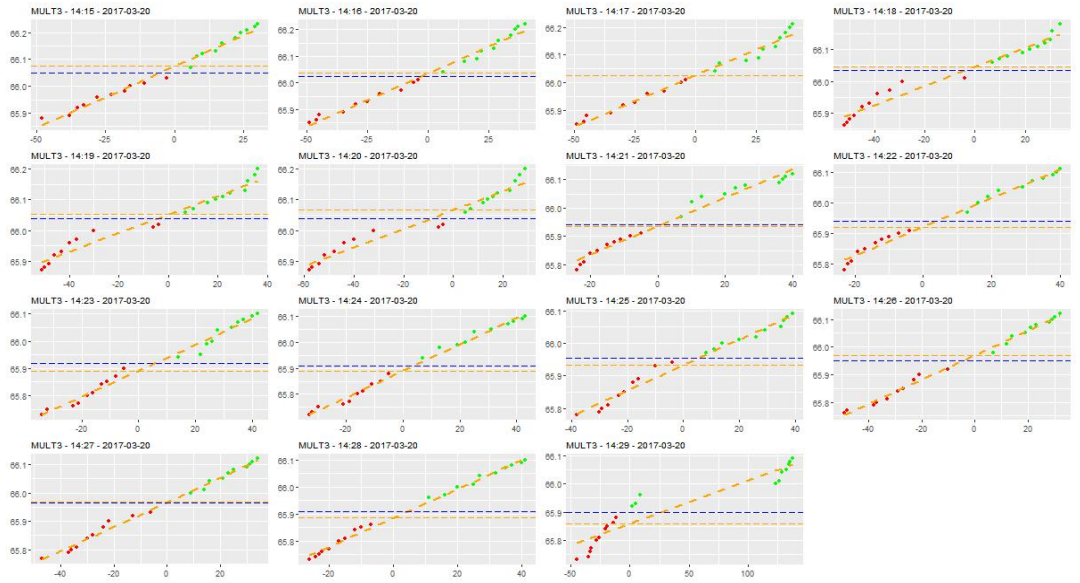


Figure A.19: SSC of MULT3 from March 20th - 19/28

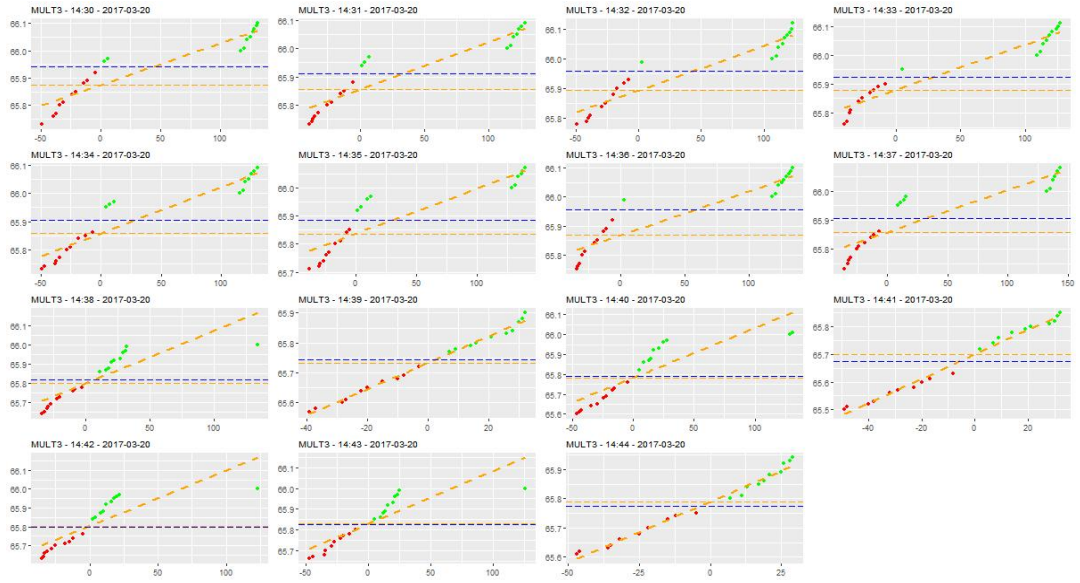


Figure A.20: SSC of MULT3 from March 20th - 20/28

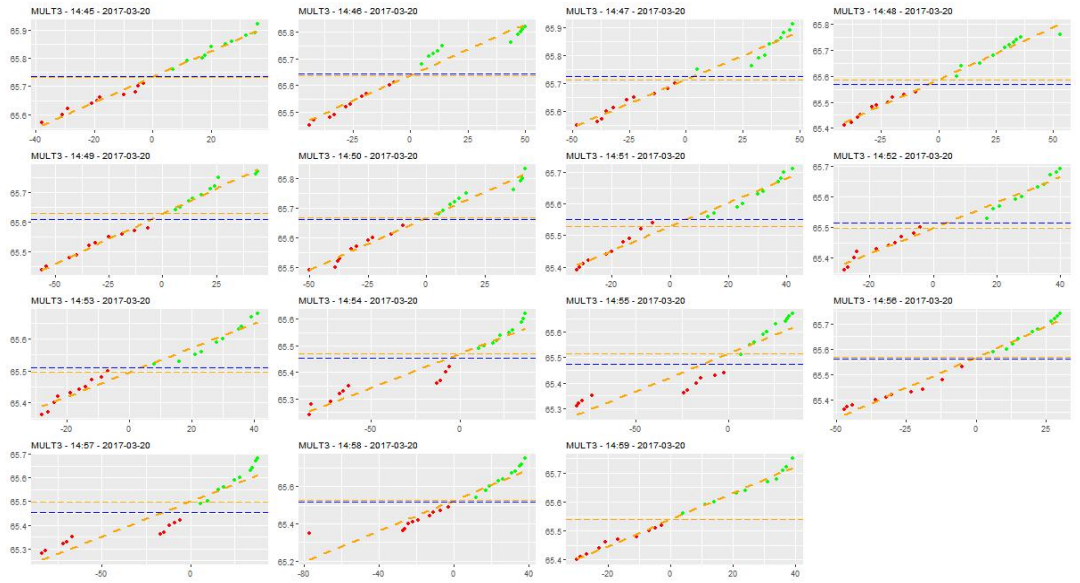


Figure A.21: SSC of MULT3 from March 20th - 21/28

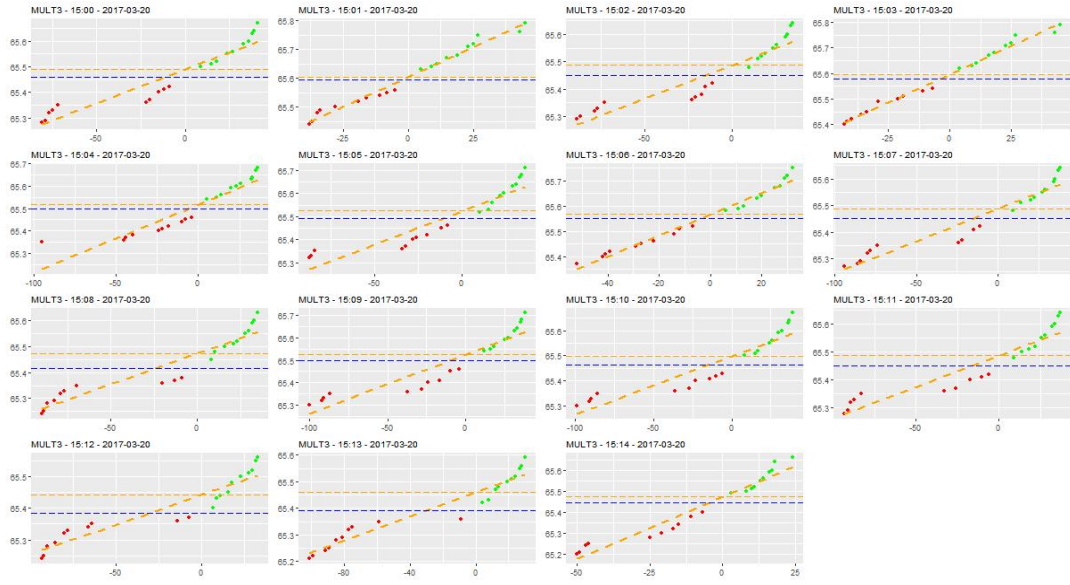


Figure A.22: SSC of MULT3 from March 20th - 22/28

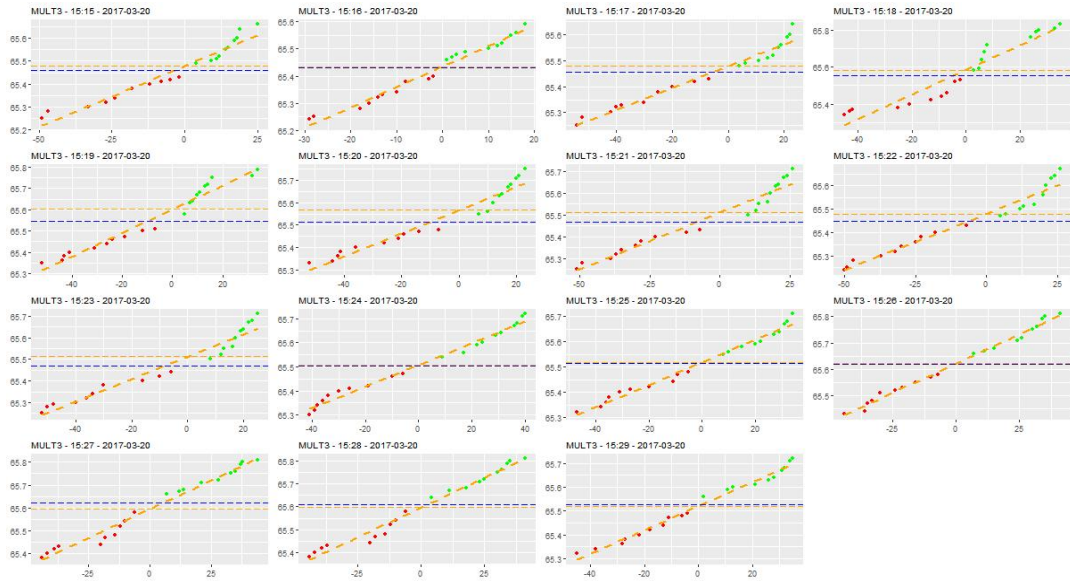


Figure A.23: SSC of MULT3 from March 20th - 23/28

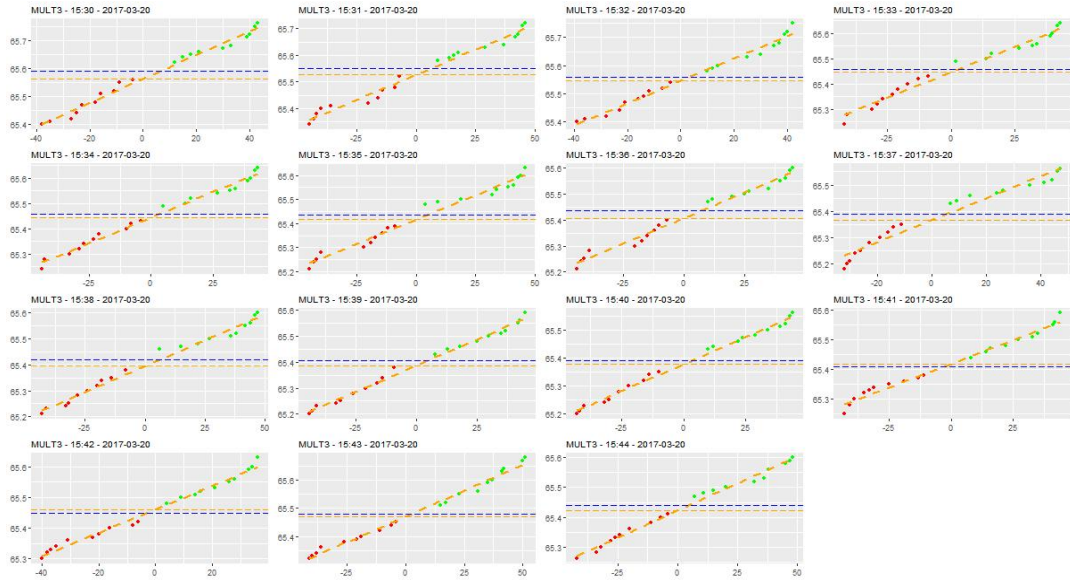


Figure A.24: SSC of MULT3 from March 20th - 24/28

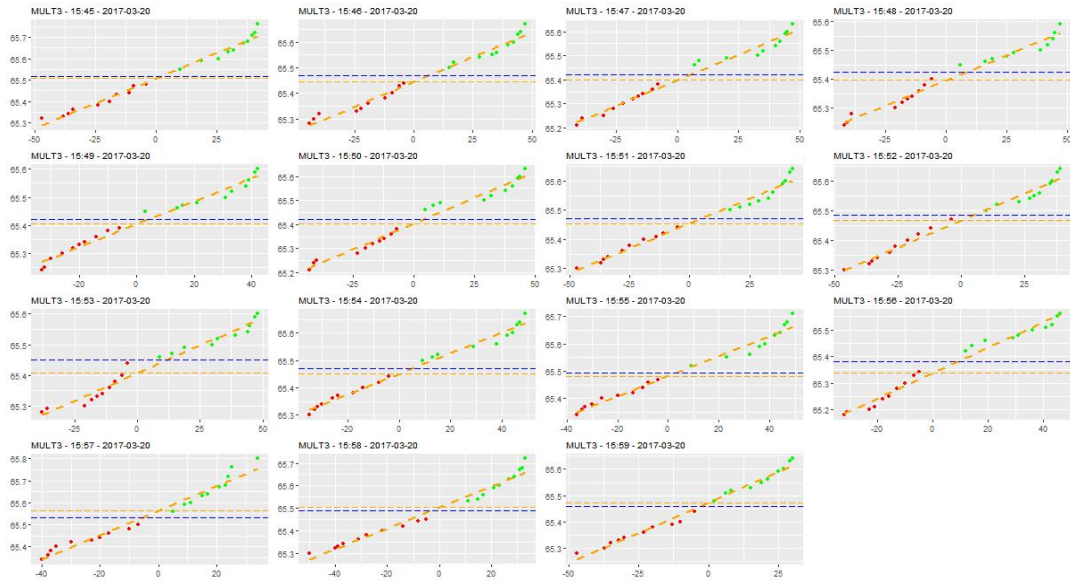


Figure A.25: SSC of MULT3 from March 20th - 25/28

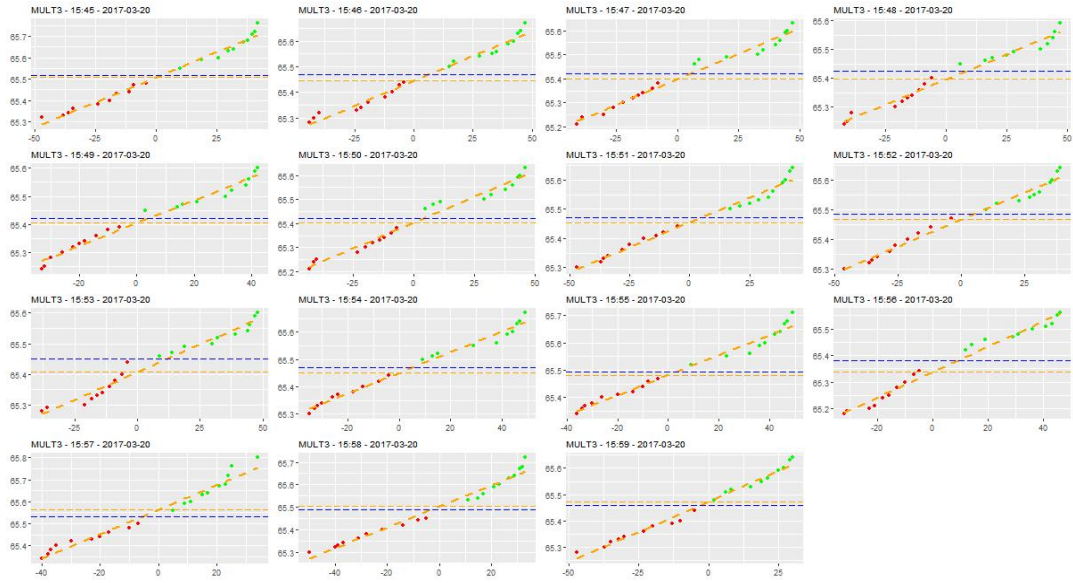


Figure A.26: SSC of MULT3 from March 20th - 26/28

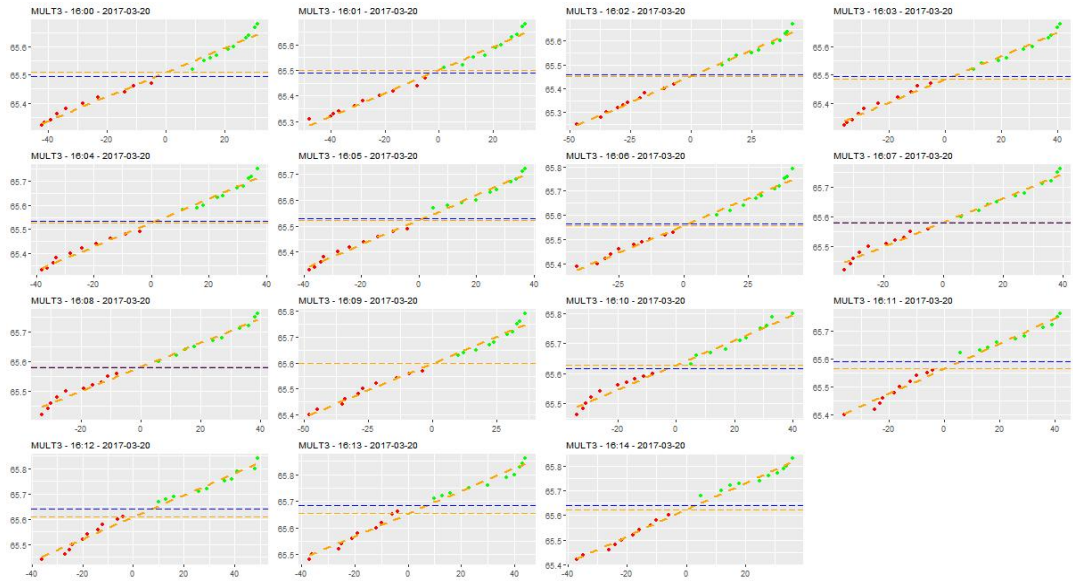


Figure A.27: SSC of MULT3 from March 20th - 27/28

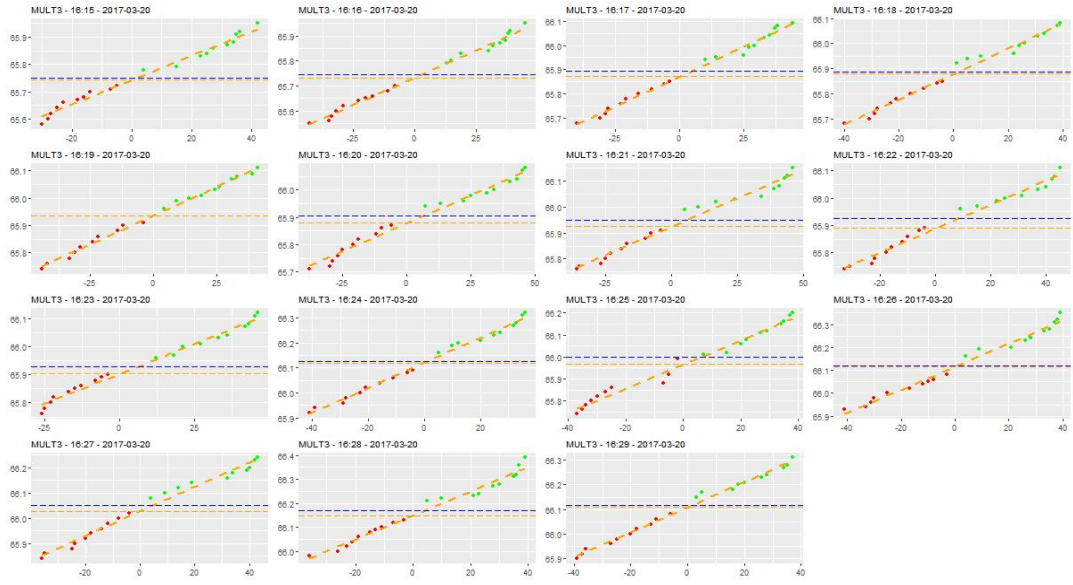
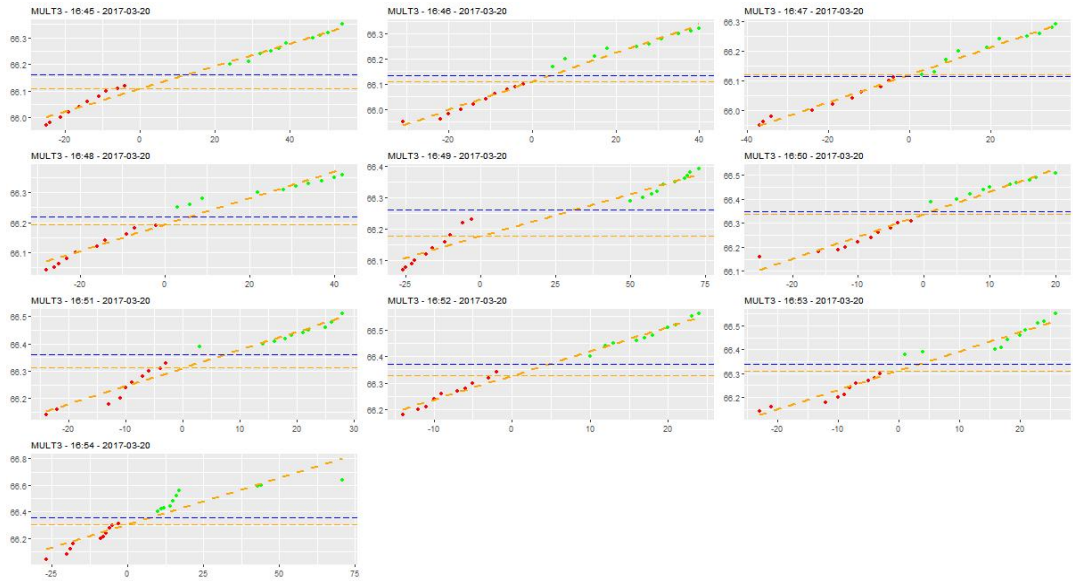
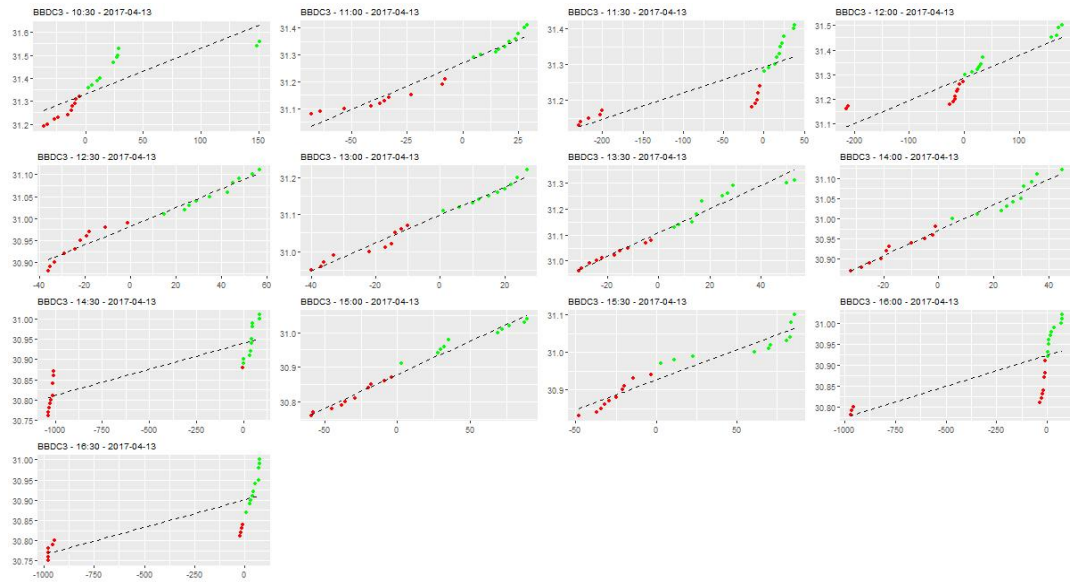


Figure A.28: SSC of MULT3 from March 20th - 28/28



As mentioned in Chapter 04: the four last LOBs have an uncommon shape incompatible with the SSC.

Figure A.29: SSC of BBDC3 from April 13th



A.3 Discrete Delta Hedging

Table A.1: Prices of European Call Option - Ku, Lee & Zhu (2012)

Initial Spot	Liquidity Parameter α				
	0 (Black-Scholes)	0.0001	0.0005	0.001	0.002
80	1.1859	1.1869	1.191	1.1959	1.2058
85	2.1613	2.1621	2.168	2.1754	2.1899
90	3.5891	3.5896	3.5973	3.6069	3.6258
95	5.5195	5.5198	5.5288	5.54	5.562
100	7.9656	7.9657	7.9753	7.9872	8.0106
105	10.9056	10.9058	10.9152	10.9268	10.9498
110	14.2920	14.2925	14.3011	14.3117	14.3327
115	18.0619	18.0627	18.0701	18.0792	18.0973

Figure A.30: Paths of Delta for Simulations without Liquidity Risk

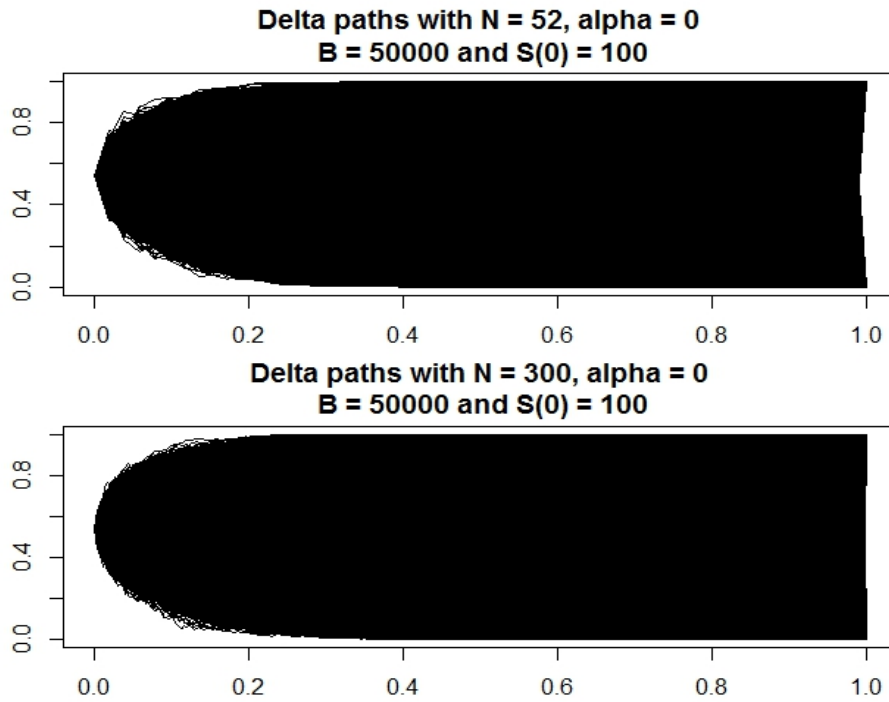


Figure A.31: Histogram of Hedging Errors for several N with $r_f = 0.05$

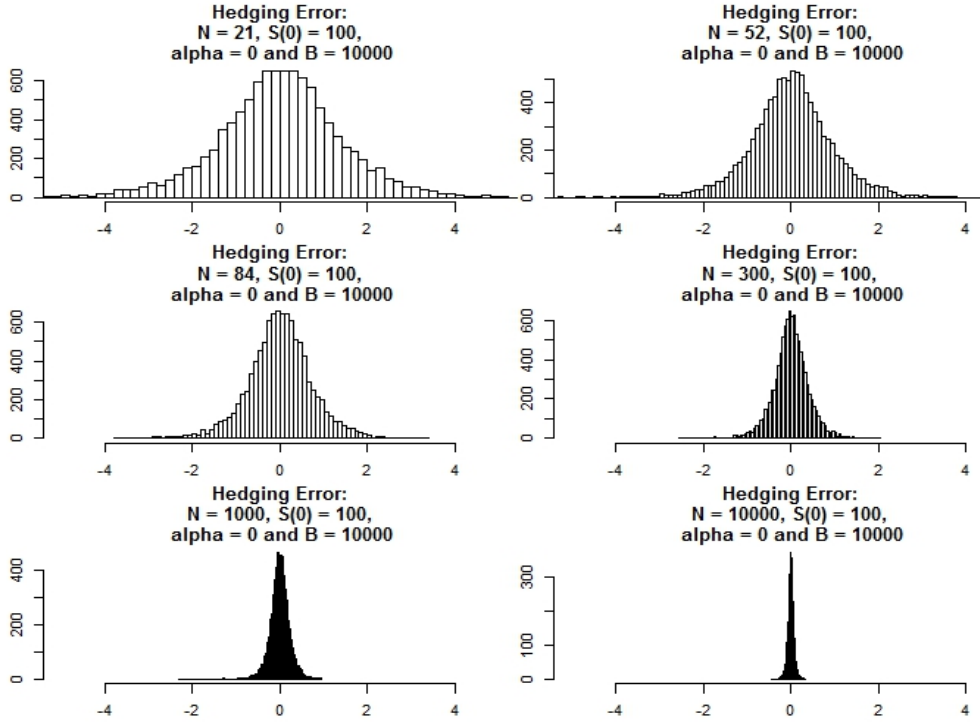


Figure A.32: Trading Prices with $r_f = 0.00$, $B = 5000$, $\alpha = 0.002$ and $N = 21, 52, 84, 300, 1000$ and 10000

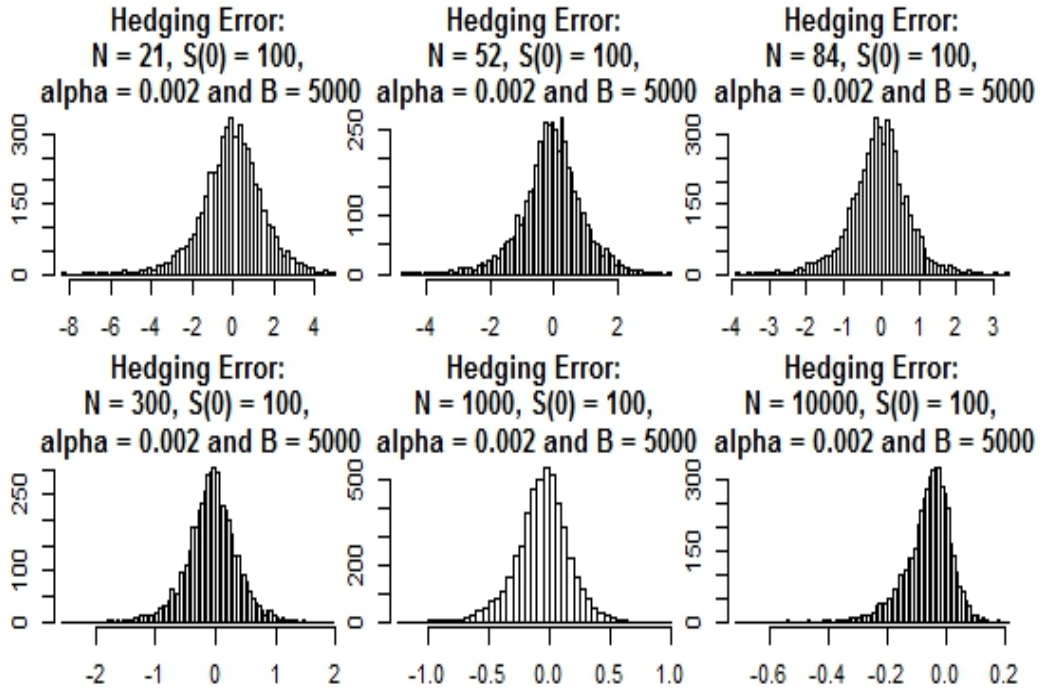
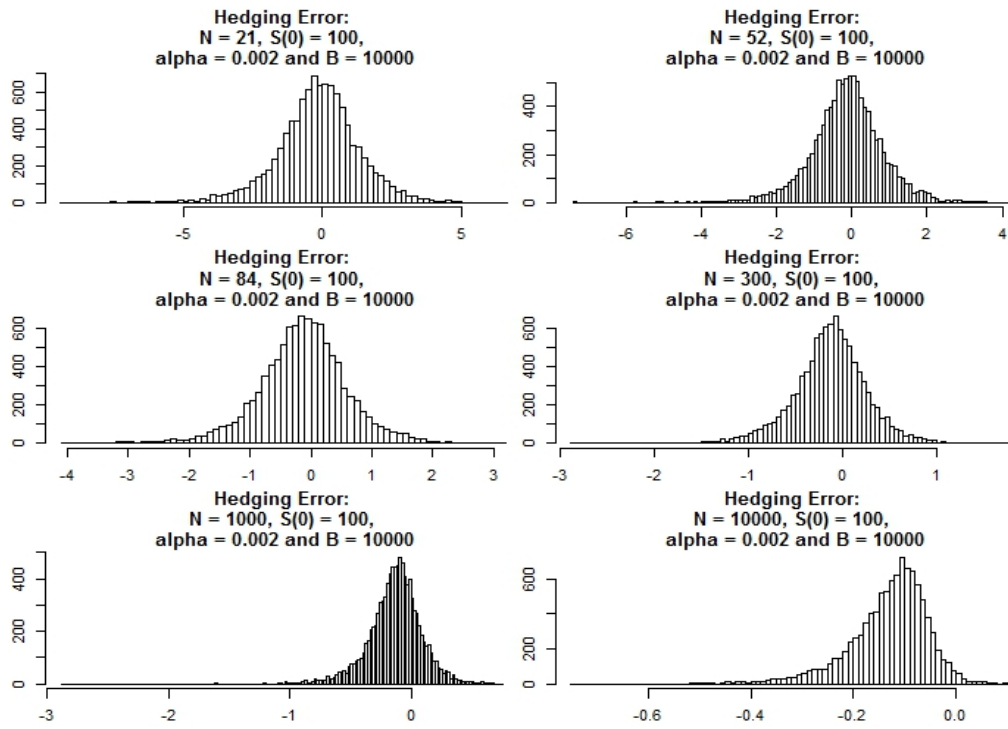


Figure A.33: Trading Prices with $r_f = 0.05$, $B = 10000$, $\alpha = 0.002$ and $N = 21$, 52, 84, 300, 1000 and 10000



A.4 Brazilian Equity Options

Table A.2: Equity Call Options with changes in underling asset in 2017

	Ticker	Expiration Date
1	BBDCL86	2017-12-18
2	VALEI1	2017-09-18
3	VALEI16	2017-09-18
4	VALEI2	2017-09-18
5	VALEI75	2017-09-18
6	VALEI86	2017-09-18
7	VALEJ40	2017-10-16
8	VALEJ50	2017-10-16
9	VALEJ59	2017-10-16
10	VALEJ6	2017-10-16
11	VALEJ60	2017-10-16
12	VALEJ61	2017-10-16
13	VALEJ68	2017-10-16
14	VALEJ85	2017-10-16
15	VALEK1	2017-11-21
16	VALEK35	2017-11-21
17	VALEL1	2017-12-18
18	VALEL12	2017-12-18
19	VALEL15	2017-12-18
20	VALEL17	2017-12-18
21	VALEL18	2017-12-18
22	VALEL20	2017-12-18
23	VALEL21	2017-12-18
24	VALEL22	2017-12-18
25	VALEL23	2017-12-18
26	VALEL26	2017-12-18
27	VALEL29	2017-12-18
28	VALEL37	2017-12-18
29	VALEL77	2017-12-18
30	VALEL9	2017-12-18

Table A.3: Equities with several types of shares in 2017

	company	Ticker	ON_PN
1	AHEB	AHEB5	PN
2	AHEB	AHEB6	PN
3	BRGE	BRGE11	PN
4	BRGE	BRGE12	PN
5	BRGE	BRGE6	PN
6	BRGE	BRGE8	PN
7	BRKM	BRKM5	PN
8	BRKM	BRKM6	PN
9	BRSR	BRSR5	PN
10	BRSR	BRSR6	PN
11	CEBR	CEBR5	PN
12	CEBR	CEBR6	PN
13	CELP	CELP5	PN
14	CELP	CELP7	PN
15	CEPE	CEPE5	PN
16	CEPE	CEPE6	PN
17	CESP	CESP5	PN
18	CESP	CESP6	PN
19	CRPG	CRPG5	PN
20	CRPG	CRPG6	PN
21	CSRN	CSRN5	PN
22	CSRN	CSRN6	PN
23	ELET	ELET5	PN
24	ELET	ELET6	PN
25	MEND	MEND5	PN
26	MEND	MEND6	PN
27	RPAD	RPAD5	PN
28	RPAD	RPAD6	PN
29	SOND	SOND5	PN
30	SOND	SOND6	PN
31	UNIP	UNIP5	PN
32	UNIP	UNIP6	PN
33	USIM	USIM5	PN
34	USIM	USIM6	PN

A.5 Almgren-Chriss Framework

This section presents more elements on the solution of Almgren-Chriss' Execution Strategy under the CARA function and presents some defining mathematical properties following Guéant (2016, p.49).

A.5.1 Optimal Strategy under CARA function

The main reason for choosing CARA utility function as the optimization criteria as in (2.14) is its independence on the agent's initial wealth. Additionally, since (6.15) gives both parameters of a Gaussian distribution, (2.14) is log-normal:

$$\begin{aligned}\mathbb{E}[-\exp(-\gamma X_T)] &= -\exp\left(-\gamma\mathbb{E}[X_T] + \frac{1}{2}\gamma^2\mathbb{V}[X_T]\right) \\ &= -\exp\left(-\gamma\left(X_0 + q_0 S_0 - \frac{k}{2}q_0^2\right)\right) \times \\ &\quad \exp\left(\gamma\left(\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt + \frac{\gamma}{2}\sigma^2 \int_0^T q_t^2 dt\right)\right)\end{aligned}\quad (\text{A.1})$$

To maximize the utility function, we note that the term that varies $(v_t)_{t \in [0, T]} \in \mathcal{A}_{\text{det}}$ is given by:

$$\left(\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt + \frac{\gamma}{2}\sigma^2 \int_0^T q_t^2 dt\right) \quad (\text{A.2})$$

which has to be minimized. Because $v_t = \frac{dq_t}{dt}$, and $\int_0^T |v_t| dt \in L^\infty(\Omega)$, the function $q : t \mapsto q_t$ is in fact absolutely continuous. Therefore, the problem boils down to a variational problem (a Bolza problem) and is solved by finding a control process minimizing

$$J(q) = \int_0^T \left(V_t L\left(\frac{q'(t)}{V_t}\right) + \frac{\gamma}{2}\sigma^2 q(t)^2 \right) dt, \quad (\text{A.3})$$

over the set of absolutely continuous functions $q \in W^{1,1}(0, T)$ satisfying the constraints $q(0) = q_0$ and $q(T) = 0$.

The functional in A.3 for q which minimizes $J(q)$ is fairly known as a Bolza problem from Variational Calculus. This discipline is mainly concerned in finding a function that fulfills all the requirements of a given space of functions.

Setting the system can be done both by Euler-Lagrange¹ or Hamiltonian characterizations. If we do not assume any regularity of $L(\rho)$, we can find a Hamilto-

¹This characterization can be found on the Annex

nian system given by:

$$\begin{cases} p'(t) &= \gamma\sigma^2 q^*(t), \\ q^{*'}(t) &= V_t H'(p(t)), \\ q^*(0) &= q_0, \\ q^*(T) &= 0, \end{cases} \quad (\text{A.4})$$

where H is the Legendre-Fenchel transform of the function L defined by

$$H(p) = \sup_{\rho} (\rho p - L(\rho)). \quad (\text{A.5})$$

Since L is said to be strictly convex, $H \in \mathcal{C}^1$, and this is the reason why we can write the Hamiltonian system in the form above. Also, if $(V_t)_t$ is continuous then we can deduce ² that q^* is a function of class $\mathcal{C}^1$. Thus, this makes $p \in \mathcal{C}^2$ under the same hypothesis.

Classically, by assuming $L(\rho)$ as of quadratic form we get

$$L(\rho) = \eta\rho^2 \quad (\text{A.6})$$

and $(V_t)_t$ is given as constant. Then the characterized optimal system in Hamiltonian terms becomes:

$$\begin{cases} p'(t) &= \gamma\sigma^2 q^*(t), \\ q^{*'}(t) &= \frac{V_t}{2\eta} p(t), \\ q^*(0) &= q_0, \\ q^*(T) &= 0. \end{cases} \quad (\text{A.7})$$

Consequently, q^* is the unique solution of the equation:

$$q^{*''} = \frac{\gamma\sigma^2 V_t}{2\eta} q^*(t), \quad (\text{A.8})$$

²Proof on the Annex

Satisfying the boundary conditions $q^*(0) = q_0$ and $q^*(T) = 0$.

$$q^*(t) = q_0 \frac{\sinh\left(\sqrt{\frac{\gamma\sigma^2 V}{2\eta}}(T-t)\right)}{\sinh\left(\sqrt{\frac{\gamma\sigma^2 V}{2\eta}}T\right)}, \quad (\text{A.9})$$

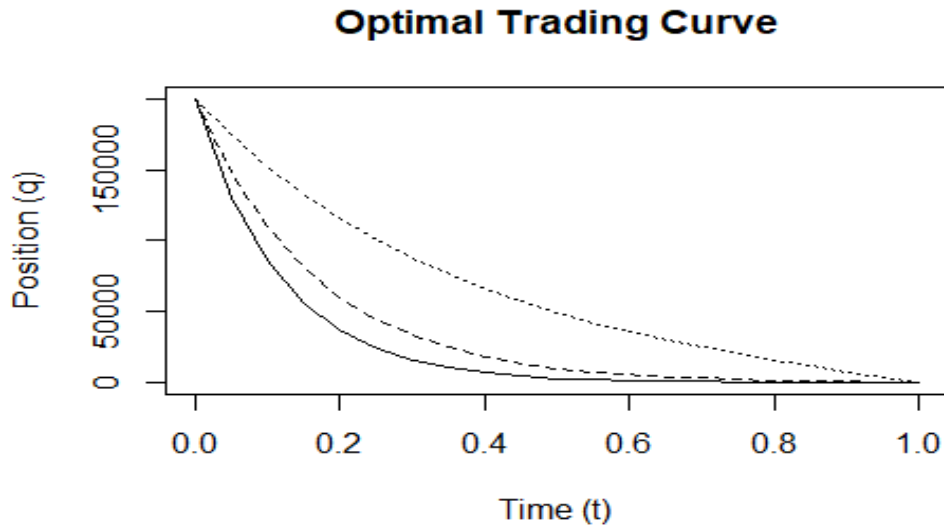
setting the optimal deterministic strategy $(v_t^*)_t$ as

$$v_t^* = \frac{dq_t^*}{dt}(t) = -q_0 \sqrt{\frac{\gamma\sigma^2 V}{2\eta}} \frac{\cosh\left(\sqrt{\frac{\gamma\sigma^2 V}{2\eta}}(T-t)\right)}{\sinh\left(\sqrt{\frac{\gamma\sigma^2 V}{2\eta}}T\right)} \quad (\text{A.10})$$

Guéant (2016, p.54) gives an example where V is of constant nature and $L(\rho) = \eta\rho^2$ as in the earlier versions of Almgren-Chriss setup³:

- $S_0 = 45BRL$;
- $q_0 = 200,000$ shares over one day ($T = 1$)
- $\sigma = 0.6BRL \cdot \text{day}^{-\frac{1}{2}} \cdot \text{share}^{-1}$, i.e., annual volatility approximately equal to 21%;
- $V = 4,000,000$ shares $\cdot \text{day}^{-1}$;
- $L(\rho) = \eta\rho^2$, with $\eta = 0.1BRL \cdot \text{share}^{-1}$.

Figure A.34: Optimal trading curve for $q_0 = 200,000$ shares over one day ($T = 1$), for different values of γ . Solid line: $\gamma = 10^{-5}BRL^{-1}$. Dashed line: $\gamma = 5 \times 10^{-6}BRL^{-1}$. Dotted line: $\gamma = 10^{-6}BRL^{-1}$.



³The code for this simple example is presented in the Annex.

```

rm(list=ls())
CARA.function <- function(gamma, X_N){
  return(-exp(-gamma*X_N))
}
L.function <- function(eta, v_t, V){
  return(eta*rho^2)
}

cont.q_star <- function(q_0, gamma, sigma, V, eta, TM, t){
  numerator <- sinh(sqrt((gamma*(sigma^2)*V)/(2*eta))*(TM-t))
  denominator <- sinh(sqrt((gamma*(sigma^2)*V)/(2*eta))*TM)
  return(q_0*(numerator/denominator))
}

cont.v_star <- function(q_0, gamma, sigma, V, eta, TM, t){
  numerator <- cosh(sqrt((gamma*(sigma^2)*V)/(2*eta))*(TM-t))
  denominator <- sinh(sqrt((gamma*(sigma^2)*V)/(2*eta))*TM)
  return(-q_0*sqrt((gamma*(sigma^2)*V)/(2*eta))
    *numerator/denominator)
}

par(mfcol = c(1,3))

q_0 <- 200000; TM <- 1; S_0 <- 45;
sigma <- 0.6; V <- 4000000;
eta <- 0.1; vec.gamma <- c(10^(-5), 5*10^(-6), 10^(-6));
N <- 100; time <- seq(0, TM, length.out = N+1)
#####
plot(x = time, y = rep(q_0/length(time), length(time)),
     type = "l", lty = 0, xlim = c(0, TM), ylim = c(0, q_0),
     xlab = "Time_(t)", ylab = "Position_(q)",
     main = "Continuous_Case")
i <- 1; list.cont_rec = list()
for(gamma in vec.gamma){
  y <- cont.q_star(q_0 = q_0, gamma = gamma, sigma = sigma,
    V = V, eta = eta, TM = TM, t = time)
  list.cont_rec[[i]] <- y
}

```

```

    lines(x = time, y = y, lty = i)
    i <- i+1
  }
#####

S_n <- function(S, sigma, TM, N, rand, k, v_n){
  dt <- TM/N
  return(S + sigma*sqrt(dt)*rand + k*v_n*dt)
}

X_n <- function(X, v_n, S, L.now, V, TM, N){
  dt <- TM/N
  return(X - v_n*S*dt-L.now(v_t = v_n)*V*dt)
}

disc.q_star <-
  function(q_0, gamma, sigma, V, eta, TM, t, N, alpha){
    dt <- TM/N
    numerator <- sinh(alpha*(TM-t))
    denominator <- sinh(alpha*TM)
    return(q_0*(numerator/denominator))
  }

disc.v_star <- function(disc.q_star, TM, N){
  size <- length(disc.q_star); dt<- TM/N
  return(
    (disc.q_star[2:size] - disc.q_star[1:(size-1)])/dt
  )
}

Terminal_X_N <-
  function(X_0, q_0, S_0, k, sigma, TM, N, B, disc.q_star, rand){
    dt <- TM/N
    return(
      X_0 + q_0*S_0 - (k/2)*(q_0^2) +
      sigma*sqrt(dt)*cumsum(disc.q_star*rand)+(k/2)*
      cumsum((disc.v_star^2)*(dt^2)) -
      cumsum(L.now(v_t = disc.v_star)*V*dt)
    )
  }

```

```
}
```

```
#####
```

```
L.now <- function(x){L.function(eta = eta, V = V, v_t = x)}
```

```
N <- 100; dt <- TM/N; time.dt <- seq(0, TM, by=dt)
```

```
plot(x = time.dt, y = rep(q_0/(N+1), (N+1)), type = "l",  
      lty = 0, xlim = c(0, TM), ylim=c(0, q_0),  
      xlab = "Time_(t)", #ylab = "Position (q)",  
      ylab="", main = "Discrete_Case")
```

```
source("BisectionMethod.R")
```

```
solve.alpha <- function(x, dt, sigma, gamma, V, eta){  
  return(2*(cosh(x*dt)-1)-((gamma*(sigma**2)*V)/(2*eta))*dt**2)  
}
```

```
i <- 1; list.disc_rec <- list()
```

```
for(gamma in vec.gamma){  
  g.x <- function(x){solve.alpha(x = x, dt = dt, sigma = sigma,  
    gamma = gamma, V= V, eta = eta )}
```

```
  alpha <- bisection.method(0, 2*pi, g.x = g.x, tol = 10**-5,  
    max.iterations = 100000)
```

```
  y <- disc.q_star(q_0 = q_0, gamma = gamma, sigma = sigma,  
    V = V, eta = eta, TM = TM, t = time.dt,  
    N = N, alpha = alpha)  
  list.disc_rec[[i]] <- y  
  lines(x = time.dt, y = y, lty = i)  
  i <- i+1  
}
```

```
#####
```

```
plot(x = time.dt, y = (list.cont_rec[[1]] - list.disc_rec[[1]]),  
      type = "l", lty = 1, xlim = c(0, TM),  
      xlab = "Time_(t)", #ylab = "Position (q)"
```



```

        ylab="Difference between Trading Curves",
        main = paste0("Continuous vs. n Discrete")
    )

    for(i in 2:3){
        y <- list.cont_rec[[i]] - list.disc_rec[[i]]
        lines(x = time.dt, y = y, lty = i)
    }
    par(mfcol = c(1,1))

    bisection.method <-
        function(a0, b0, g.x = FUN, tol = 10**-5,
            max.iterations = 100000){

        i <- 0; an <- a0; bn <- b0
        g_a0 <- g.x(x = a0); g_b0 <- g.x(x = b0)

        while(i < max.iterations){
            cn <- an + (bn-an)/2
            g_cn <- g.x(x = cn)

            if(g_cn==0 | (bn-an)/2 < tol){
                root <- cn
                return(root)
                break
            }
            i = i + 1
            if(g_a0*g_cn > 0){an <- cn} else{bn <- cn}

            if(i>=max.iterations){
                print(
                    paste0("Error after ", max.iterations, " iterations.")
                )
            }
        }
    }
}

```

A.5.2 Deterministic vs. Stochastic Optimality

A further topic subjected to extension on Almgren-Chriss framework is the “deterministic” nature of the execution schedule: the whole trading curve over $[0, T]$ is obtained before the execution, thus a deterministic trading curve. The question that arises is if $v \in \mathcal{A}_{\text{det}}$ is really the ultimate best strategy in comparison to, say, a stochastic one.

Indeed, for the present framework no liquidation strategy can do better than the best deterministic strategy. Guéant (2016, p.47-48) shows what this can be proven by showing that:

$$\sup_{v \in \mathcal{A}} \mathbb{E}[-\exp(-\gamma X_T)] = \sup_{v \in \mathcal{A}_{\text{det}}} \mathbb{E}[-\exp(-\gamma X_T)] \quad (\text{A.11})$$

Proof. Stating that

$$\sup_{v \in \mathcal{A}} \mathbb{E}[-\exp(-\gamma X_T)] \leq \sup_{v \in \mathcal{A}_{\text{det}}} \mathbb{E}[-\exp(-\gamma X_T)] \quad (\text{A.12})$$

Since X_T is defined as:

$$X_T = X_0 + q_0 S_0 - \frac{k}{2} q_0^2 + \sigma \int_0^T q_t dW_t - \int_0^T V_t L \left(\frac{v_t}{V_t} \right) dt \quad (\text{A.13})$$

It is found that:

$$\begin{aligned} \mathbb{E}[-\exp(-\gamma X_T)] &= -\exp \left(-\gamma \left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 \right) \right) \times \\ &\quad \mathbb{E} \left[\exp \left(\gamma \left(\int_0^T V_t L \left(\frac{v_t}{V_t} \right) dt \right) \right) \exp \left(-\gamma \sigma \int_0^T q_t dW_t \right) \right] \end{aligned} \quad (\text{A.14})$$

The result above differs from the deterministic strategy because the utility function is no more evaluated by the concept of “indifference price” or “certainty equivalent” utility. To evaluate the stochastic component we can use the probability measure $\mathbb{Q}$ defined by the Radon-Nikodym derivative:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(-\gamma \sigma \int_0^T q_t dW_t - \frac{1}{2} \gamma^2 \sigma^2 \int_0^T q_t^2 dt \right), \quad (\text{A.15})$$

which modifies the utility function:

$$\begin{aligned} \mathbb{E}[-\exp(-\gamma X_T)] &= -\exp\left(-\gamma\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2\right)\right) \times \\ &\quad \mathbb{E}^{\mathbb{Q}}\left[\exp\left(\gamma\left(\int_0^T V_t L\left(\frac{v_t}{V_t}\right) dt\right)\right) \exp\left(\frac{1}{2}\gamma^2 \sigma^2 \int_0^T q_t^2 dt\right)\right] \end{aligned} \quad (\text{A.16})$$

Therefore,

$$\mathbb{E}[-\exp(-\gamma X_T)] = \mathbb{E}^{\mathbb{Q}}\left[-\exp\left(-\gamma\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - J(q)\right)\right)\right], \quad (\text{A.17})$$

where $J(q) = \int_0^T \left(V_t L\left(\frac{q'(t)}{V_t}\right) + \frac{\gamma}{2} \sigma^2 q(t)^2\right) dt$ is defined almost surely because, by definition, q is almost surely in the set of absolutely continuous functions $q \in W^{1,1}(0, T)$.

Using the fact that q^* , the optimal strategy, is unique is possible to write:

$$\mathbb{E}[-\exp(-\gamma X_T)] \leq -\exp\left(-\gamma\left(X_0 + q_0 S_0 - \frac{k}{2} q_0^2 - J(q^*)\right)\right) \leq \sup_{v \in \mathcal{A}_{\det}} \mathbb{E}[-\exp(-\gamma X_T)]. \quad (\text{A.18})$$

And, finally, by taking the supremum on the left-hand side, we get:

$$\sup_{v \in \mathcal{A}} \mathbb{E}[-\exp(-\gamma X_T)] \leq \sup_{v \in \mathcal{A}_{\det}} \mathbb{E}[-\exp(-\gamma X_T)], \quad (\text{A.19})$$

Finally proving the true optimality of a deterministic strategy, specially over a stochastic one. It is quite surprising that in a dynamic and unstable locus such as intra-day trading a deterministic strategy would be superior to more flexible, conditional, adaptive policy. And that is so because each step of the mathematical formulation is firmed in assumptions such as the maximization criterion and the nature deterministic nature of the liquidity cost function (2.11).